

# Analysis of OMICS data

## Lecture #6 - Proteomics

Identification of proteins

Part 1/3: Introduction

20mar-2026

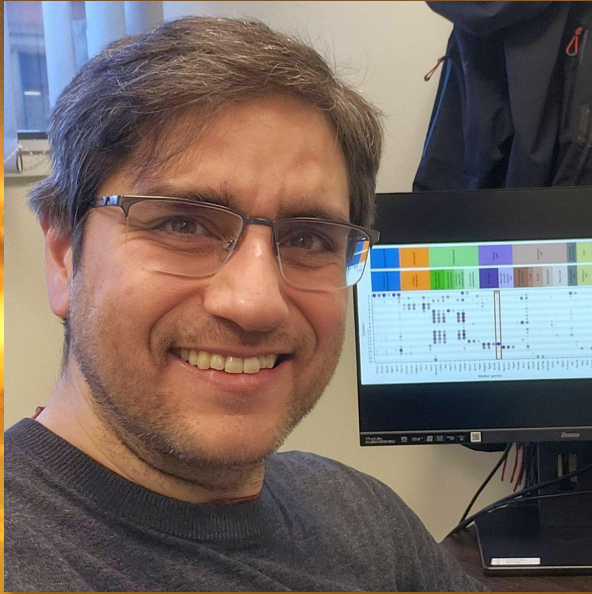


**Marco Trevisan-Herraz**

marco.trevisan@newcastle.ac.uk

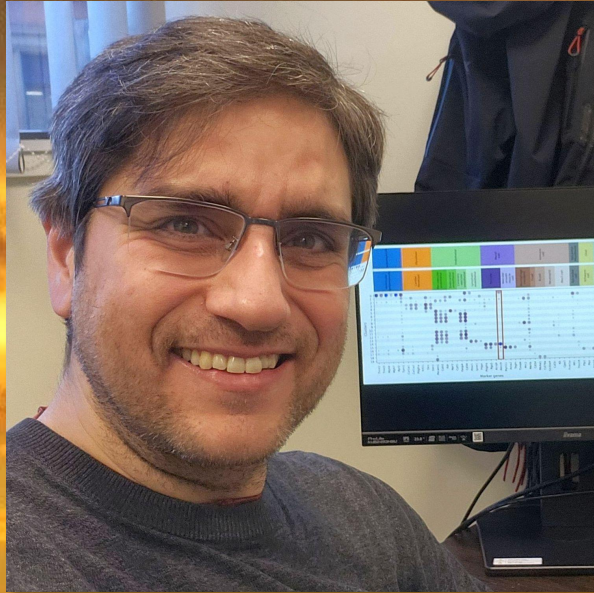
# Marco Trevisan-Herraz, PhD

Researcher in  
computational biology





# Marco Trevisan-Herraz, PhD

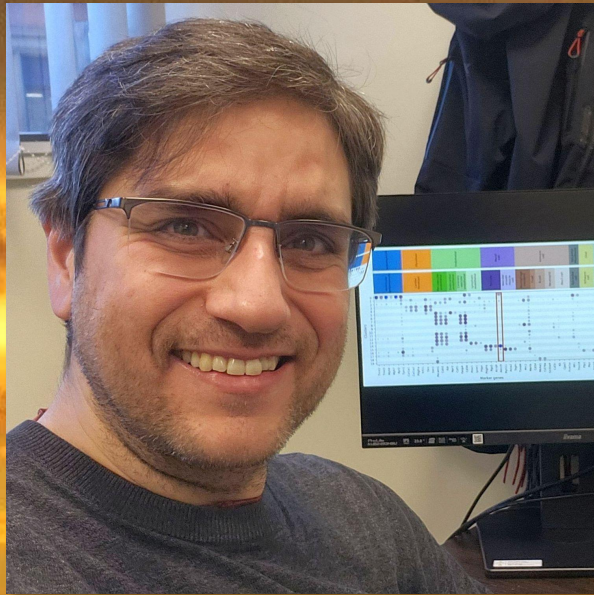


**Researcher in  
computational biology**

**with a background in physics,  
proteomics and epigenomics**



# Marco Trevisan-Herraz, PhD



**Researcher in  
computational biology**

**with a background in physics,  
proteomics and epigenomics**

**interested in understanding  
ageing in cardiovascular models.**



## First part: short intro

### 10 interesting things to know about proteomics

- 1) the five omics
- 2) why proteomics
- 3) some recent wins in proteomics
- 4) bioinformatics in proteomics is quite different than other bioinformatics
- 5) the primary structure of proteins

## Second part: identification

- 6) Western blot
- 7) first approach to identify peptides
- 8) digestion of proteins: from proteins to peptides
- 9) missed cleavages
- 10) mass spectra of peptides
- 11) peptide mass fingerprinting
- 12) MSMS or fragmentation
- 13) protein databases
- 14) monoisotopic and average mass
- 15) variable and fixed modifications
- 16) isotopic envelopes and the binomial distribution
- 17) XCorr and DeltaCn

## Third part: search your own spectrum

- 18) Overview of methods to quantify proteins
- 19) Silac labelling
- 20)  $^{18}\text{O}$  labelling
- 21) iTRAQ / TMT

## (Fourth part: quantitative proteomics - only if enough time)

### 4 interesting things to do about proteomics

- I. getting a sequence of proteins from a public database
- II. making an *in silico* digestion
- III. calculating the  $\text{MH}^+$  of a peptide
- IV. identifying the peptide in a fragmentation spectrum
- V. playing with iTRAQ data using MSnbase

# Part I

A very short introduction  
to bioinformatics in proteomics



# the Central Dogma of Molecular Biology

genes

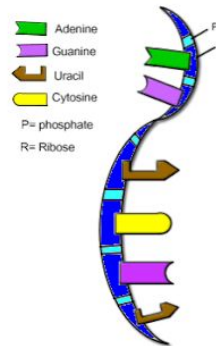


genomics

**what could the  
cell do?**



mRNA

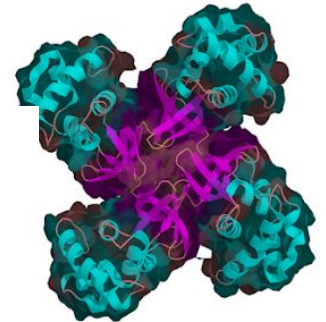


transcriptomics

**what is the cell  
going to do?**



proteins



proteomics

**what is the cell  
*actually* doing?**

# ~~the Central Dogma of Molecular Biology~~

## the five omics

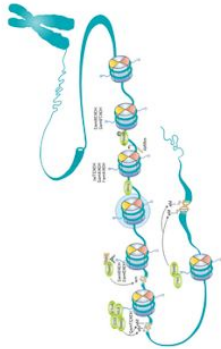
genes



genomics

**what could the  
cell do?**

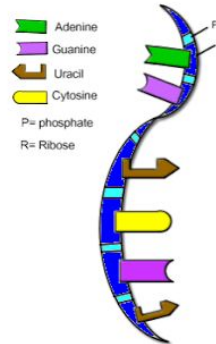
chromatin



epigenomics

**what intends  
the cell to do?**

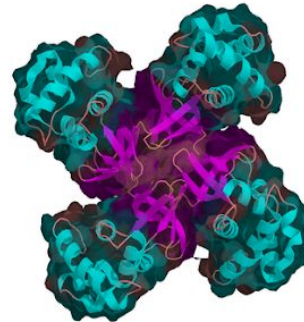
mRNA



transcriptomics

**what is the cell  
going to do?**

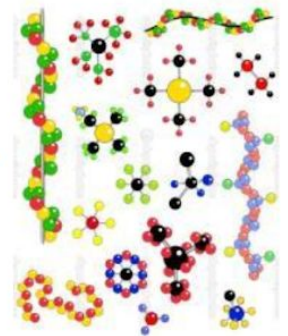
proteins



proteomics

**what is the cell  
actually doing?**

metabolites



metabolomics

**what did the  
cell do?**

Is this sportsman  
cheating?

What happens to  
this mouse when  
we give it this  
drug?

Why did this  
person have a  
cardiac arrest?

Has this person  
this disease?

How can we  
prevent  
Alzheimer's  
disease?

Is this plant  
using this  
metabolic  
pathway to  
produce this  
drug?

Was this person  
killed using this  
poison?



Is this sportsman  
cheating?

What happens to  
this mouse when  
we give it this  
drug?

Why did this  
person have a  
cardiac arrest?

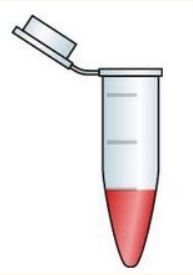
Has this person  
this disease?

# what are the cells doing?

How can we  
prevent  
Alzheimer's  
disease?

Is this plant  
using this  
metabolic  
pathway to  
produce this  
drug?

Was this person  
killed using this  
poison?



Is this sportsman  
cheating?

What happens to  
this mouse when  
we give it this  
drug?

Why did this  
person have a  
cardiac arrest?

Has this person  
this disease?

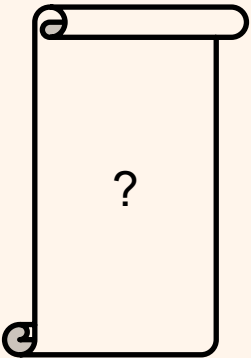
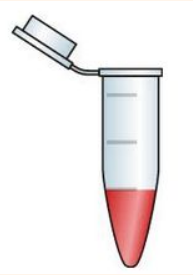
**what are the  
cells doing?**

How can we  
prevent  
Alzheimer's  
disease?

Is this plant  
using this  
metabolic  
pathway to  
produce this  
drug?

Was this person  
killed using this  
poison?

**what is in  
that sample?**



# Some recent proteomics “wins”

 Frontiers in Immunology

TYPE Original Research  
PUBLISHED 08 August 2025  
DOI 10.3389/fimmu.2025.1641289

 Check for updates

**OPEN ACCESS**

EDITED BY  
Laurent Réna, Nanyang Technological University, Singapore

REVIEWED BY  
Paulo J.G. Bettencourt, Faculty of Medicine, Portugal  
Zi Wei Chang, A\*STAR Infectious Disease Labs, Singapore

\*CORRESPONDENCE  
Robert W. Siegel  
✉ siegel\_robert@lilly.com  
Laurent P. Malherbe  
✉ malherbe\_laurent@lilly.com

RECEIVED 04 June 2025  
ACCEPTED 24 July 2025  
PUBLISHED 08 August 2025

CITATION  
Balasubramanian A, Prachar M, Klaproth B, Copeland V, Justesen S, Wen Y, Siegel RW and Malherbe LP (2025) Epitope mapping using immunopeptidomics reveals novel immunodominant CD8 T cell epitopes of the AAV9 capsid. *Front. Immunol.* 16:1641289. doi: 10.3389/fimmu.2025.1641289

COPYRIGHT  
© 2025 Balasubramanian, Prachar, Klaproth, Copeland, Justesen, Wen, Siegel and Malherbe

## Epitope mapping using immunopeptidomics reveals novel immunodominant CD8 T cell epitopes of the AAV9 capsid

Akhila Balasubramanian<sup>1</sup>, Marek Prachar<sup>2</sup>, Birgit Klaproth<sup>2</sup>, Victoria Copeland<sup>1</sup>, Sune Justesen<sup>2</sup>, Yi Wen<sup>1</sup>, Robert W. Siegel<sup>3\*</sup> and Laurent P. Malherbe<sup>3\*</sup>

<sup>1</sup>Lilly Research Laboratories, Eli Lilly and Company, Indianapolis, IN, United States, <sup>2</sup>Lilly Oncology, Eli Lilly and Company, Copenhagen, Denmark

**Introduction:** Adeno-associated virus (AAV)-mediated gene therapy is a promising approach to treat genetic disorders, offering advantages such as high transduction efficiency, diverse tissue tropism, and an acceptable safety profile. However, the immunogenicity of AAV vectors, particularly the activation of AAV capsid-specific CD8 T cells, significantly impacts therapeutic efficacy and safety. Capsid-specific T cell responses are currently monitored in the clinic using overlapping peptides, which do not represent the naturally presented capsid immunopeptidome. Our previous work identified the naturally presented peptides of AAV capsids using MHC-associated peptide proteomics (MAPPs).

**Methods:** In this study, we compared overlapping and MAPPs-derived peptides

## #1 - Balasubramanian 2025

<https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2025.1641289/full>

**Safer gene therapies:** they found nine previously unknown signs of a virus used to deliver gene therapy that the immune system can detect.

Proteomics was essential because other techniques captured different T cell responses; only immunopeptidomics directly measured the peptides actually presented on leukocyte antigens.

# Some recent proteomics “wins”



RESEARCH ARTICLE



## Chemoproteomics validates selective targeting of *Plasmodium* M1 alanyl aminopeptidase as an antimalarial strategy

Carlo Giannangelo<sup>1†</sup>, Matthew P Challis<sup>1†</sup>, Ghizal Siddiqui<sup>1</sup>, Rebecca Edgar<sup>2,3</sup>, Tess R Malcolm<sup>4,5</sup>, Chaille T Webb<sup>4,5</sup>, Nyssa Drinkwater<sup>4,5</sup>, Natalie Vinh<sup>6</sup>, Christopher Macraill<sup>1</sup>, Natalie Counihan<sup>2,3</sup>, Sandra Duffy<sup>7</sup>, Sergio Wittlin<sup>8,9</sup>, Shane M Devine<sup>10,11</sup>, Vicky M Avery<sup>7,12</sup>, Tania De Koning-Ward<sup>2,3</sup>, Peter Scammells<sup>6</sup>, Sheena McGowan<sup>4,5</sup>, Darren J Creek<sup>1\*</sup>

<sup>1</sup>Drug Delivery, Disposition and Dynamics, Monash Institute of Pharmaceutical Sciences, Monash University, Parkville, Australia; <sup>2</sup>School of Medicine, Deakin University, Geelong, Australia; <sup>3</sup>The Institute for Mental and Physical Health and Clinical Translation, Deakin University, Geelong, Australia; <sup>4</sup>Monash Biomedicine Discovery Institute and Department of Microbiology, Monash University, Clayton, Australia; <sup>5</sup>Centre to Impact AMR, Monash University, Clayton, Australia; <sup>6</sup>Medicinal Chemistry, Monash Institute of Pharmaceutical Sciences, Monash University, Parkville, Australia; <sup>7</sup>Discovery Biology, Centre for Cellular Phenomics, Griffith University, Nathan, Australia; <sup>8</sup>Swiss Tropical and Public Health Institute, Allschwil, Switzerland; <sup>9</sup>University of Basel, Basel, Switzerland; <sup>10</sup>The Walter and Eliza Hall Institute of Medical Research, Parkville, Australia; <sup>11</sup>Department of Medical Biology, The University of Melbourne, Parkville, Australia; <sup>12</sup>School of Environment and Science, Griffith University, Nathan, Australia

\*For correspondence:  
darren.creek@monash.edu

<sup>†</sup>These authors contributed  
equally to this work

**Competing interest:** The authors  
declare that no competing  
interests exist.

**Funding:** See page 26

Preprint posted  
17 August 2023

Sent for Review  
27 October 2023

**Abstract** New antimalarial drug candidates that act via novel mechanisms are urgently needed to combat malaria drug resistance. Here, we describe the multi-omic chemical validation of *Plas-*

## #2 - Giannangelo 2024

<https://elifesciences.org/articles/92990>

**New strategies to beat malaria:**  
protein analysis confirms that malaria parasites can be selectively killed by blocking a crucial enzyme they need to survive.

Proteomics was essential for the target profiling; crystallography and metabolomics helped afterward, but were secondary.



# Some recent proteomics “wins”

npj | precision oncology

Published in partnership with The Horrel Institute, University of Minnesota

Case report



<https://doi.org/10.1038/s41698-024-00696-6>

## Proteomics based selection achieves complete response to HER2 therapy in HER2 IHC 0 breast cancer

Check for updates

Laura E. Johnston<sup>1</sup>, Jamie Randall<sup>1</sup>, Safae Chouraichi<sup>1</sup>, Mary Luu<sup>2</sup>, Allison L. Hunt<sup>2</sup>, Lauren Mauro<sup>1</sup>, Claudius Mueller<sup>3</sup>, Justin B. Davis<sup>3</sup>, Emanuel F. Petricoin<sup>4</sup>, Thomas P. Conrads<sup>2</sup>, Timothy L. Cannon<sup>1</sup> & Jasmine Huynh<sup>1</sup>✉

Recent trials have shown the efficacy of trastuzumab deruxtecan (T-DXd) in HER2-negative patients, but there is not yet a way to identify which patients will best respond, especially with the inability of current HER2 IHC and FISH assays to accurately determine HER2 expression in the unamplified setting. Here, we present a heavily pre-treated patient with triple-negative breast cancer (HER2 IHC 0 who had a complete response to T-DXd. In this case, we used a CLIA-certified reverse-phase protein array-based proteomic assay (RPPA) to determine that the patient had moderate HER2 protein expression (HER2<sup>total</sup> 2+, 42%) and activation (HER2<sup>Y1248</sup> 1+, 23%). Using these results, we determined that the patient may benefit from T-DXd despite being traditionally qualified as HER2 IHC 0. These findings highlight the potential for proteomics-based assays that may more accurately quantitate HER2 and (its activation) in the *HER2* unamplified/IHC 0 setting to better select patients whose tumors are classically molecularly defined as HER2 IHC 0, but still could respond to HER2-directed therapy, and give patients access to therapies which for which they otherwise would not be eligible.

### #3 - Johnston 2024

<https://www.nature.com/articles/s41698-024-00696-6>

**Fighting breast cancer:** Standard testing suggested this breast cancer would not respond to the drug, but a more detailed protein analysis showed the drug target was still active.

Proteomics was essential for this case because standard other techniques missed the actionable signal.

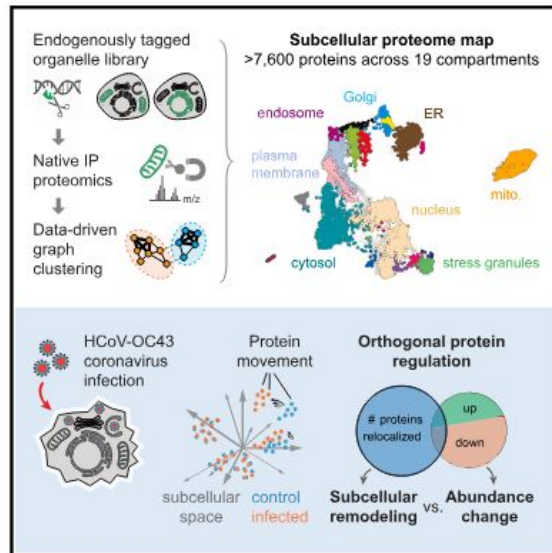
# Some recent proteomics “wins”

Cell

Resource

## Global organelle profiling reveals subcellular localization and remodeling at proteome scale

### Graphical abstract



### Highlights

- Native immunoprecipitation proteomics deployed to a cell-wide organelle collection
- >7,600 proteins resolved across 19 subcellular structures and organellar interfaces

### Authors

Marco Y. Hein, Duo Peng, Verina Todorova, ..., Daniel N. Itzhak, Joshua E. Elias, Manuel D. Leonetti

### Correspondence

marco.hein@maxperutzlabs.ac.at (M.Y.H.), duo.peng@czbiohub.org (D.P.), josh.elias@czbiohub.org (J.E.E.), manuel.leonetti@czbiohub.org (M.D.L.)

### In brief

Organelle proteomics defines the cellular landscape of protein localization and highlights the role of subcellular remodeling in driving responses to perturbations such as viral infections.

#4 - Hein 2024

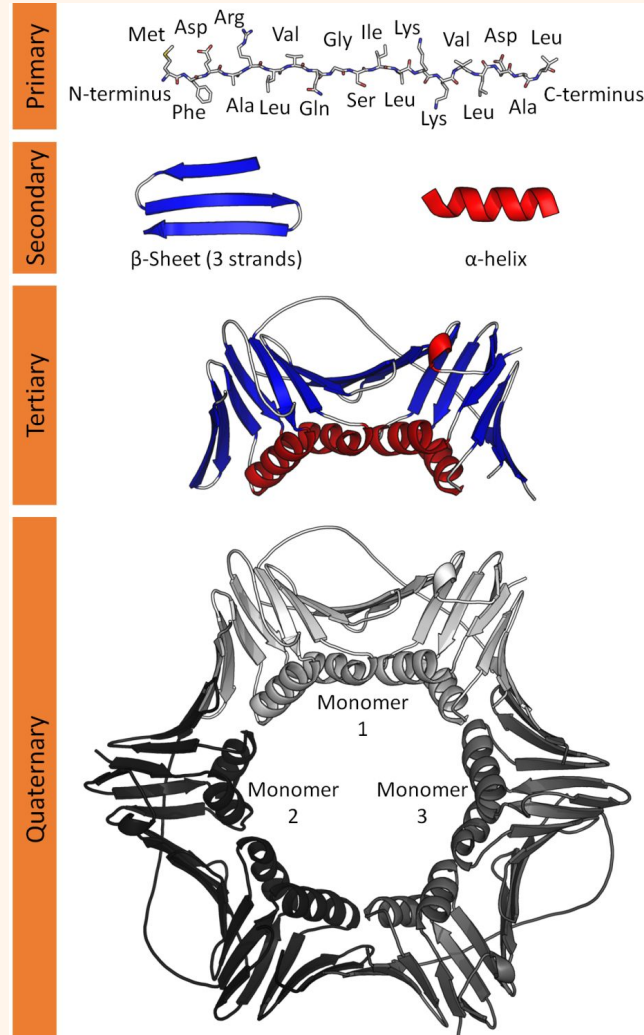
<https://www.cell.com/cell/fulltext/S0092-8674%2824%2901344-8>

**Understanding how eukaryotic cells work:** showed that eukaryotic cells are organized by large-scale protein relocalisation between compartments (rather than protein abundance), a proteome-wide principle that had not previously been mapped

Proteomics was essential because it's the only way to assign thousands of proteins to compartments and track whole-proteome relocalization in parallel.

What is "important" in  
a protein?

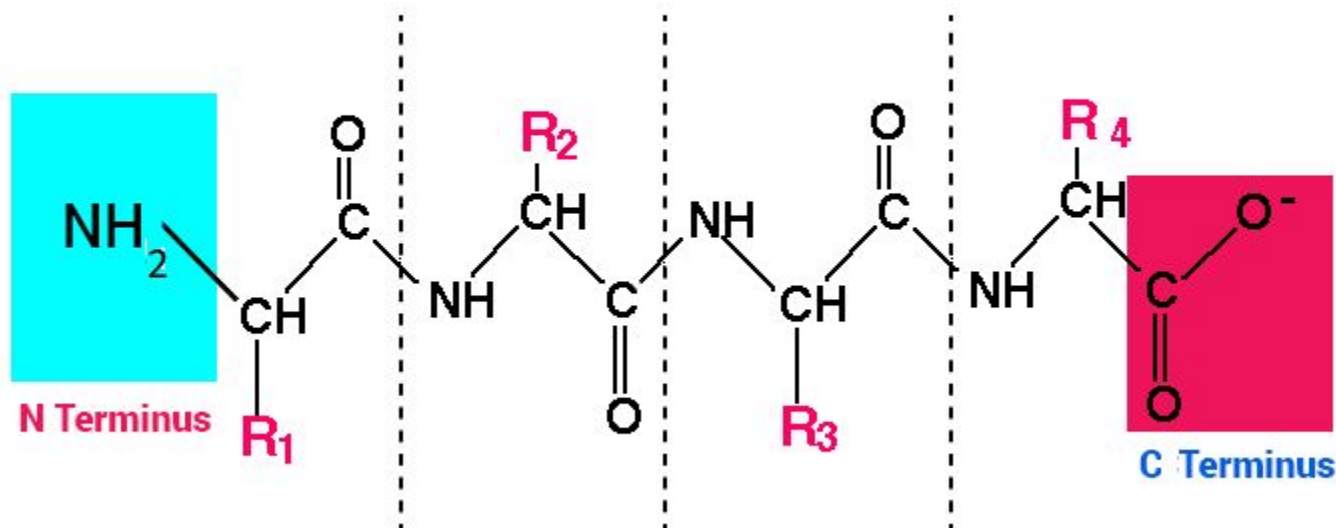
# the primary structure of proteins





**the primary structure of proteins:  
not just a sequence of amino acids!**

## PEPTIDES



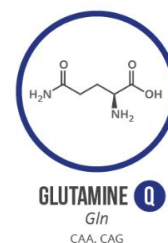
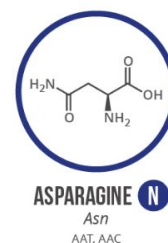
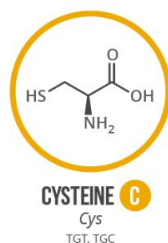
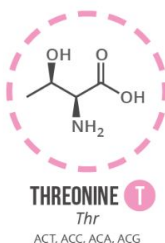
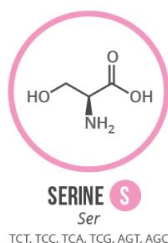
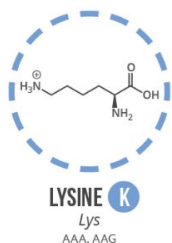
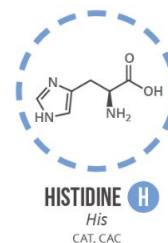
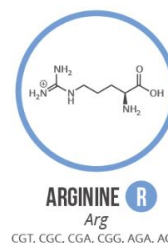
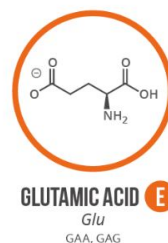
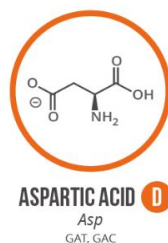
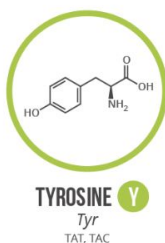
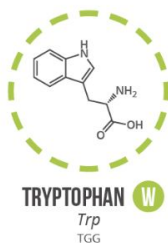
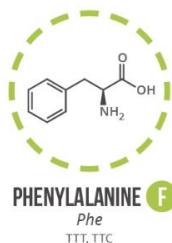
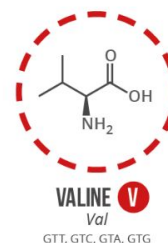
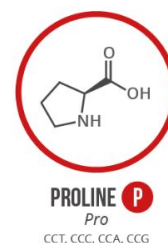
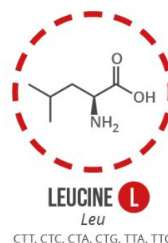
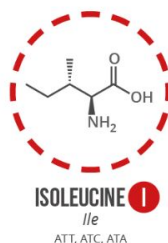
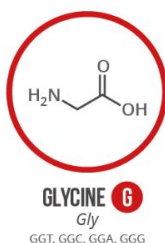
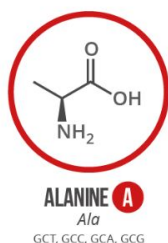
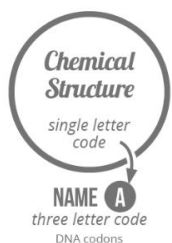
Source:  
<https://byjus.com/chemistry/peptides/>

# the primary structure of proteins: not just a sequence of amino acids!

## A GUIDE TO THE TWENTY COMMON AMINO ACIDS

AMINO ACIDS ARE THE BUILDING BLOCKS OF PROTEINS IN LIVING ORGANISMS. THERE ARE OVER 500 AMINO ACIDS FOUND IN NATURE - HOWEVER, THE HUMAN GENETIC CODE ONLY DIRECTLY ENCODES 20. 'ESSENTIAL' AMINO ACIDS MUST BE OBTAINED FROM THE DIET, WHILST NON-ESSENTIAL AMINO ACIDS CAN BE SYNTHESISED IN THE BODY.

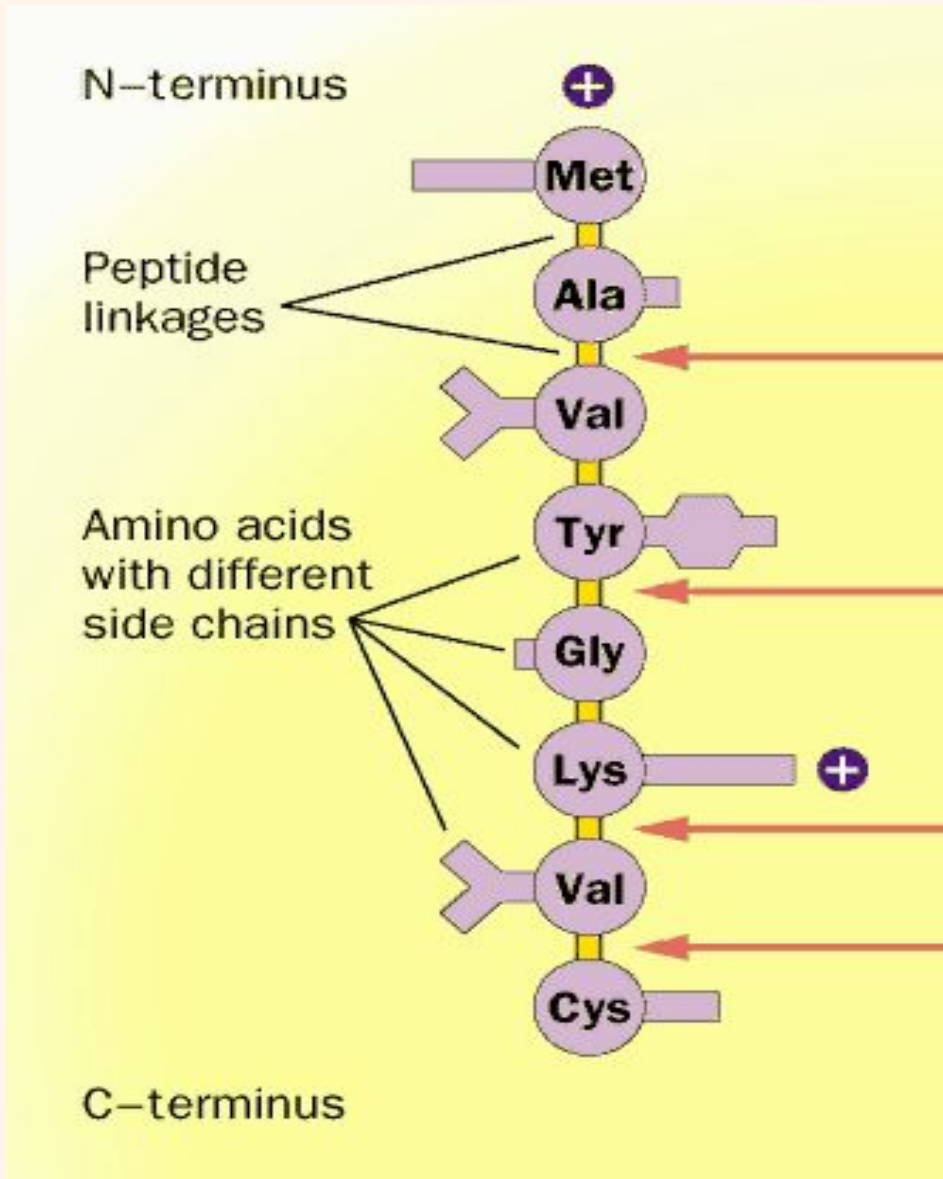
**Chart Key:** ● ALIPHATIC ● AROMATIC ● ACIDIC ● BASIC ● HYDROXYLIC ● SULFUR-CONTAINING ● AMIDIC ○ NON-ESSENTIAL ○ ESSENTIAL



**Note:** This chart only shows those amino acids for which the human genetic code directly codes for. Selenocysteine is often referred to as the 21st amino acid, but is encoded in a special manner. In some cases, distinguishing between asparagine/aspartic acid and glutamine/glutamic acid is difficult. In these cases, the codes asx (B) and glx (Z) are respectively used.

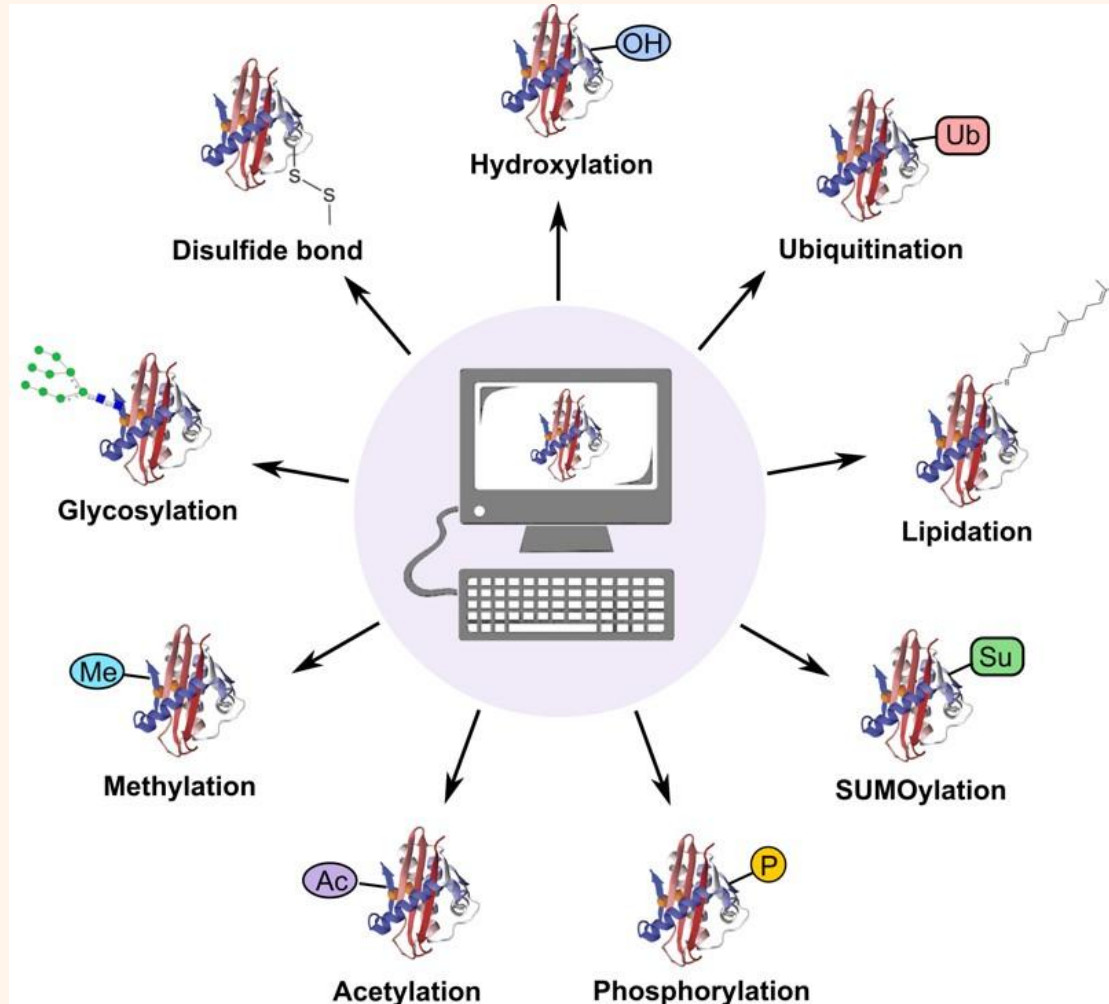


# the primary structure of proteins: not just a sequence of amino acids!



by convention, from  
Nterm to Cterm

# the primary structure of proteins: not just a sequence of amino acids!



Post-translational modifications  
the "dark matter of proteomics"

image from  
profacgen.com

# End of Part I

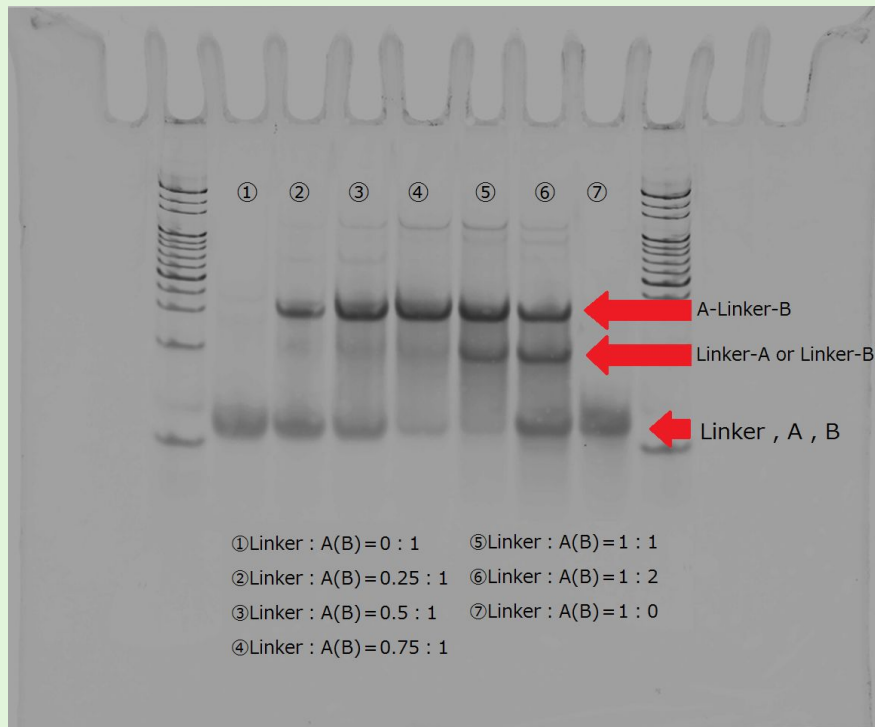
Part II will be about  
the challenge of protein identification



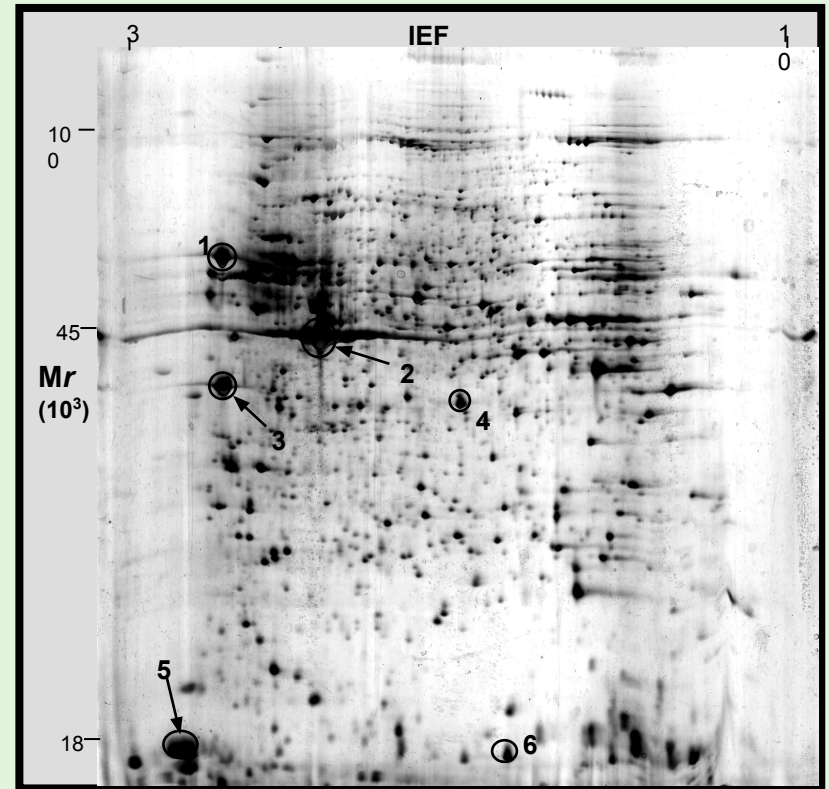
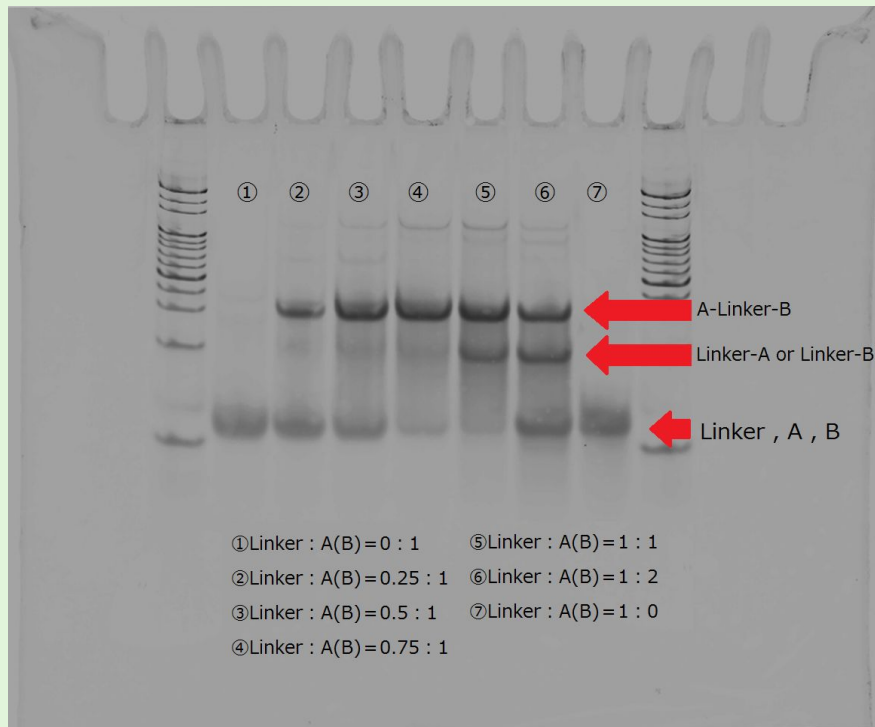
# Part II

The challenge of protein identification

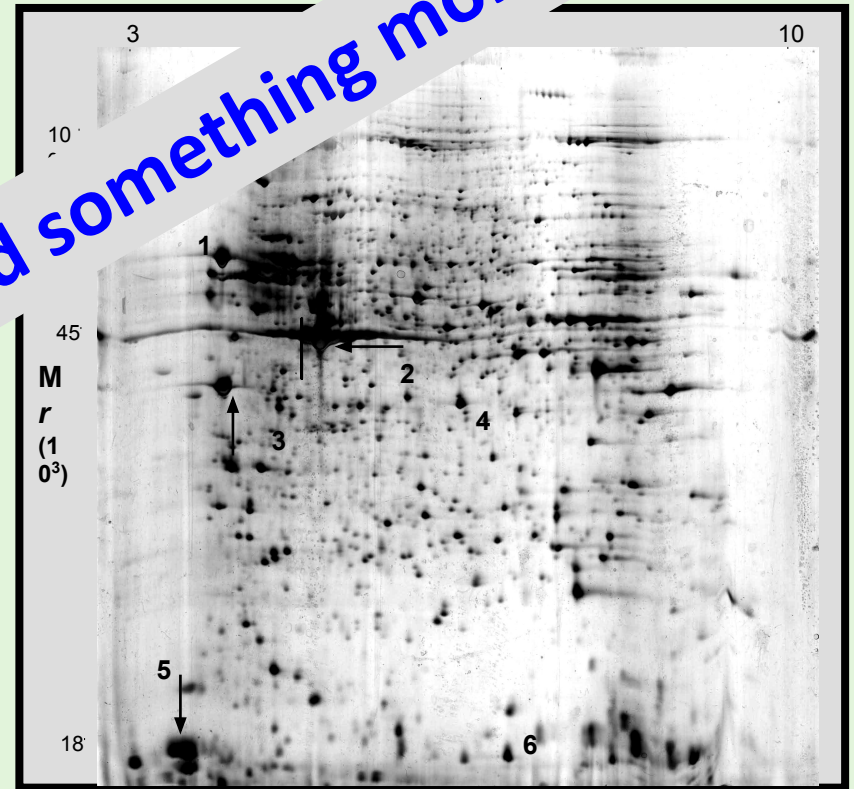
# Western blot



# Western blot and 2D gel electrophoresis



# Western blot and 2D gel electrophoresis



But what if we need something more?

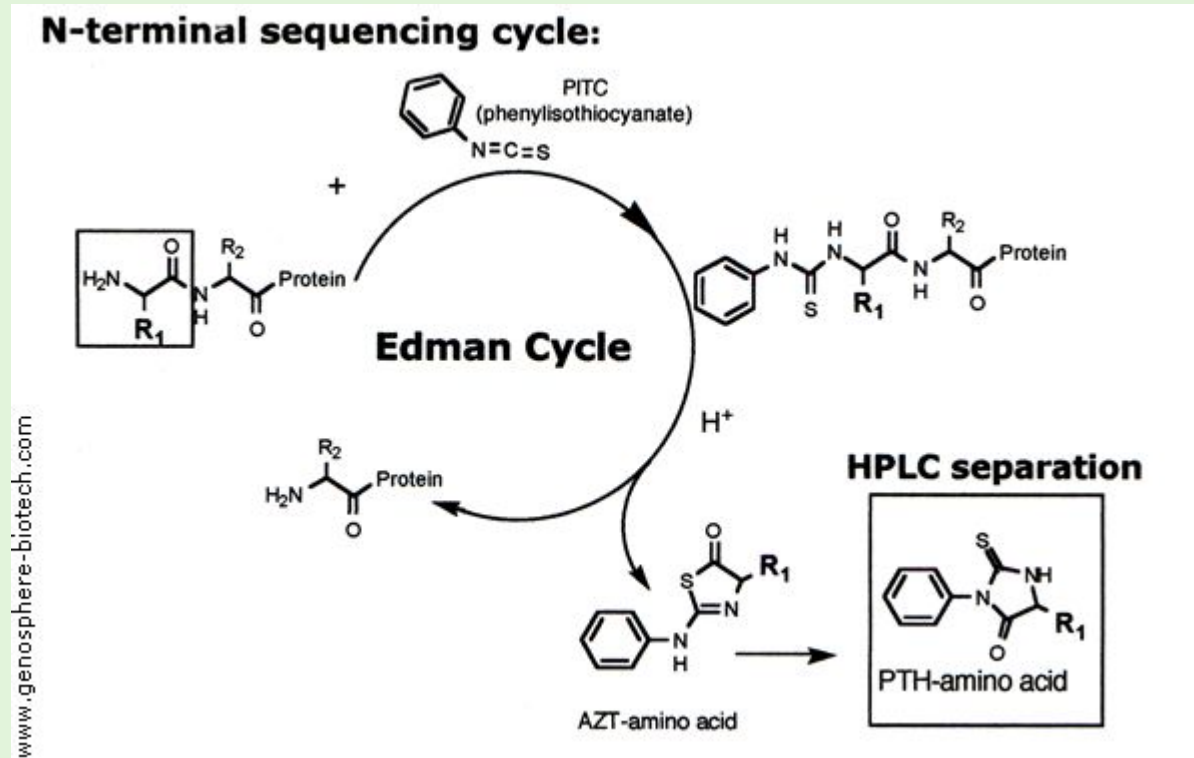
# #1) How do we identify proteins?

- A) We sequence their amino acids with Edman sequencing
- B) We use infrared spectroscopy
- C) We use X-ray crystallography
- D) We check the weight of their components
- E) None of the above



# first approach to identify peptides

## Edman degradation



# Some mass spectrometres



LTQ Orbitrap XL



Q Exactive



MALDI TOF



Orbitrap Fusion



TripleTOF

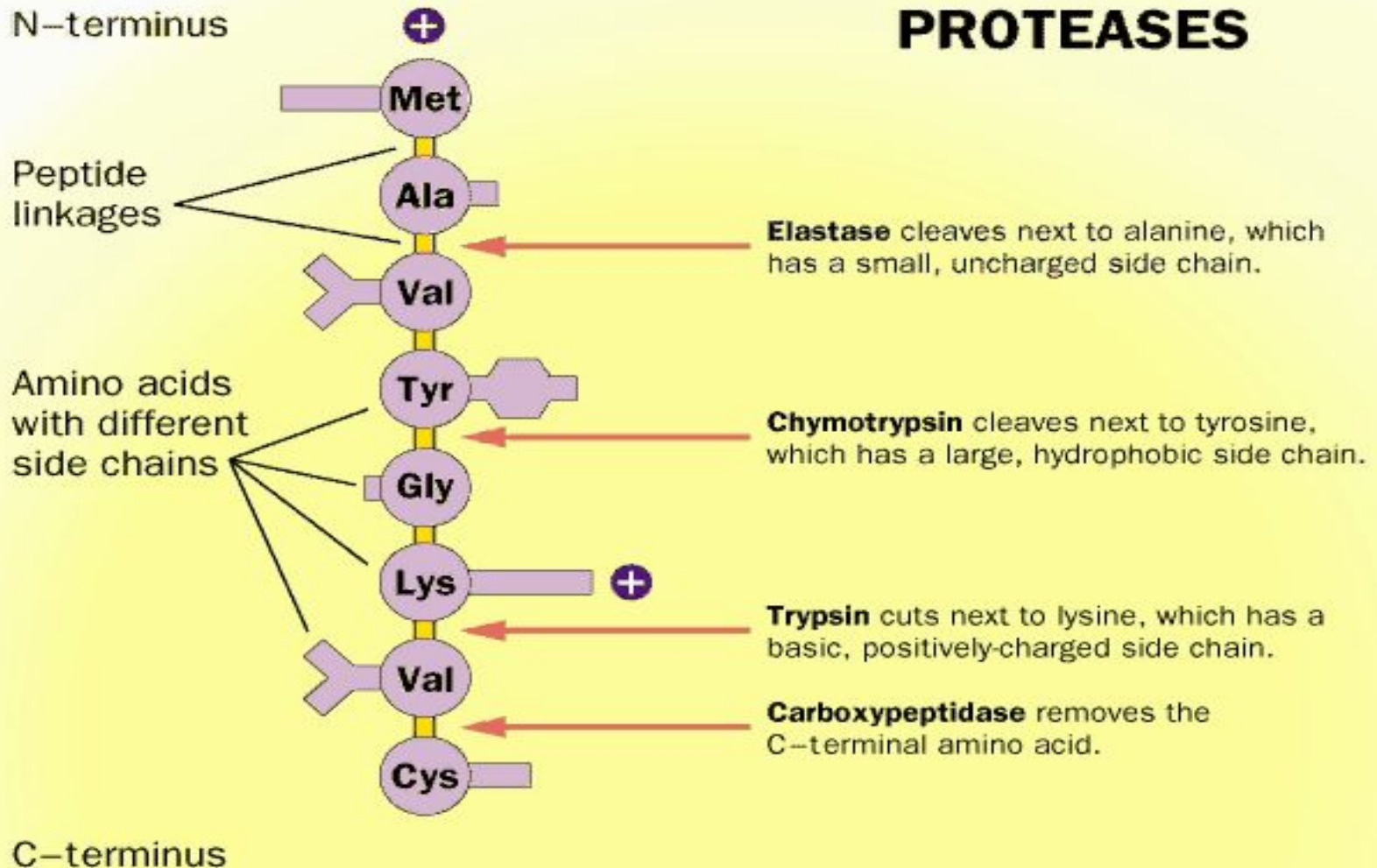
# #2) How do we get proteins into the mass spectrometer?

- A) We freeze them with liquid nitrogen to maintain their structure, then we make small clusters that can enter the mass spectrometer
- B) We dilute the proteins in water or benzene (depending on their polar properties)
- C) We use porcine-made nanobots that cut proteins always at arginine and proline, so we can work with shorter peptides instead
- D) We heat the proteins at 70 °C, and add formic acid 0.3M, so they are denatured; then, they enter the mass spectrometer using a thin tube
- E) None of the above

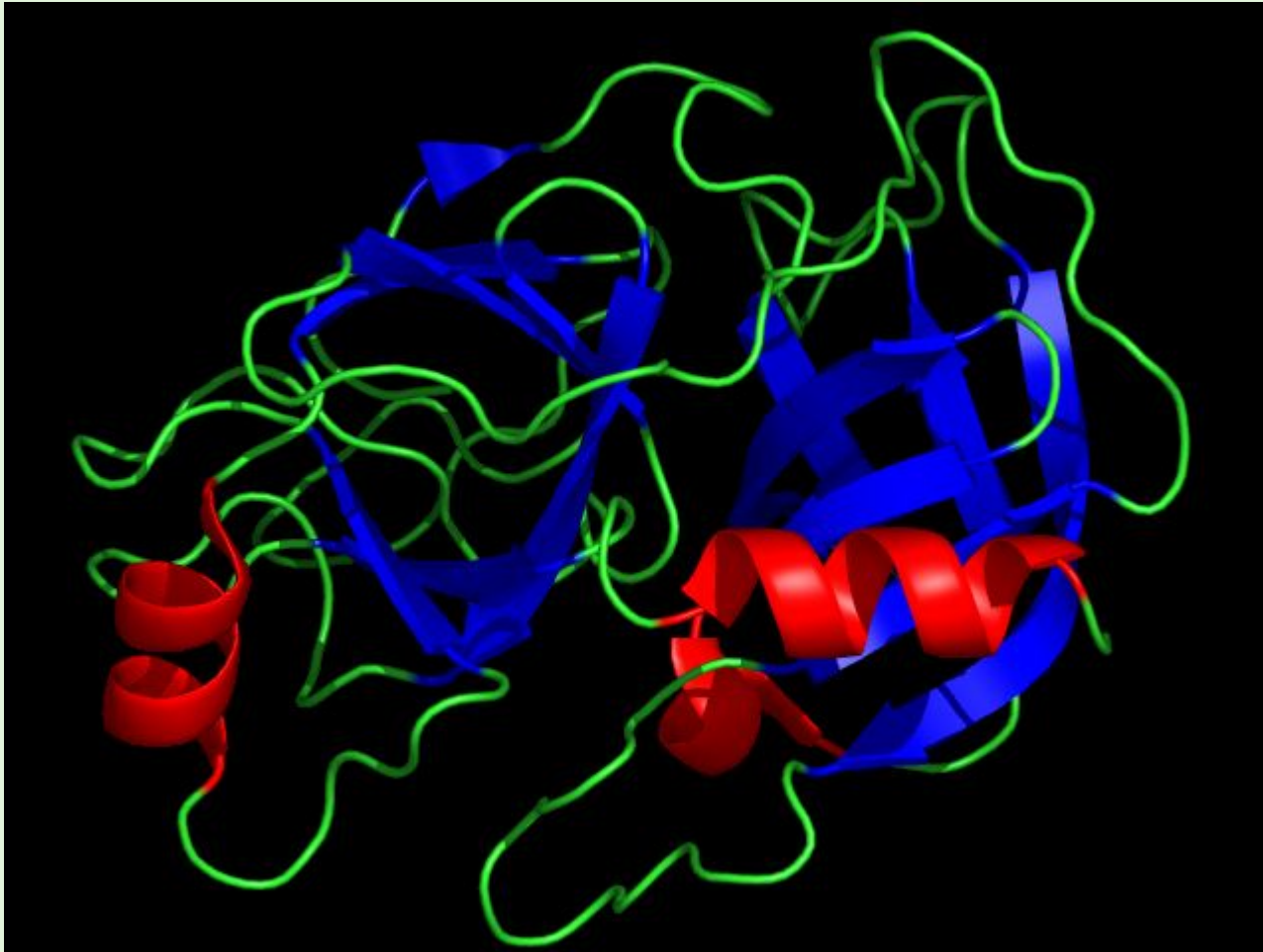
# digestion of proteins: from proteins to peptides

N-terminus

## PROTEASES



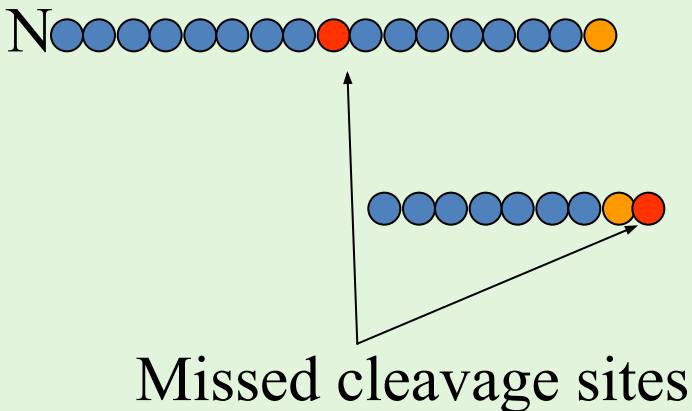
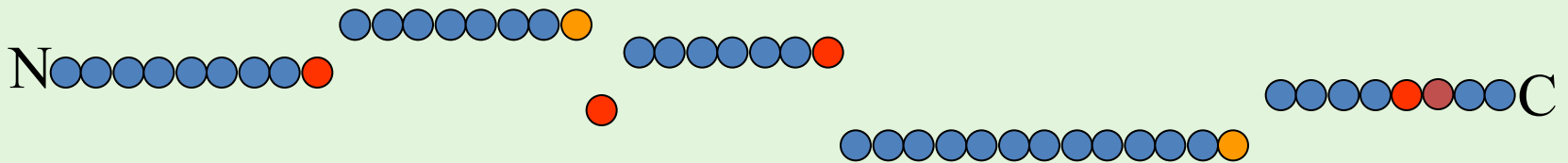
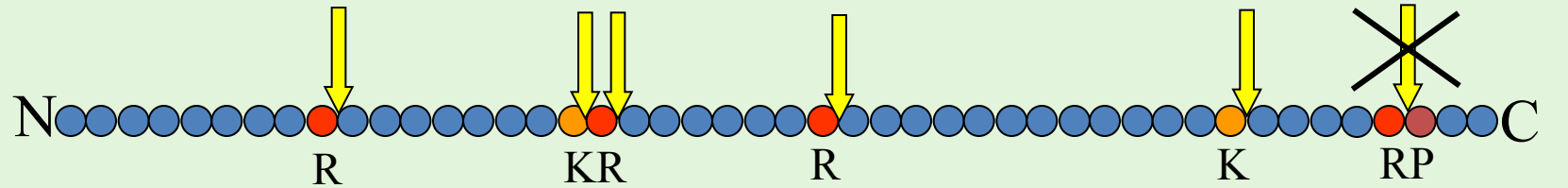
# digestion of proteins: from proteins to peptides



trypsin



# digestion of proteins: from proteins to peptides



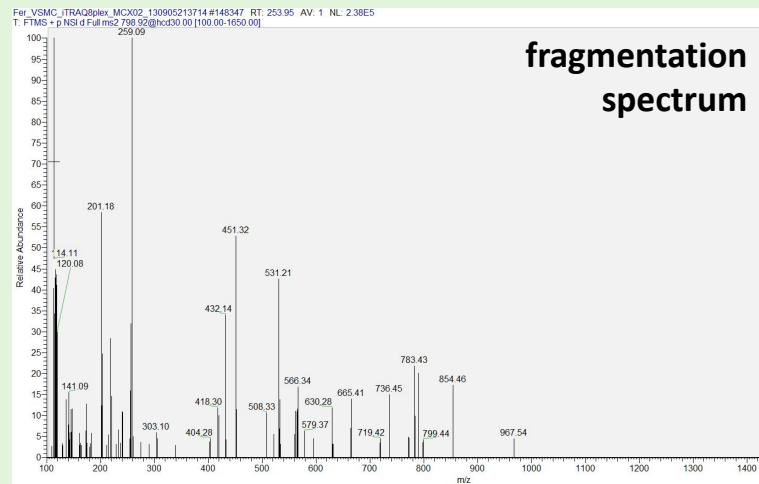
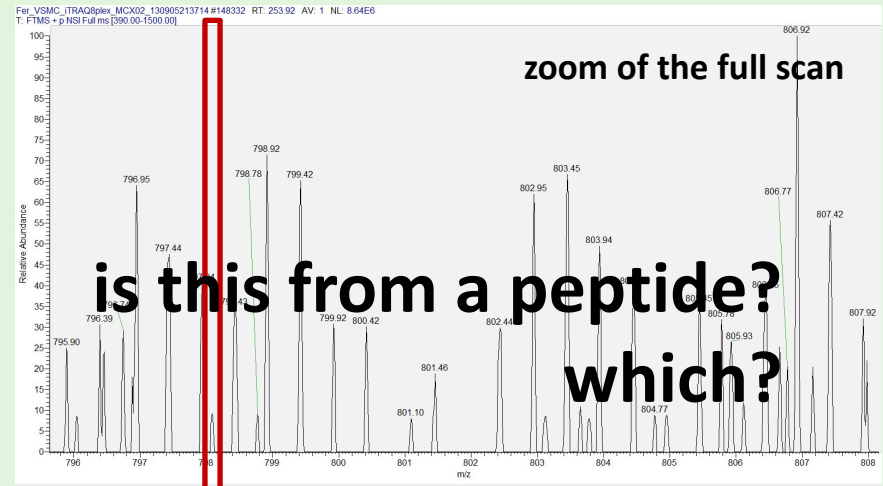
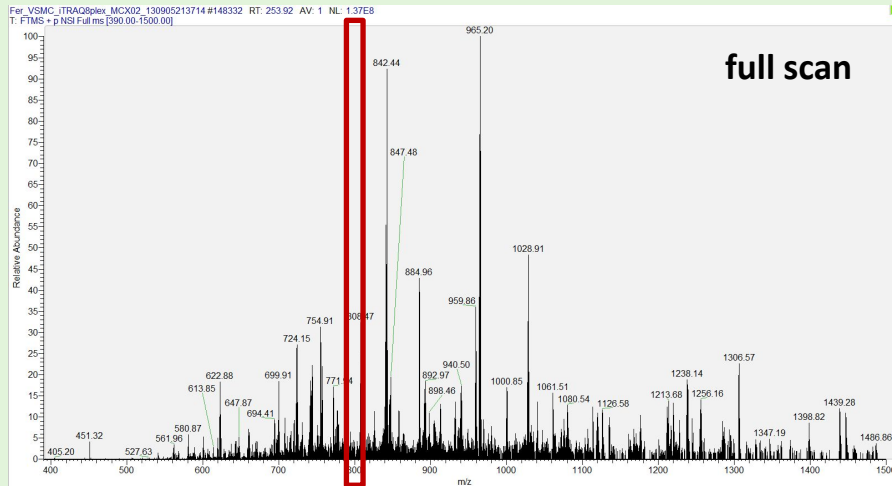
Digestion using trypsin

# **digestion of proteins: from proteins to peptides**

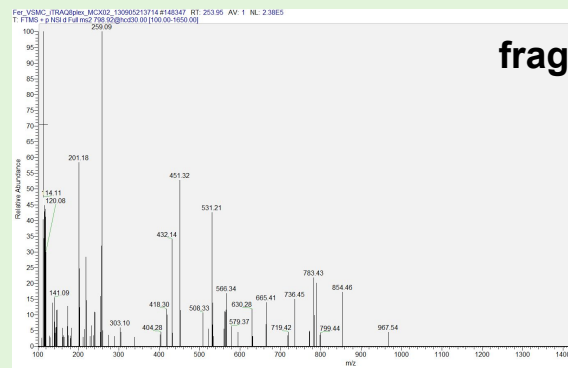
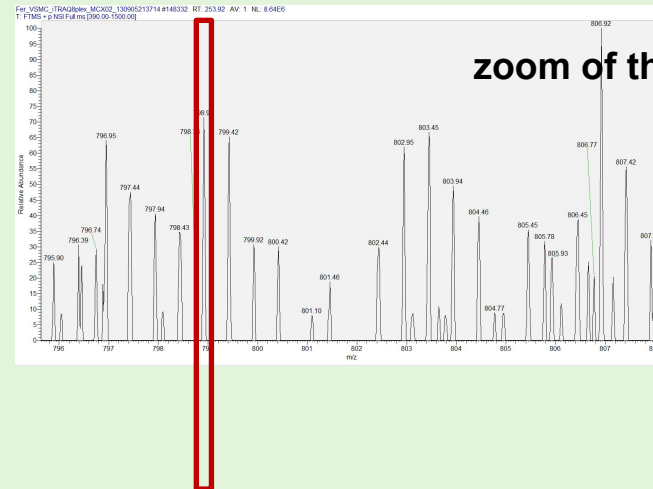
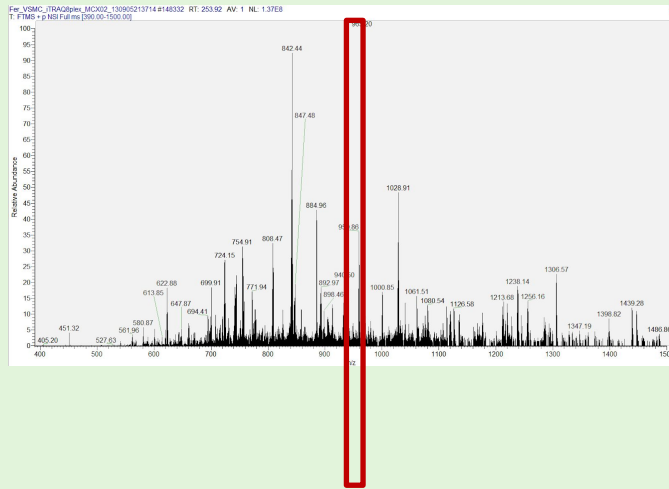
## Missed Cleavages

- Digests are usually not perfect
- Cleavage sites may be missed by an enzyme
- These partially cleaved peptides are known as partials
- Reduce the discrimination of a search

# mass spectra of peptides

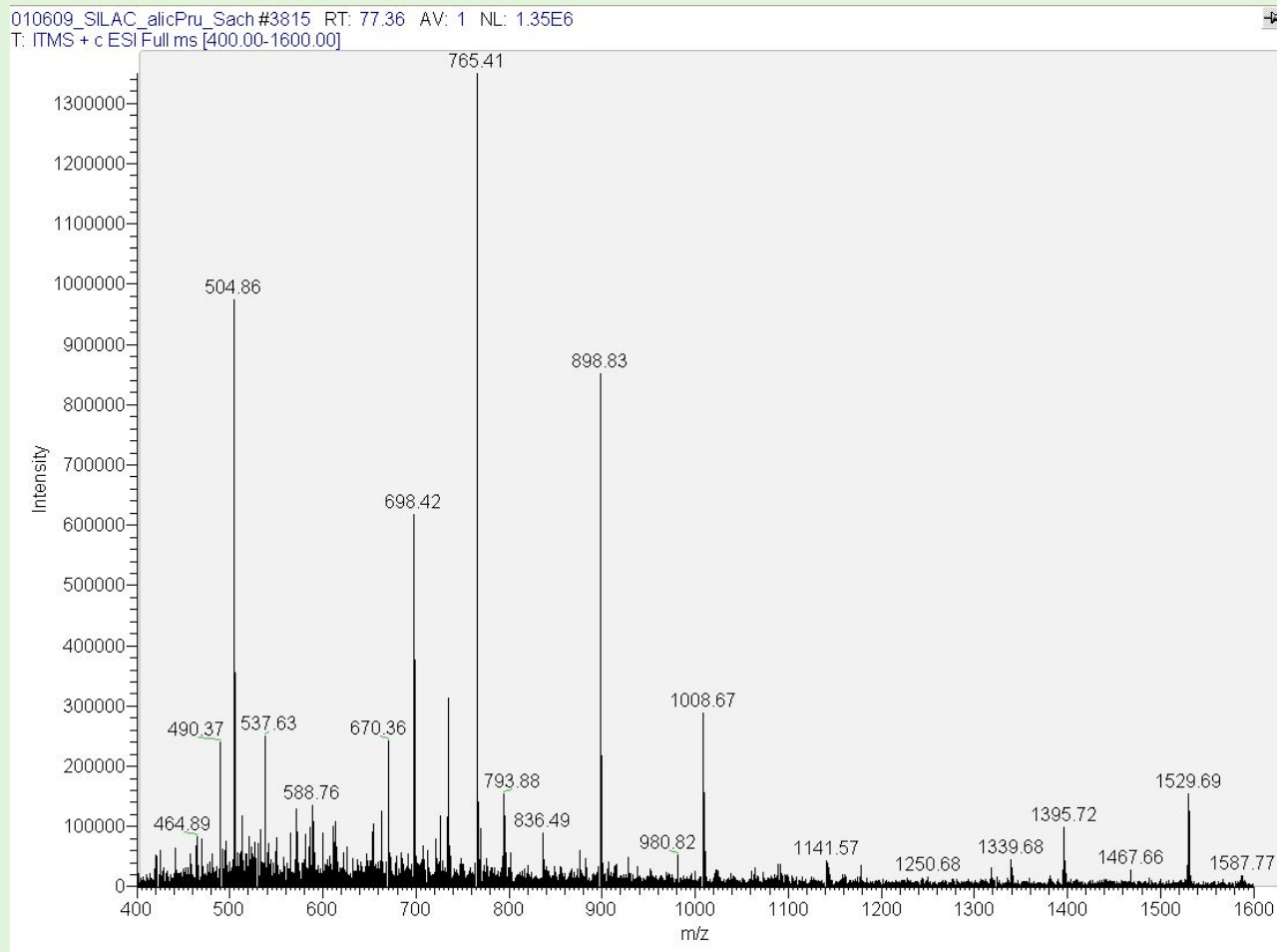


# Mass spectra of peptides



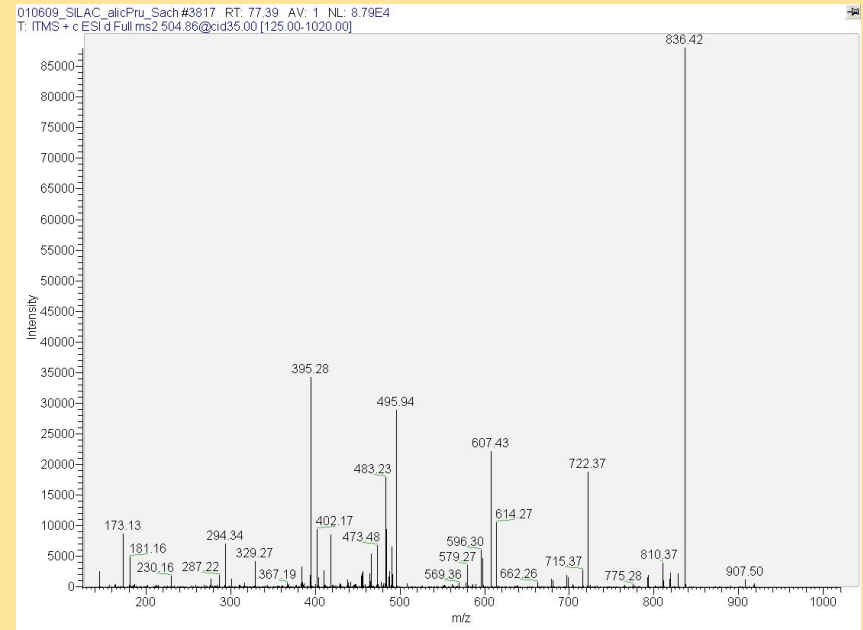
is this from a peptide? which?

# Full scan (MS1)



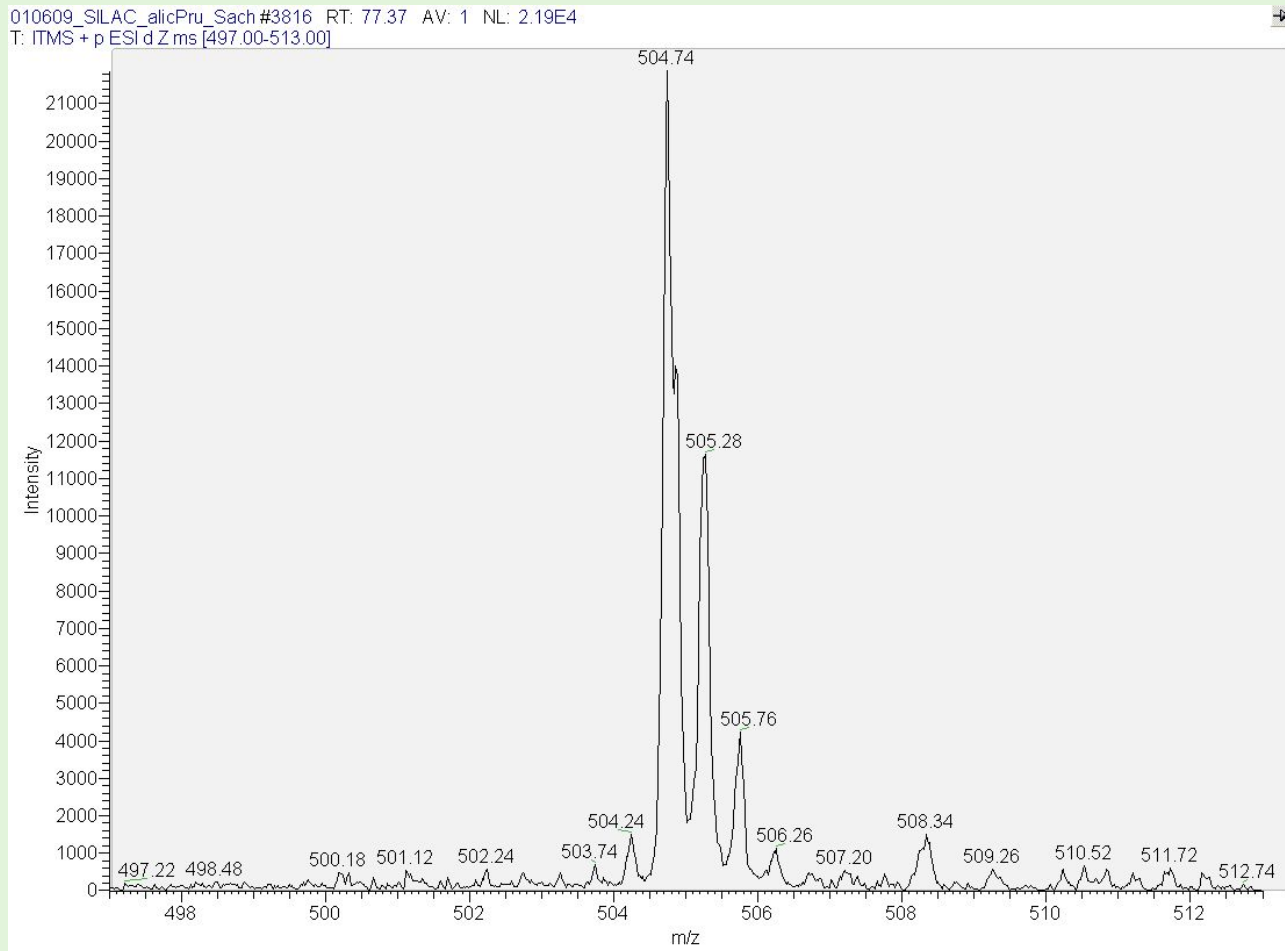


# #3) What are these peaks?

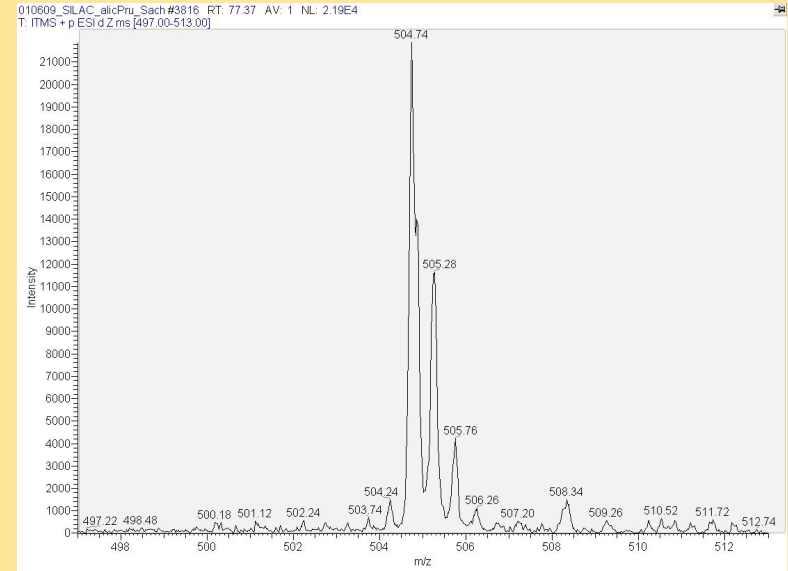


- A) The X-axis is the mass/amino acid ratio, the Y-axis is the intensity of the light emitted.
- B) The X-axis are the picograms of each protein in every cell, the Y axis is the intensity of the protein measured
- C) The X-axis is mass/charge ratio, the Y-axis is amount of ions counted
- D) None of the above

# Zoom scan (MS1)



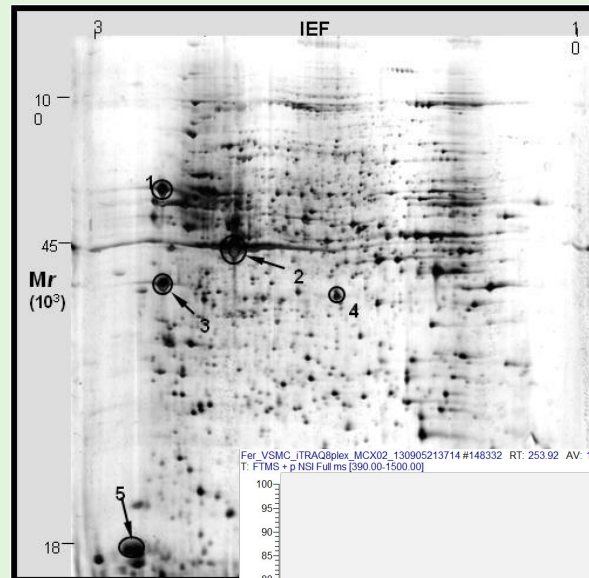
# #4) What are these peaks with very similar m/z?



- A) Different peptides that just by chance have very similar mass
- B) The same peptide, but with some contaminant
- C) Different peptides, with the same amino acids, but in different order
- D) The same peptide, but with different isotopic combinations

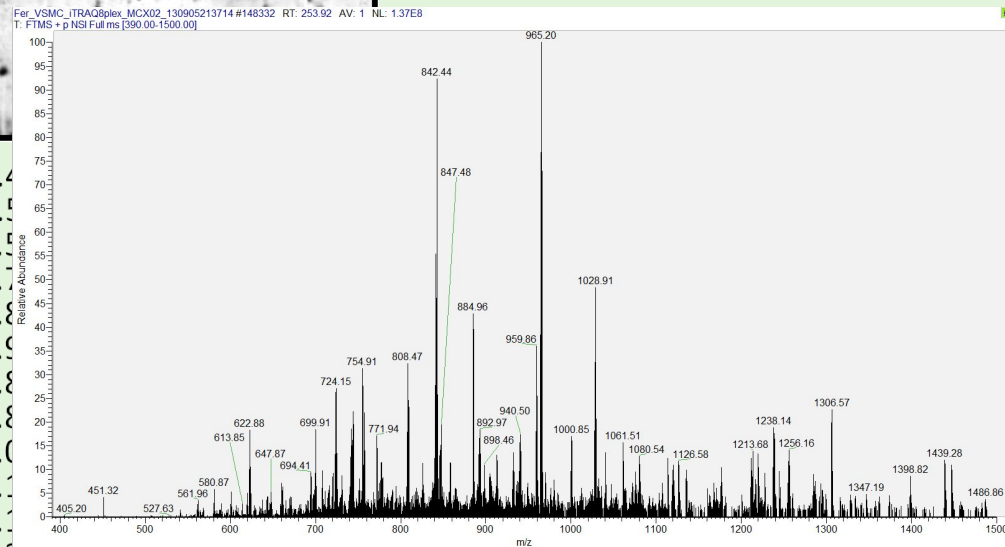
# **from the spectrum to the identification: peptide mass fingerprinting**

# from the spectrum to the identification: peptide mass fingerprinting



## List of masses $(M+H)^+$

|              |              |              |              |
|--------------|--------------|--------------|--------------|
| 784.3418140  | 786.3842737  | 791.3324814  | 802.4        |
| 868.0212272  | 920.5367162  | 930.5019057  | 980.5        |
| 1002.5186292 | 1030.0570454 | 1035.9943063 | 1104.5       |
| 1157.5212345 | 1223.0031412 | 1257.7214116 | 1292.7       |
| 1320.6705505 | 1336.7215360 | 1378.6730204 | 1380.8       |
| 1400.8334328 | 1417.7440014 | 1439.7449235 | 1443.9       |
| 1452.8004541 | 1475.8468711 | 1478.7756116 | 1515.8       |
| 1557.8880162 | 1618.9662799 | 1640.8552052 | 1695.8       |
| 1853.9439990 | 1868.9176883 | 1890.8955786 | 1972.0       |
| 2004.9713883 | 2013.9972541 | 2105.0723419 | 2211.1       |
| 2280.1405386 | 2348.2985678 | 2352.2265273 | 2374.1       |
| 2429.2204631 | 2452.1348603 | 2586.1954157 | 2907.3866417 |
| 2940.5052161 | 2961.3846998 | 2977.3692220 | 3084.3946593 |



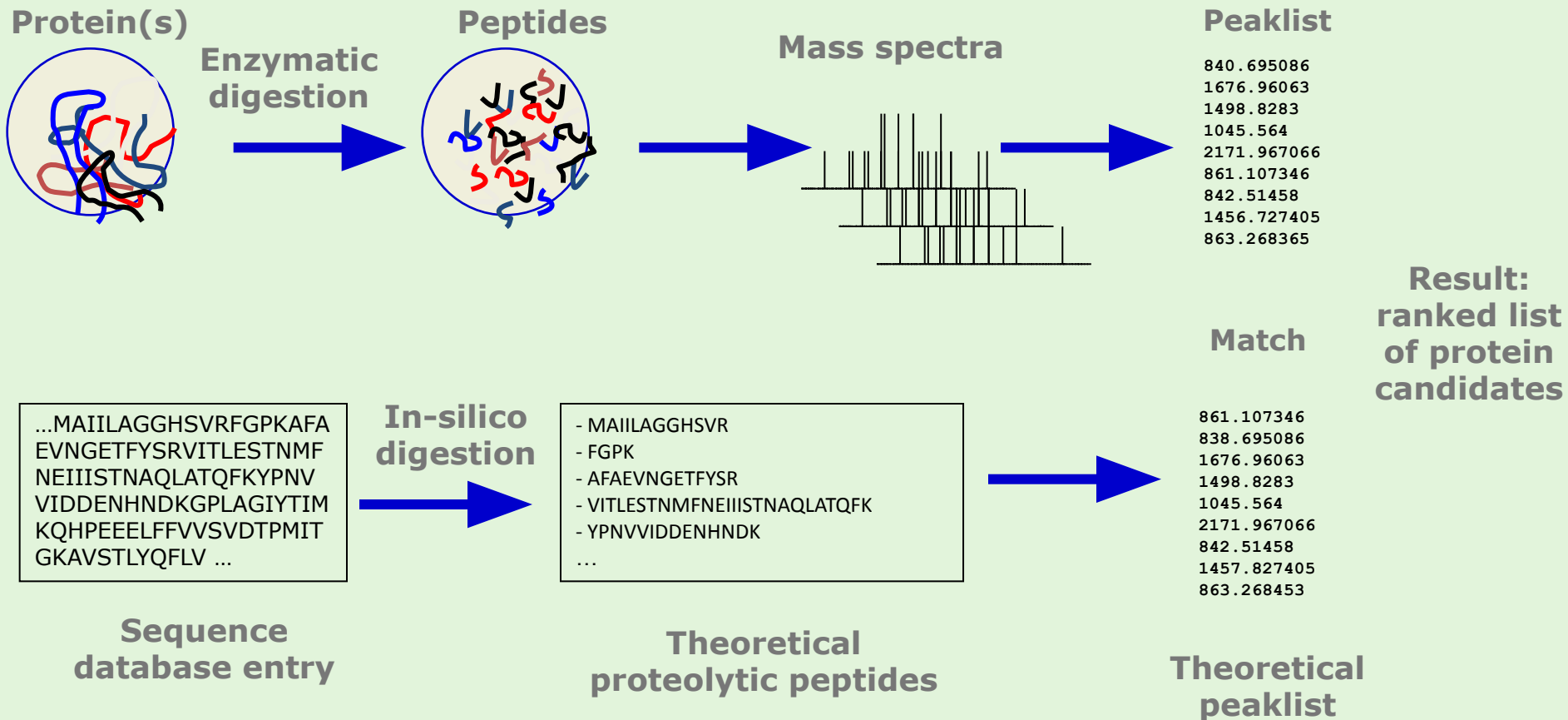
2929.3468854

# **from the spectrum to the identification: peptide mass fingerprinting**

- Produce a theoretical digest of all the proteins in a database with a specific enzyme
- Compare these theoretical masses with experimentally observed masses
- Assign a score to matching peptides/proteins



# from the spectrum to the identification: peptide mass fingerprinting



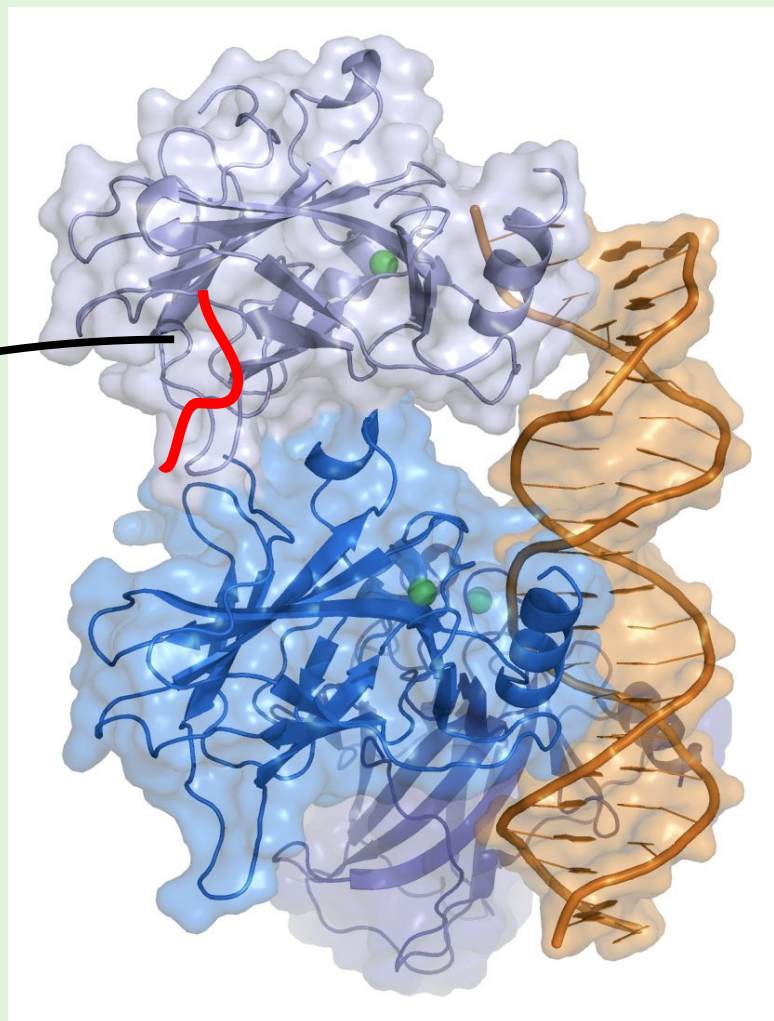
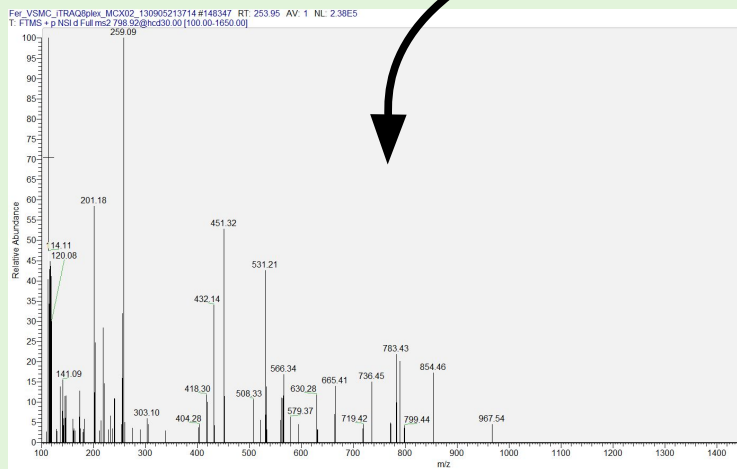
But PMF has a big  
problem...

But PMF has a big  
problem...

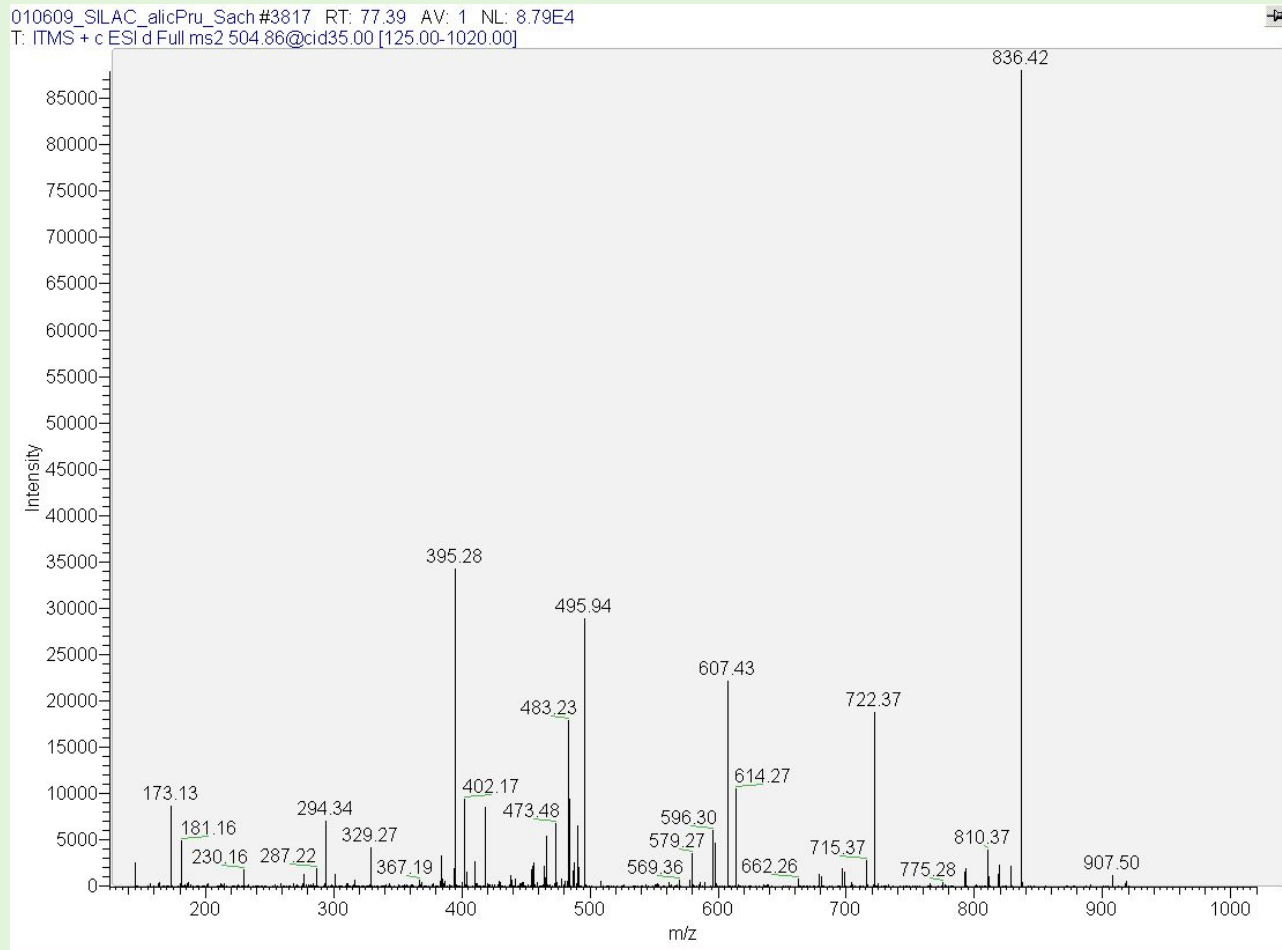
When you have more than  
one protein in the sample, it's  
a huge computational mess

**from the spectrum  
to the identification:  
fragmentation**

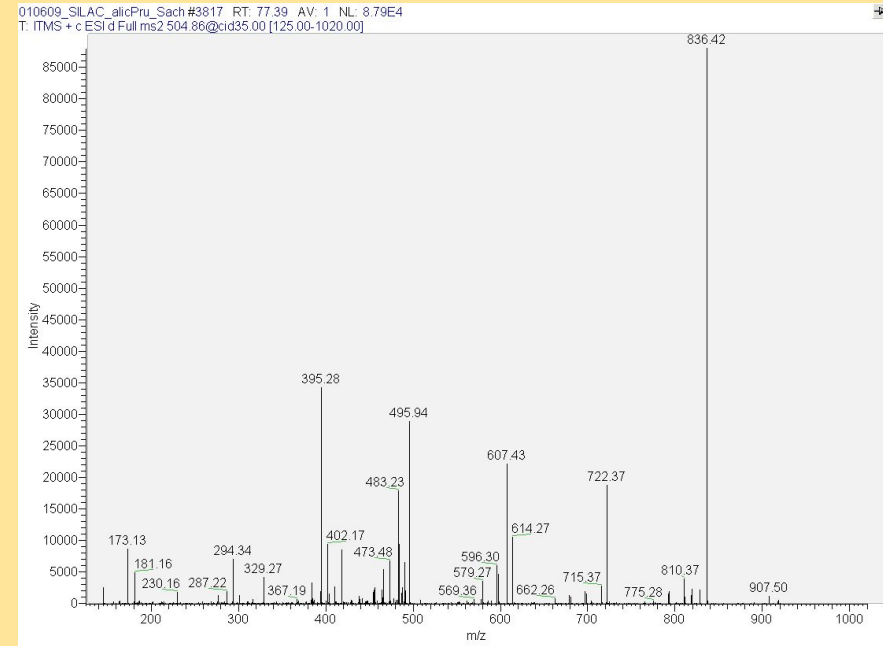
# Interpreting fragmentation spectra



# Fragmentation spectrum (MS2)



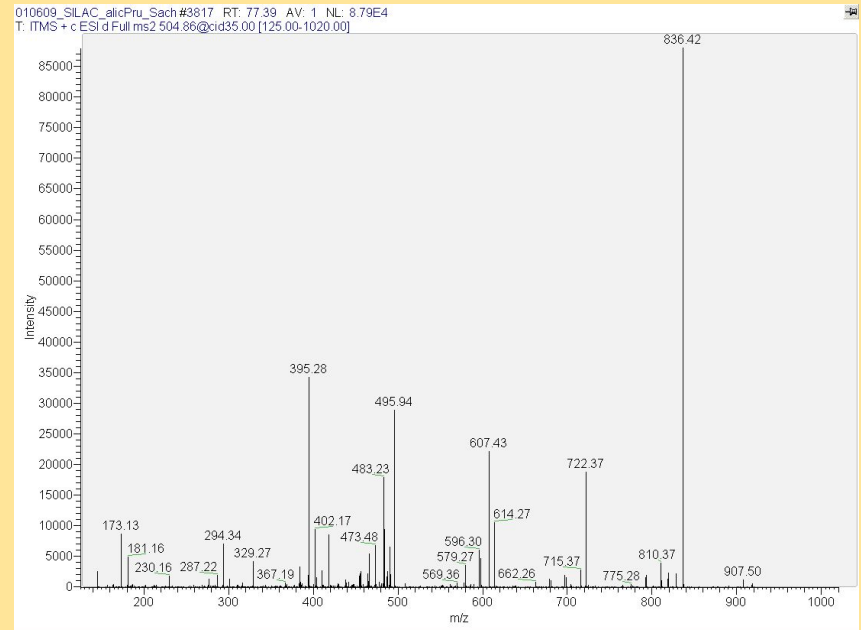
# #5) How do we break peptides?



- A) We use helium bullets that break them into two pieces
- B) We use an ultraviolet laser
- C) We use strong magnetic fields (9.4 Tesla) that bend them to their breaking point
- D) All of the above



# #6) OK, we have the spectra, now what?

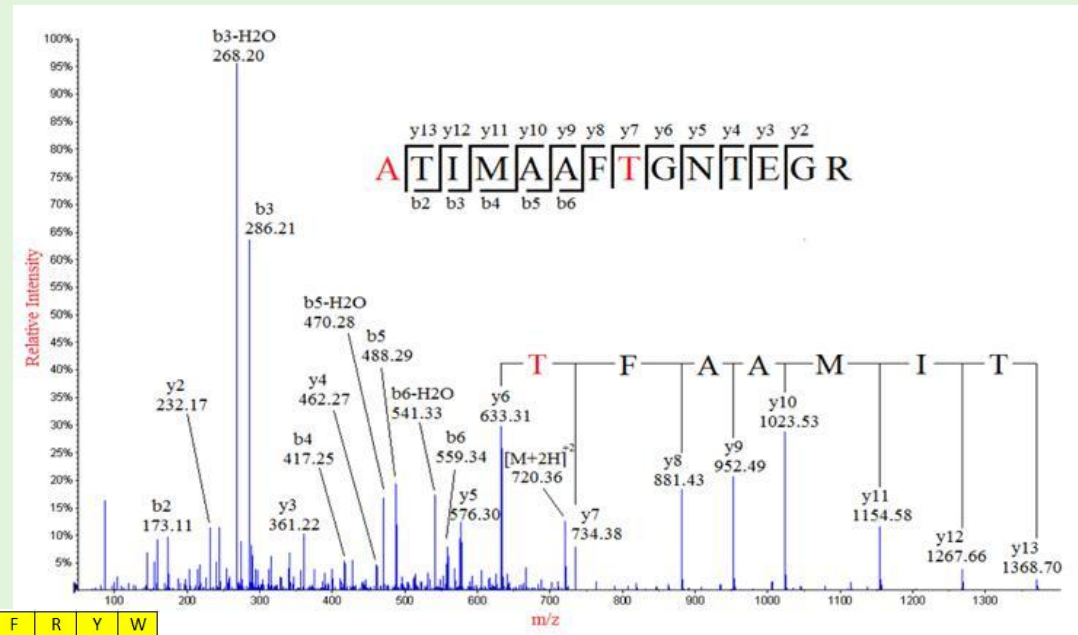


- A) We get a huge database with many peptides, simulate the spectra that we would obtain from them, and check which is the most similar to the experimental one
- B) We measure the distances between peaks, and check which amino acids would fit in between, reconstructing the peptide sequence
- C) We check the experimental spectra from previous analyses, and see what was identified when those were sequenced
- D) All of the above

<https://tinyurl.com/omics2025-q6>

# mass spectra of peptides

## De novo sequencing

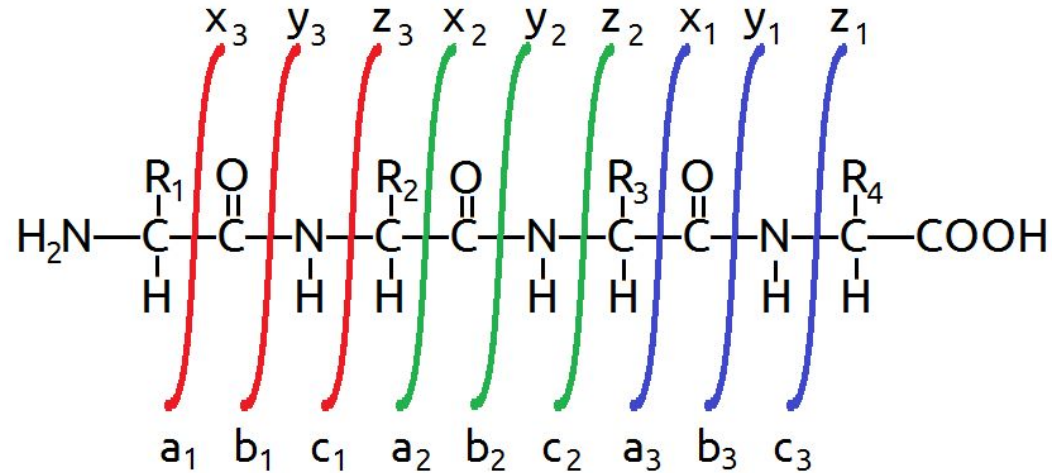


|                | G   | A   | S   | P   | V   | T   | C   | I/L | N   | D   | K/Q | E   | M   | H   | F   | R   | Y   | W   |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| G              | 115 |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| A              | 129 | 143 |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| S              | 145 | 159 | 175 |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| P              | 155 | 169 | 185 | 195 |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| V              | 157 | 171 | 187 | 197 | 199 |     |     |     |     |     |     |     |     |     |     |     |     |     |
| T              | 159 | 173 | 189 | 199 | 201 | 203 |     |     |     |     |     |     |     |     |     |     |     |     |
| C              | 161 | 175 | 191 | 201 | 203 | 205 | 207 |     |     |     |     |     |     |     |     |     |     |     |
| I/L            | 171 | 185 | 201 | 211 | 213 | 215 | 217 | 227 |     |     |     |     |     |     |     |     |     |     |
| N              | 172 | 186 | 202 | 212 | 214 | 216 | 218 | 228 | 229 |     |     |     |     |     |     |     |     |     |
| D              | 173 | 187 | 203 | 213 | 215 | 217 | 219 | 229 | 230 | 231 |     |     |     |     |     |     |     |     |
| K/Q            | 186 | 200 | 216 | 226 | 228 | 230 | 232 | 242 | 243 | 244 | 257 |     |     |     |     |     |     |     |
| E              | 187 | 201 | 217 | 227 | 229 | 231 | 233 | 243 | 244 | 245 | 258 | 259 |     |     |     |     |     |     |
| M              | 189 | 203 | 219 | 229 | 231 | 233 | 235 | 245 | 246 | 247 | 260 | 261 | 263 |     |     |     |     |     |
| H              | 195 | 209 | 225 | 235 | 237 | 239 | 241 | 251 | 252 | 253 | 266 | 267 | 269 | 275 |     |     |     |     |
| F <sup>b</sup> | 205 | 219 | 235 | 245 | 247 | 249 | 251 | 261 | 262 | 263 | 276 | 277 | 279 | 285 | 295 |     |     |     |
| R              | 214 | 228 | 244 | 254 | 256 | 258 | 260 | 270 | 271 | 272 | 285 | 286 | 288 | 294 | 304 | 313 |     |     |
| Y              | 221 | 235 | 251 | 261 | 263 | 265 | 267 | 277 | 278 | 279 | 292 | 293 | 295 | 301 | 311 | 320 | 327 |     |
| W              | 244 | 258 | 274 | 284 | 286 | 288 | 290 | 300 | 301 | 302 | 315 | 316 | 318 | 324 | 334 | 343 | 350 | 373 |

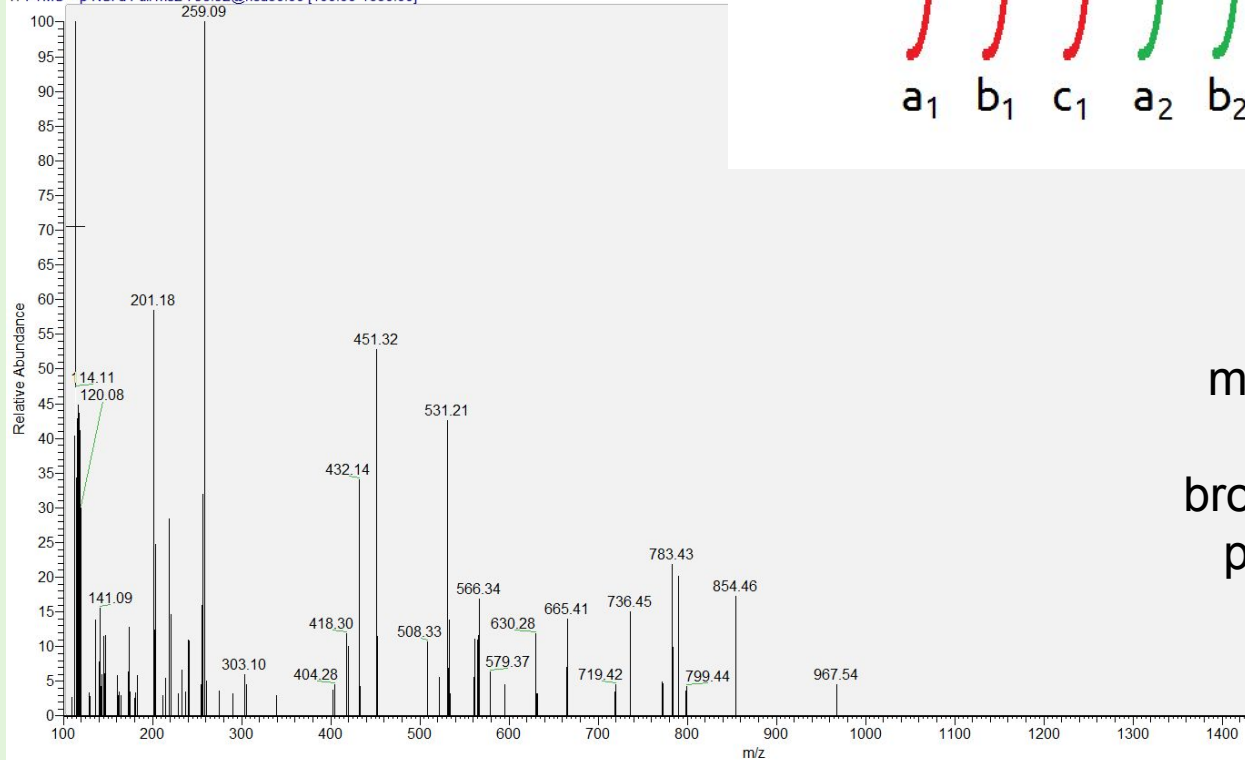
GG=N=114; GA=K/Q=128; GV=R=156; GE=AD=SV=W=186.

# The Roepstorff-Fohlman notation

a notation to identify the fragment ions of a peptide



Fer\_VSMC\_ITRAQ8plex\_MCX02\_130905213714 #148347 RT: 253.95 AV: 1 NL: 2.38E5  
T: FTMS + p NSI d Full ms2 798.92@hcd30.00 [100.00-1650.00]



most frequently (but not always) peptides are broken at peptide bonds, producing **b** and **y** ions

# **mass spectra of peptides**

in practice, we identify peptides  
that are already in databases!

# protein databases

I. getting a sequence of proteins from a public database

II. making an *in silico* digestion

# protein databases

 [BLAST](#) [Align](#) [Peptide search](#) [ID mapping](#) [SPARQL](#) [Release 2026\\_01](#) | [Statistics](#)    [Help](#)

## Find your protein

UniProtKB ▾

human pyruvate kinase

Advanced | List

Search

Examples: Insulin, APP, Human, P05067, organism\_id:9606

UniProt is the world's leading high-quality, comprehensive and freely accessible resource of protein sequence and functional information. [Cite UniProt](#)”

 Our Proteomes and UniProtKB/TrEMBL resources are undergoing a significant transition. Please read our [help page](#), view [affected entries and proteomes](#) , or [contact us](#) with any questions.

Proteins

UniProt Knowledgebase



Reviewed (Swiss-Prot)  
574,627



Unreviewed (TrEMBL)  
202,556,313

Species Proteomes

Protein sets for species with sequenced genomes from across the tree of life

Protein Clusters

UniRef

Clusters of protein sequences at 100%, 90% & 50% identity

Sequence archive

UniParc

Non-redundant archive of publicly available protein sequences seen across different databases



# protein databases

**UniProt** [BLAST](#) [Align](#) [Peptide search](#) [ID mapping](#) [SPARQL](#) [UniProtKB](#)  [Advanced](#) | [List](#) [Search](#) [Home](#) [Help](#)

**Status**

- Reviewed (Swiss-Prot) (175)
- Unreviewed (TrEMBL) (6,037)

**Popular organisms**

- Human (134)
- Zebrafish (18)
- Mouse (16)
- Rat (9)
- Fruit fly (4)

**Taxonomy**[Filter by taxonomy](#)

**Group by**

- Taxonomy
- Keywords
- Gene Ontology
- Enzyme Class

**Proteins with**

- 3D structure (73)
- Active site (1,573)
- Activity regulation (25)

**The unreviewed UniProtKB/TrEMBL database will be reduced in size in release 2026\_02 (first half of 2026).**

Entries to be retained in UniProtKB:

- Entries from reference proteomes
- All reviewed (Swiss-Prot) entries
- Selected unreviewed (TrEMBL) entries with experimental or biologically important data

Entries to be removed: Unreviewed (TrEMBL) entries that are not part of a reference proteome

Entries removed from unreviewed UniProtKB/TrEMBL will remain accessible in the UniParc sequence archive. Please read our [help page](#), view [affected entries and proteomes](#), or [contact us](#) with any questions.

## UniProtKB 6,212 results

Tools Download (6k) Add View: Cards ☐ Table ☒ Customize columns Share

| <input type="checkbox"/> Entry                  | Entry Name | Protein Names                                                                       | Gene Names     | Organism             | Length |
|-------------------------------------------------|------------|-------------------------------------------------------------------------------------|----------------|----------------------|--------|
| <input type="checkbox"/> <a href="#">P20613</a> | KPYR_HUMAN | Pyruvate kinase PKLR[...]                                                           | PKLR, PK1, PKL | Homo sapiens (Human) | 574 AA |
| <input type="checkbox"/> <a href="#">Q15120</a> | PDK3_HUMAN | [Pyruvate dehydrogenase (acetyl-transferring)] kinase isozyme 3, mitochondrial[...] | PDK3, PDHK3    | Homo sapiens (Human) | 406 AA |
| <input type="checkbox"/> <a href="#">Q16654</a> | PDK4_HUMAN | [Pyruvate dehydrogenase (acetyl-transferring)] kinase isozyme 4, mitochondrial[...] | PDK4, PDHK4    | Homo sapiens (Human) | 411 AA |
| <input type="checkbox"/> <a href="#">Q15118</a> | PDK1_HUMAN | [Pyruvate dehydrogenase (acetyl-transferring)] kinase isozyme 1, mitochondrial[...] | PDK1, PDHK1    | Homo sapiens (Human) | 436 AA |
| <input type="checkbox"/> <a href="#">Q15119</a> | PDK2_HUMAN | [Pyruvate dehydrogenase (acetyl-transferring)] kinase isozyme 2, mitochondrial[...] | PDK2, PDHK2    | Homo sapiens (Human) | 407 AA |

Help



# protein databases

 [BLAST](#) [Align](#) [Peptide search](#) [ID mapping](#) [SPARQL](#) [UniProtKB](#)  [Advanced](#) | [List](#) [Search](#)    [Help](#)

[Function](#)  
[Names & Taxonomy](#)  
[Subcellular Location](#)  
[Disease & Variants](#)  
[PTM/Processing](#)  
[Expression](#)  
[Interaction](#)  
[Structure](#)  
[Family & Domains](#)  
[Sequence & Isoform](#)  
[Similar Proteins](#)

## P30613 · KPYR\_HUMAN

|                       |                                                                                                                   |                                |                                                                                         |
|-----------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------|
| Protein <sup>i</sup>  | Pyruvate kinase PKLR                                                                                              | Amino acids                    | 574 ( <a href="#">go to sequence</a> )                                                  |
| Gene <sup>i</sup>     | PKLR                                                                                                              | Protein existence <sup>i</sup> | Evidence at protein level                                                               |
| Status <sup>i</sup>   |  UniProtKB reviewed (Swiss-Prot) | Annotation score <sup>i</sup>  |  5/5 |
| Organism <sup>i</sup> | Homo sapiens (Human)                                                                                              |                                |                                                                                         |

[Entry](#) [Variant viewer](#) [826](#) [Feature viewer](#) [Genomic coordinates](#) [Publications](#) [External links](#) [History](#)

 [Tools](#)  [Download](#)  [Add](#) [Add a publication](#) [Entry feedback](#)

### Function<sup>i</sup>

Pyruvate kinase that catalyzes the conversion of phosphoenolpyruvate to pyruvate with the synthesis of ATP, and which plays a key role in glycolysis.  1 Publication

### Miscellaneous

There are 4 isozymes of pyruvate kinase in mammals: L, R, M1 and M2. L type is major isozyme in the liver, R is found in red cells, M1 is the main form in muscle, heart and brain, and M2 is found in early fetal tissues.

### Catalytic activity<sup>i</sup>

**Rhea:18157** 

pyruvate + ATP = phosphoenolpyruvate + ADP + H<sup>+</sup>  1 Publication

This reaction proceeds in the backward direction.  1 Publication

EC:2.7.1.40 ([UniProtKB](#) | [ENZYME](#)  | [Rhea](#)  )

[Hide reaction details](#) 

[Reaction](#) [Atom map](#)

[Help](#)

# protein databases

## Sequence & Isoform<sup>i</sup>

[Align isoforms \(2\)](#) [Add isoforms](#)

Sequence status<sup>i</sup> | Complete

This entry describes 2 isoforms<sup>i</sup> produced by **Alternative splicing**.

### P30613-1

This isoform has been chosen as the **canonical** sequence. All positional information in this entry refers to it. This is also the sequence that appears in the downloadable versions of the entry.

Name R-type

Synonyms PKR

See also sequence in [UniParc](#) or sequence clusters in [UniRef](#)

Tools [Download](#) [Add](#) [Highlight](#) [Copy sequence](#)

Length 574

Mass (Da) 61,830

Last updated 2000-05-30 v2

MD5 Checksum<sup>i</sup> 9E485BFD0D795CAEA2EE212ABF8C92A

|            |             |            |            |            |             |            |            |            |            |             |            |
|------------|-------------|------------|------------|------------|-------------|------------|------------|------------|------------|-------------|------------|
| MSIQENISSL | QLRSWWSKSQ  | RDLAKSILIG | APGGPAGYLR | RASVAQLTQE | LGTAFFQQQQ  | LPAAMADTFL | EHLCLLDIDS | EPVAARSTSI | IATIGPASRS | VERLKEMIKA  | GMNIARLNFS |
| HGSHEYHAES | IANVREAVES  | FAGSPLSYRP | VAIALDTKGP | EIRTGILQGG | PESEVELVKG  | SQVLTVDPDA | FRTRGNANTV | WDYPNIVRV  | VPVGGRIVID | DGLISLVVQK  | IGPEGLVTQV |
| ENGGVLGSRK | GVNLPGAQVD  | LPGLSEQDVR | DLRFGVEHGV | DIVFASFVRK | ASDVAAVRAA  | LGPEGHGIKI | ISKIENHEGV | KRFDEILEVS | DGIMVARGDL | GIEIPAEEKVF | LAQKMMIGRC |
| NLAGKPVVCA | TQMLESIMITK | PRPTRAETSD | VANAVLDGAD | CIMLSGETAK | GNFPEAVEKVM | QHAIAREAEA | AVYHRQLFEE | LRRAPLSRD  | PTEVTAIGAV | EAAFKCCAAA  | IIVLTTTGRS |
| AQLLSRYRPR | AAVIAVTRSA  | QAARQVHLCR | GVFPLLYREP | PEAIWADDVD | RRVQFGIESG  | KLRGFLRVGD | LVIVVTGWRP | GSGYTNIMRV | LSIS       |             |            |

### P30613-2

Name L-type

Synonyms PKL

See also sequence in [UniParc](#) or sequence clusters in [UniRef](#)

Differences from canonical 1-33: MSIQENISSLQLRSWWSKSQRDLAKSILIGAPG → ME

[Curated](#)

# protein databases

```
>sp|P30613|KPYR_HUMAN Pyruvate kinase PKLR OS=Homo sapiens OX=9606 GN=PKLR PE=1 SV=2
MSIQENISSLQLRSWVSKSQRDLAKSILIGAPGGPAGYLRRASVAQLTQELGTAFFQQQQ
LPAAMADTFLEHLCLLDIDSEPVAARSTSIATIGPASRSVERLKEMIKAGMNIARLNFS
HGSHEYHAESIANVREAVESFAGSPLSYRPVAIALDTKGPEIRTGILQGGPESEVELVKG
SQVLVTVDPAFRTRGNANTVWVDYPNIVRVVPVGGRIYIDDGLISLVVQKIGPEGLVTQV
ENGGVLGSRKGVNLPGAQVDLPGLSEQDVRDLRFGVEHGVDIVFASFVRKASDVAAVRAA
LGPEGHGIKIISKIENHEGVKRFDEILEVSDGIMVARGDLGIEIPAENVFLAQKMMIGRC
NLAGKPVVCATQMLESMITKPRPTRAETSDVANAVLDGADCIMLSGETAKGNFPVEAVKM
QHAIAREAEAAVYHRQLFEELRRAAPLSRDPTEVTAIGAVEAAFKCCAAAIIVLTTTGRS
AQLLSRYRPRAAVIAVTRSAQAARQVHLCRGVFPLLYREPPEAIWADDVDRRVQFGIESG
KLRGFLRVGDLVIVVTGWRPGSGYTNIMRVLSIS
```

# protein databases

```
>sp|P30613|KPYR_HUMAN Pyruvate kinase PKLR OS=Homo sapiens OX=9606 GN=PKLR PE=1 SV=2
MSIQENISSLQLRSWVSKSQRDLAKSILIGAPGGPAGYLRASVAQLTQELGTAFFQQQQ
LPAAMADTFLEHLCLLDIDSEPVAARSTSIATIGPASRSVERLKEMIKAGMNIARLNFS
HGSHEYHAESIANVREAVESFAGSPLSYRPVAIALDTKGPEIRTGILQGGPESEVELVKG
SQVLVTVDPAFRTRGNANTVWVDYPNIVRVVPVGGRIYIDDGLISLVVQKIGPEGLVTQV
ENGGVLGSRKGVNLPGAQVDLPGLSEQDVRDLRFGVEHGVDIVFASFVRKASDVAAVRAA
LGPEGHGIKIISKIENHEGVKRFDEILEVSDGIMVARGDLGIEIPA EKVF LAQKMMIGRC
NLAGKPVVCATQMLESMITKPRPTRAETSDVANAVLDGADCIMLSGETAKGNFPVEAVKM
QHAIAREAEAAVYHRQLFEELRRAAPLSRDPTEVTAIGAVEAAFKCCAAAIIVLTTTGRS
AQLLSRYRPRAAVIAVTRSAQAARQVHLCRGVFP LLYREPPEAIWADDVDRRVQFGIESG
KLRGFLRVGDLVIVVTGWRPGSGYTNIMRVLSIS
```

# protein databases

MSIQENISSLQLR

SWVSK

SQR

DLAK

SILIGAPGGPAGYLR

R

ASVAQLTQELGTAFFQQQQLPAAMADTFLEHLCLL

DIDSEPVAAR

STSIIATIGPASR

SVER

LK

EMIK

AGMNIAR

LNFSHGSHEYHAESIANVR

EAVESFAGSPLSYR

PVAIALDTK

GPEIR

TGILQGGPESEVELVK

GSQVLVTVDPAFR

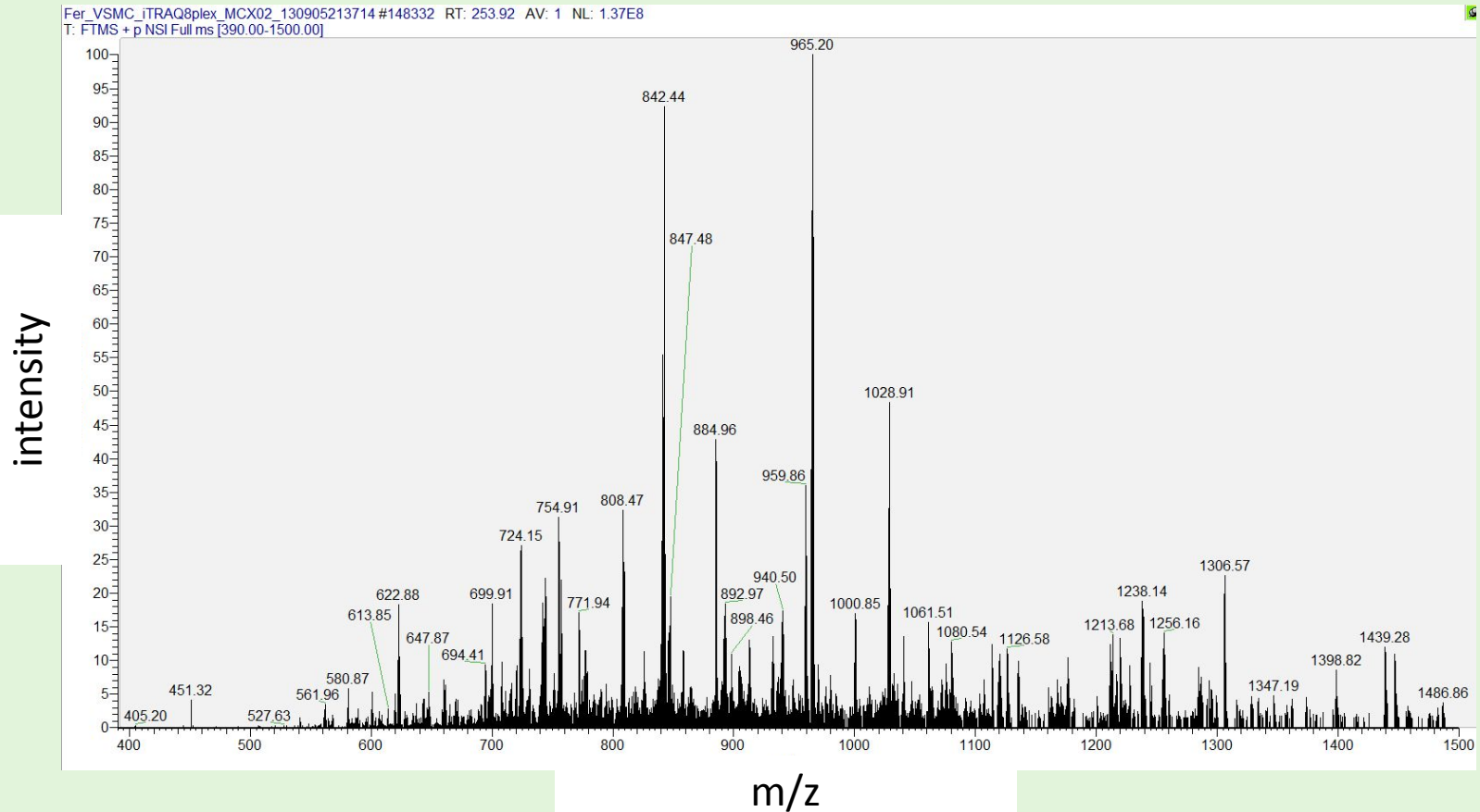
TR

GNANTVWVDYPNIVR

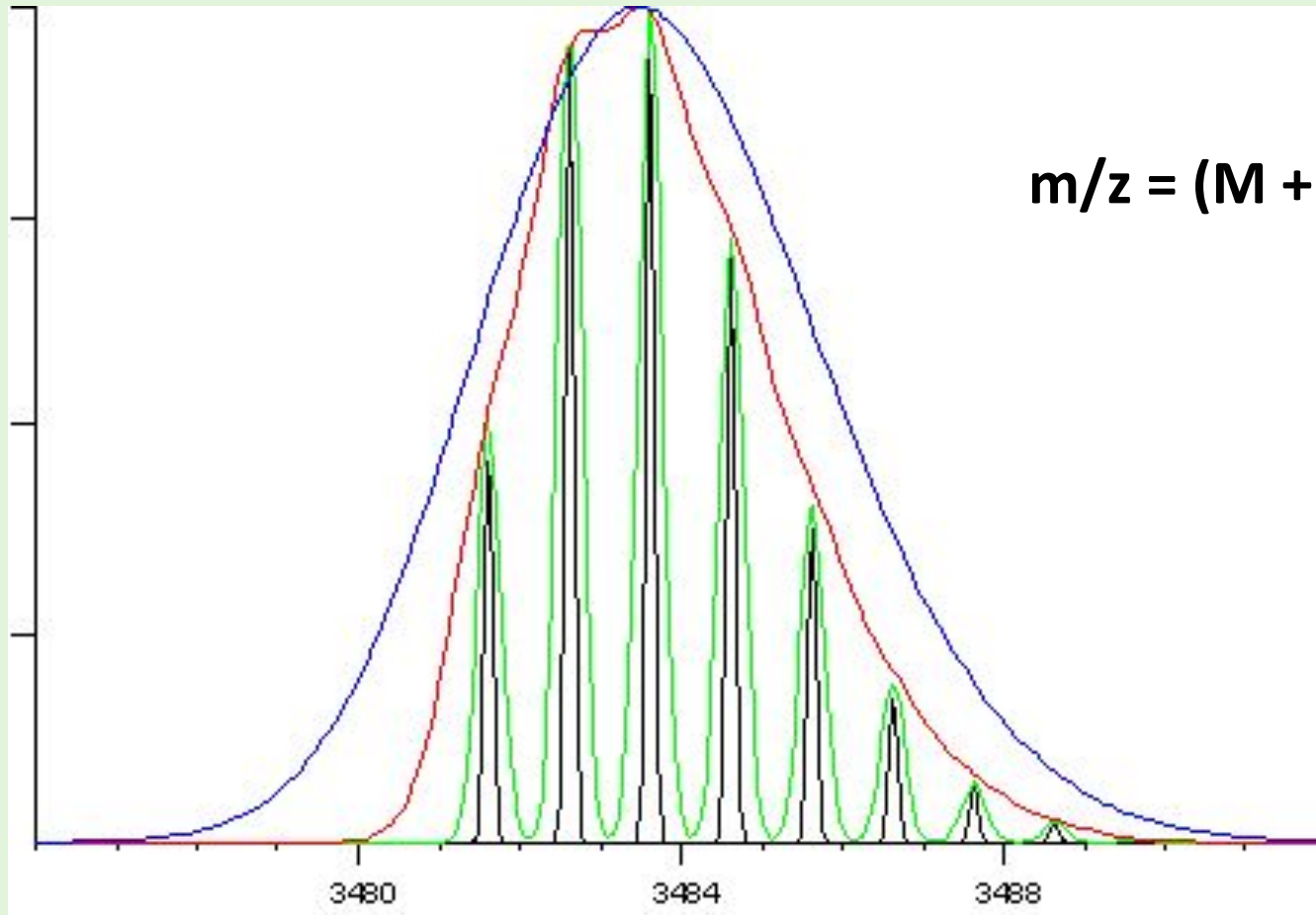
VMPVGGP



# mass spectra of peptides



# monoisotopic and average mass



molecular ion of glucagon at resolutions of  
1000 (blue), 3000 (red), 10,000 (green) and  
30,000 (black)



# monoisotopic and average mass

MSIQENISSLQLR

SWVSK

SQR

DLAK

**SILIGAPGGPAGYLR**

R

ASVAQLTQELGTAFFQQQQLPAAMADTFLEHLCLL

DIDSEPVAAR

STSIIATIGPASR

SVER

LK

EMIK

AGMNIAR

LNFSHGSHEYHAESIANVR

EAVESFAGSPLSYR

PVAIALDTK

GPEIR

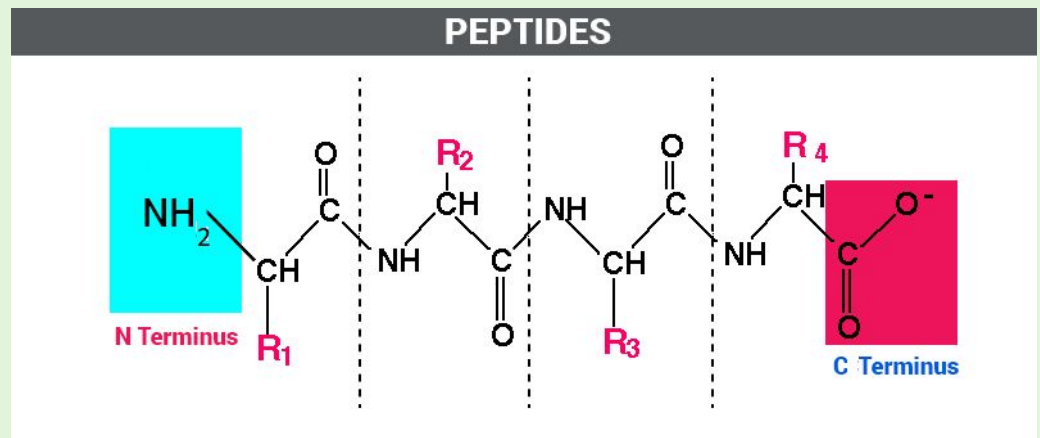
TGILQGGPESEVELVK

GSQVLVTVDPAFR

TR

GNANTVWVDYPNIVR

| 1-letter | 3-letter | Amino acid              | Mono isotopic mass | Average mass |
|----------|----------|-------------------------|--------------------|--------------|
| A        | Ala      | Alanine                 | 71.03711           | 71.0788      |
| C        | Cys      | Cysteine                | 103.00919          | 103.1388     |
| D        | Asp      | Aspartic acid           | 115.02694          | 115.0886     |
| E        | Glu      | Glutamic acid           | 129.04259          | 129.1155     |
| F        | Phe      | Phenylalanine           | 147.06841          | 147.1766     |
| G        | Gly      | Glycine                 | 57.02146           | 57.0519      |
| H        | His      | Histidine               | 137.05891          | 137.1411     |
| I        | Ile      | Isoleucine              | 113.08406          | 113.1594     |
| K        | Lys      | Lysine                  | 128.09496          | 128.1741     |
| L        | Leu      | Leucine                 | 113.08406          | 113.1594     |
| M        | Met      | Methionine              | 131.04049          | 131.1926     |
| N        | Asn      | Asparagine              | 114.04293          | 114.1038     |
| P        | Pro      | Proline                 | 97.05276           | 97.1167      |
| Q        | Gln      | Glutamine               | 128.05858          | 128.1307     |
| R        | Arg      | Arginine                | 156.10111          | 156.1875     |
| S        | Ser      | Serine                  | 87.03203           | 87.0782      |
| T        | Thr      | Threonine               | 101.04768          | 101.1051     |
| V        | Val      | Valine                  | 99.06841           | 99.1326      |
| W        | Trp      | Tryptophan              | 186.07931          | 186.2132     |
| Y        | Tyr      | Tyrosine                | 163.06333          | 163.176      |
| —        | —        | N-terminus (H)          | 1.007825           | 1.00794      |
| —        | —        | C-terminus (OH)         | 17.00274           | 17.00734     |
| —        | —        | H <sup>+</sup> (proton) | 1.007276           | 1.00739      |



source: [https://web.expasy.org/findmod/findmod\\_masses.html](https://web.expasy.org/findmod/findmod_masses.html)

| 1-letter | 3-letter | Amino acid      | Mono isotopic mass | Average mass |
|----------|----------|-----------------|--------------------|--------------|
| A        | Ala      | Alanine         | 71.03711           | 71.0788      |
| C        | Cys      | Cysteine        | 103.00919          | 103.1388     |
| D        | Asp      | Aspartic acid   | 115.02694          | 115.0886     |
| E        | Glu      | Glutamic acid   | 129.04259          | 129.1155     |
| F        | Phe      | Phenylalanine   | 147.06841          | 147.1766     |
| G        | Gly      | Glycine         | 57.02146           | 57.0519      |
| H        | His      | Histidine       | 137.05891          | 137.1411     |
| I        | Ile      | Isoleucine      | 113.08406          | 113.1594     |
| K        | Lys      | Lysine          | 128.09496          | 128.1741     |
| L        | Leu      | Leucine         | 113.08406          | 113.1594     |
| M        | Met      | Methionine      | 131.04049          | 131.1926     |
| N        | Asn      | Asparagine      | 114.04293          | 114.1038     |
| P        | Pro      | Proline         | 97.05276           | 97.1167      |
| Q        | Gln      | Glutamine       | 128.05858          | 128.1307     |
| R        | Arg      | Arginine        | 156.10111          | 156.1875     |
| S        | Ser      | Serine          | 87.03203           | 87.0782      |
| T        | Thr      | Threonine       | 101.04768          | 101.1051     |
| V        | Val      | Valine          | 99.06841           | 99.1326      |
| W        | Trp      | Tryptophan      | 186.07931          | 186.2132     |
| Y        | Tyr      | Tyrosine        | 163.06333          | 163.176      |
| —        | —        | N-terminus (H)  | 1.007825           | 1.00794      |
| —        | —        | C-terminus (OH) | 17.00274           | 17.00734     |
| —        | —        | H+ (proton)     | 1.007276           | 1.00739      |

## Fragment Ion Calculator

- The calculator takes protein sequences in single-letter code (not including ambiguous amino acids).
- Each sequence should be written on its own line.
- Whitespace and numbers are ignored within the sequence.

### Peptide Sequence

Peptide:

PEPTIDE

Submit

Reset

Mass type:

☒ MONO

☐ AVG

Charge state:

☒ +1

☐ +2

☐ +3

Ion types:

☐ A

☐ X

☒ B

☒ Y

☐ C

☐ Z

### Modifications (optional)

Add to N- or C-terminus:

N-terminus

C-terminus

0.0

Add to all AA residues and/or specific location:

AA or Pos

Value

e.g. C 57.0

3 80.0

(add +57 to all Cys and  
add +80 to 3rd AA residue)

| 1-letter | 3-letter | Amino acid      | Mono isotopic mass | Average mass |
|----------|----------|-----------------|--------------------|--------------|
| A        | Ala      | Alanine         | 71.03711           | 71.0788      |
| C        | Cys      | Cysteine        | 103.00919          | 103.1388     |
| D        | Asp      | Aspartic acid   | 115.02694          | 115.0886     |
| E        | Glu      | Glutamic acid   | 129.04259          | 129.1155     |
| F        | Phe      | Phenylalanine   | 147.06841          | 147.1766     |
| G        | Gly      | Glycine         | 57.02146           | 57.0519      |
| H        | His      | Histidine       | 137.05891          | 137.1411     |
| I        | Ile      | Isoleucine      | 113.08406          | 113.1594     |
| K        | Lys      | Lysine          | 128.09496          | 128.1741     |
| L        | Leu      | Leucine         | 113.08406          | 113.1594     |
| M        | Met      | Methionine      | 131.04049          | 131.1926     |
| N        | Asn      | Asparagine      | 114.04293          | 114.1038     |
| P        | Pro      | Proline         | 97.05276           | 97.1167      |
| Q        | Gln      | Glutamine       | 128.05858          | 128.1307     |
| R        | Arg      | Arginine        | 156.10111          | 156.1875     |
| S        | Ser      | Serine          | 87.03203           | 87.0782      |
| T        | Thr      | Threonine       | 101.04768          | 101.1051     |
| V        | Val      | Valine          | 99.06841           | 99.1326      |
| W        | Trp      | Tryptophan      | 186.07931          | 186.2132     |
| Y        | Tyr      | Tyrosine        | 163.06333          | 163.176      |
| —        | —        | N-terminus (H)  | 1.007825           | 1.00794      |
| —        | —        | C-terminus (OH) | 17.00274           | 17.00734     |
| —        | —        | H+ (proton)     | 1.007276           | 1.00739      |

## Fragment Ion Calculator

- The calculator takes protein sequences in single-letter code (not including ambiguous amino acids).
- Each sequence should be written on its own line.
- Whitespace and numbers are ignored within the sequence.

### Peptide Sequence

Peptide:




Mass type:

☒ MONO  
☐ AVG

Charge state:

☒ +1  
☐ +2  
☐ +3

Ion types:

☐ A ☐ X  
☒ B ☒ Y  
☐ C ☐ Z

### Modifications (optional)

Add to N- or C-terminus:

N-terminus

C-terminus

Add to all AA residues and/or specific location:

AA or Pos

Value

e.g. C 57.0  
3 80.0

(add +57 to all Cys and  
add +80 to 3rd AA residue)



| 1-letter | 3-letter | Amino acid              | Mono isotopic mass | Average mass |
|----------|----------|-------------------------|--------------------|--------------|
| A        | Ala      | Alanine                 | 71.03711           | 71.0788      |
| C        | Cys      | Cysteine                | 103.00919          | 103.1388     |
| D        | Asp      | Aspartic acid           | 115.02694          | 115.0886     |
| E        | Glu      | Glutamic acid           | 129.04259          | 129.1155     |
| F        | Phe      | Phenylalanine           | 147.06841          | 147.1766     |
| G        | Gly      | Glycine                 | 57.02146           | 57.0519      |
| H        | His      | Histidine               | 137.05891          | 137.1411     |
| I        | Ile      | Isoleucine              | 113.08406          | 113.1594     |
| K        | Lys      | Lysine                  | 128.09496          | 128.1741     |
| L        | Leu      | Leucine                 | 113.08406          | 113.1594     |
| M        | Met      | Methionine              | 131.04049          | 131.1926     |
| N        | Asn      | Asparagine              | 114.04293          | 114.1038     |
| P        | Pro      | Proline                 | 97.05276           | 97.1167      |
| Q        | Gln      | Glutamine               | 128.05858          | 128.1307     |
| R        | Arg      | Arginine                | 156.10111          | 156.1875     |
| S        | Ser      | Serine                  | 87.03203           | 87.0782      |
| T        | Thr      | Threonine               | 101.04768          | 101.1051     |
| V        | Val      | Valine                  | 99.06841           | 99.1326      |
| W        | Trp      | Tryptophan              | 186.07931          | 186.2132     |
| Y        | Tyr      | Tyrosine                | 163.06333          | 163.176      |
| —        | —        | N-terminus (H)          | 1.007825           | 1.00794      |
| —        | —        | C-terminus (OH)         | 17.00274           | 17.00734     |
| —        | —        | H <sup>+</sup> (proton) | 1.007276           | 1.00739      |

## Fragment Ion Calculator Results

Sequence: **SILIGAPGGPAGYLR**, pI: **8.46288**

### Fragment Ion Table, monoisotopic masses

| Seq | #  | B          | Y          | # (+1) |
|-----|----|------------|------------|--------|
| S   | 1  | 88.03935   | 1441.81622 | 15     |
| I   | 2  | 201.12341  | 1354.78419 | 14     |
| L   | 3  | 314.20748  | 1241.70012 | 13     |
| I   | 4  | 427.29154  | 1128.61606 | 12     |
| G   | 5  | 484.31300  | 1015.53200 | 11     |
| A   | 6  | 555.35012  | 958.51053  | 10     |
| P   | 7  | 652.40288  | 887.47342  | 9      |
| G   | 8  | 709.42434  | 790.42066  | 8      |
| G   | 9  | 766.44581  | 733.39919  | 7      |
| P   | 10 | 863.49857  | 676.37773  | 6      |
| A   | 11 | 934.53569  | 579.32497  | 5      |
| G   | 12 | 991.55715  | 508.28785  | 4      |
| Y   | 13 | 1154.62048 | 451.26639  | 3      |
| L   | 14 | 1267.70454 | 288.20306  | 2      |
| R   | 15 | 1423.80565 | 175.11900  | 1      |

### Mass/Charge Table

|                      | Mass       |            |
|----------------------|------------|------------|
|                      | Mono       | Avg        |
| (M)                  | 1441.81622 | 1441.69336 |
| (M+H) <sup>+</sup>   | 1441.81622 | 1442.70064 |
| (M+2H) <sup>2+</sup> | 721.41177  | 721.85398  |
| (M+3H) <sup>3+</sup> | 481.27695  | 481.57176  |
| (M+4H) <sup>4+</sup> | 361.20955  | 361.43065  |

# Variable and fixed modifications

Either for technical or biological reasons, many peptides have post-translational modifications which affect their mass.

The algorithm takes into account two kinds of modifications:

**fixed modifications:** those affecting all the amino acids of the same type.

example: the carbamidomethylation of cysteine  
(57.021464 Da)

**variable modifications:** those affecting part of the amino acids of the same type, but not all.

example: the oxidation of methionine, (15.994915 Da)

# Variable and fixed modifications

**UNIMOD**

protein modifications for mass spectrometry

[Help](#)

Unimod

Logged as **Guest** [Log out](#)

[Change password](#)

[Advanced search](#)

|             |                                |                               |                                        |                                                  |
|-------------|--------------------------------|-------------------------------|----------------------------------------|--------------------------------------------------|
| Search for: | Any field <input type="text"/> | Contains <input type="text"/> | Details found:<br>1425<br>Page 1 of 72 | Records Per<br>Page::<br>20 <input type="text"/> |
|             | <input type="text"/>           | <input type="text"/>          | <input type="button" value="Search"/>  | <input type="button" value="Show all"/>          |

|  | Accession # | PSI-MS Name | Interim name | Description                                             | Monoisotopic mass | Average mass | Composition                   |
|-----------------------------------------------------------------------------------|-------------|-------------|--------------|---------------------------------------------------------|-------------------|--------------|-------------------------------|
| <a href="#">View</a>                                                              | 1287        |             | Arg-loss     | Loss of arginine due to transpeptidation                | -156.101111       | -156.1857    | H(-12) C(-6) N(-4) O(-1)      |
| <a href="#">View</a>                                                              | 765         |             | Met-loss     | Removal of initiator methionine from protein N-terminus | -131.040485       | -131.1961    | H(-9) C(-5) N(-1) O(-1) S(-1) |
| <a href="#">View</a>                                                              | 676         |             | Trp->Gly     | Trp->Gly substitution                                   | -129.057849       | -129.1586    | H(-7) C(-9) N(-1)             |
| <a href="#">View</a>                                                              | 313         | Lys-loss    | -K           | Loss of C-terminal K from Heavy Chain of MAb            | -128.094963       | -128.1723    | H(-12) C(-6) N(-2) O(-1)      |
| <a href="#">View</a>                                                              | 1224        |             | Trp->Ala     | Trp->Ala substitution                                   | -115.042199       | -115.1320    | H(-5) C(-8) N(-1)             |
| <a href="#">View</a>                                                              | 1239        |             | Tyr->Gly     | Tyr->Gly substitution                                   | -106.041865       | -106.1219    | H(-6) C(-7) O(-1)             |
| <a href="#">View</a>                                                              | 646         |             | Arg->Gly     | Arg->Gly substitution                                   | -99.079647        | -99.1344     | H(-9) C(-4) N(-3)             |
| <a href="#">View</a>                                                              | 673         |             | Trp->Ser     | Trp->Ser substitution                                   | -99.047285        | -99.1326     | H(-5) C(-8) N(-1)             |



MSIQENISSLQLR

SWVSK

SQR

DLAK

**SILIGAPGGPAGYLR -> 1441.81622**

R

ASVAQLTQELGTAFFQQQLPAAMADTFLEHLCLL

DIDSEPVAAR

STSIATIGPASR -> 1273.71108

SVER

LK

EMIK

AGMNIAR

LNFSHGSHEYHAESIANVR -> 2168.02683

EAVESFAGSPLSYR

PVAIALDTK

GPEIR

TGILQGGPESEVELVK -> 1655.88508

GSQVLVTVDPAFR

TR

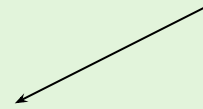
GNANTVWVDYPNIVR

VVPVGGR -> 683.41993

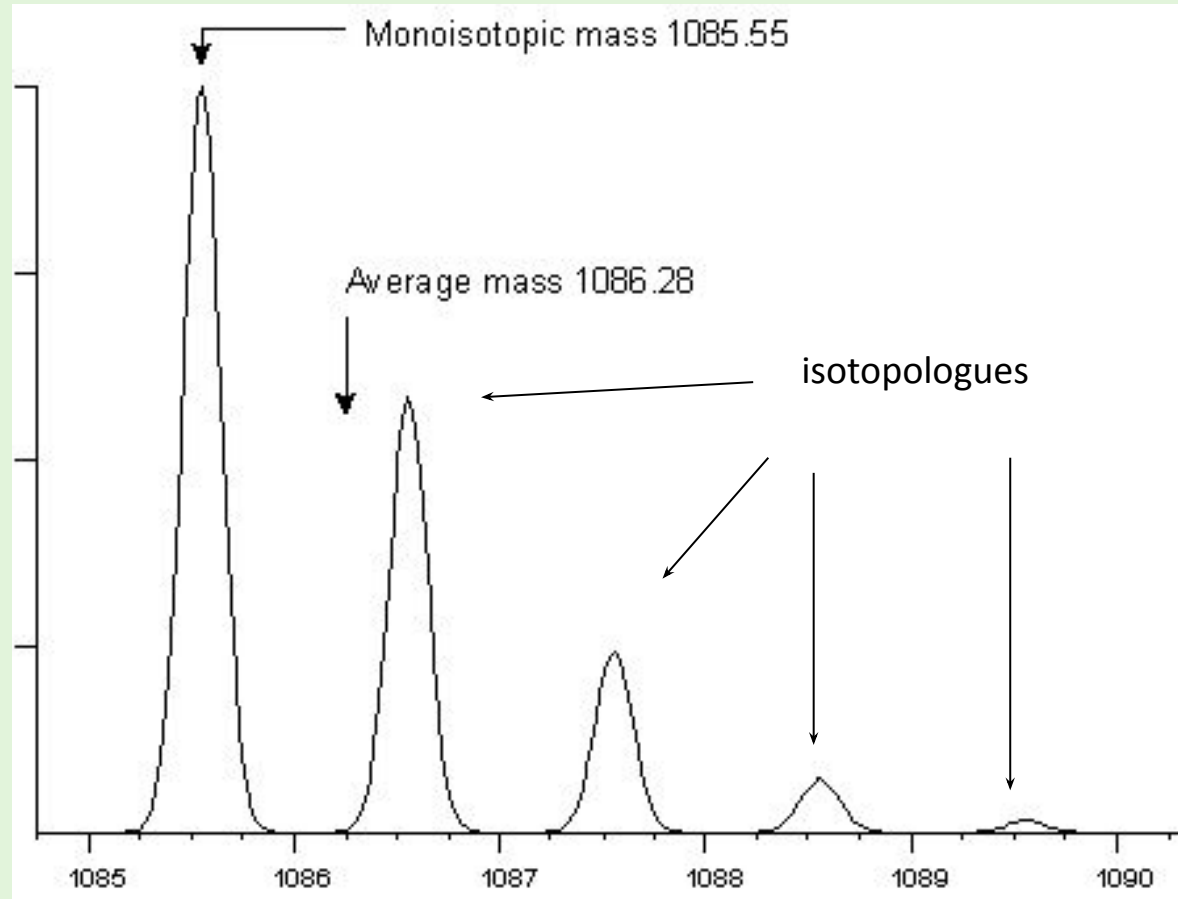
IYIDDGLISLVVQK -> 1575.89928

IGREGIVTOVENCCVLCSP

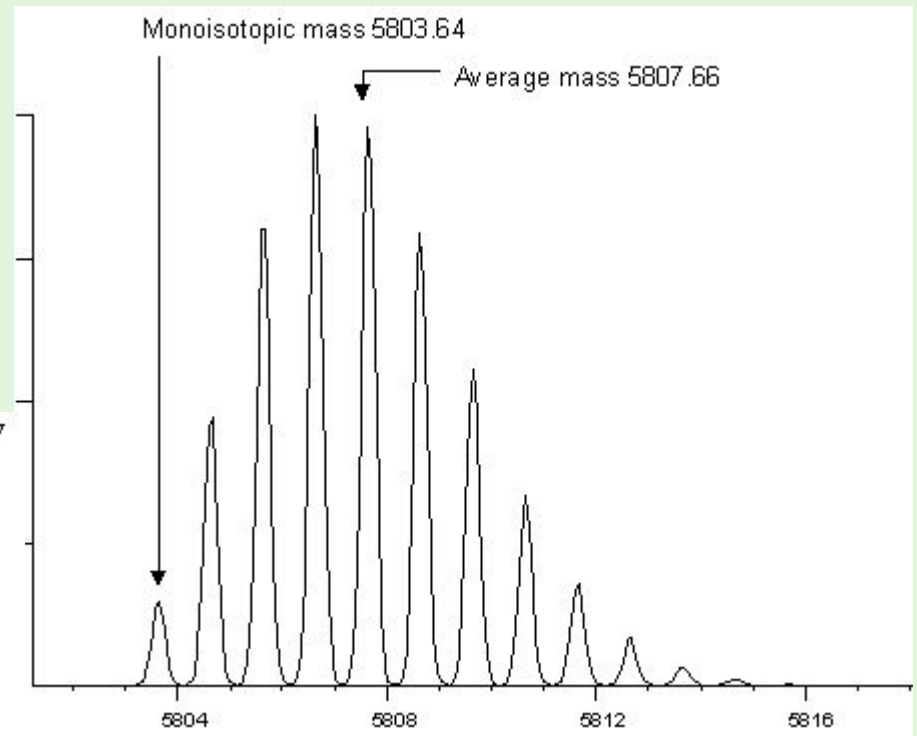
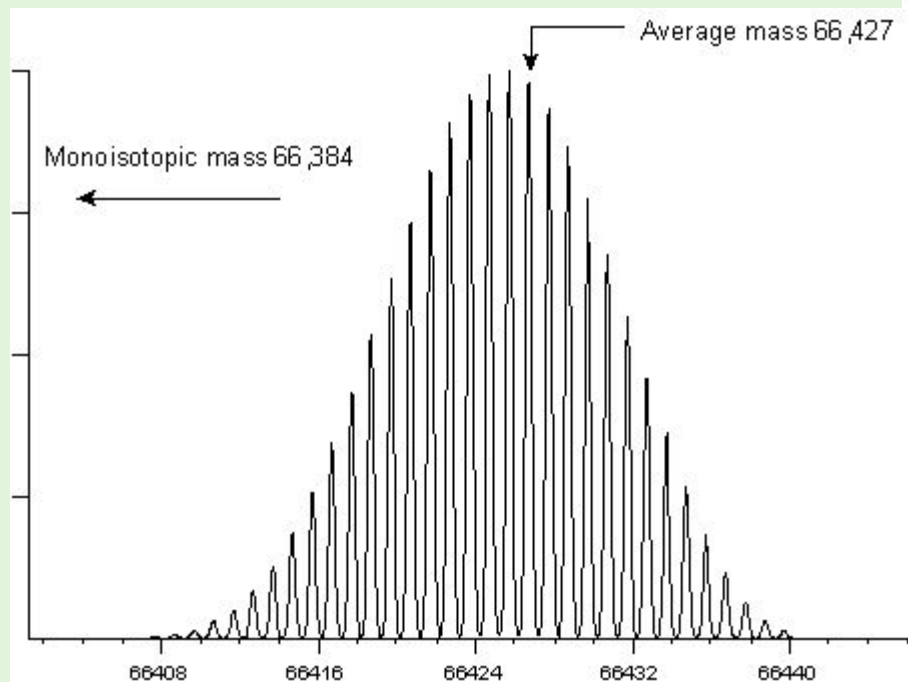
**MH+**



# isotopic envelopes and the binomial distribution



# isotopic envelopes and the binomial distribution



# isotopic envelopes and the binomial distribution

**Mass and Abundance Values**

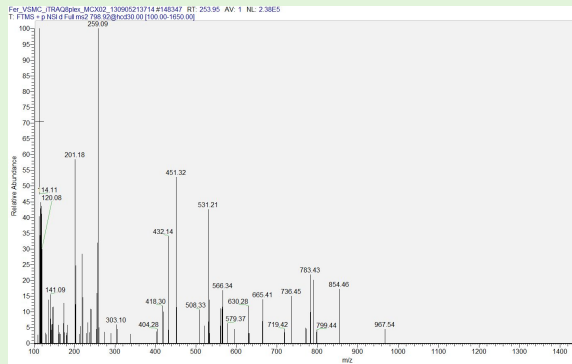
| Element | Isotopically averaged mass <sup>a</sup> | Isotope          | Isotopic mass          | Isotopic abundance, % |
|---------|-----------------------------------------|------------------|------------------------|-----------------------|
| H       | 1.00794                                 | <sup>1</sup> H   | 1.007835 <sup>b</sup>  | 99.985                |
|         |                                         | <sup>2</sup> H   | 2.01410                | 0.015                 |
| C       | 12.011                                  | <sup>12</sup> C  | 12.00000               | 98.90                 |
|         |                                         | <sup>13</sup> C  | 13.003355 <sup>b</sup> | 1.10                  |
| N       | 14.00674                                | <sup>14</sup> N  | 14.00307               | 99.634                |
|         |                                         | <sup>15</sup> N  | 15.00011               | 0.366                 |
| O       | 15.9994                                 | <sup>16</sup> O  | 15.99491               | 99.762                |
|         |                                         | <sup>17</sup> O  | 16.99913               | 0.038                 |
|         |                                         | <sup>18</sup> O  | 17.99916               | 0.200                 |
| F       | 18.99840                                | <sup>19</sup> F  | 18.99840               | 100                   |
| Na      | 22.98977                                | <sup>23</sup> Na | 22.98977               | 100                   |
| Si      | 28.0855                                 | <sup>28</sup> Si | 27.97693               | 92.23                 |
|         |                                         | <sup>29</sup> Si | 28.97649               | 4.67                  |
|         |                                         | <sup>30</sup> Si | 29.97377               | 3.10                  |
| P       | 30.97376                                | <sup>31</sup> P  | 30.97376               | 100                   |
| S       | 32.066                                  | <sup>32</sup> S  | 31.97207               | 95.02                 |
|         |                                         | <sup>33</sup> S  | 32.97146               | 0.75                  |
|         |                                         | <sup>34</sup> S  | 33.96787               | 4.21                  |
|         |                                         | <sup>36</sup> S  | 35.96708               | 0.02                  |
| Cl      | 35.4527                                 | <sup>35</sup> Cl | 34.96885               | 75.77                 |
|         |                                         | <sup>37</sup> Cl | 36.96590               | 24.23                 |
| K       | 39.0983                                 | <sup>39</sup> K  | 38.96371               | 93.258                |
|         |                                         | <sup>40</sup> K  | 39.96400               | 0.012                 |
|         |                                         | <sup>41</sup> K  | 40.96183               | 6.730                 |

<sup>a</sup>Equivalent to atomic weight.

<sup>b</sup>Not rounded off where result is equivocal.

# Interpreting fragmentation spectra

```
>sp|P04637|P53_HUMAN Cellular tumor antigen p53 OS=Homo sapiens GN=TP53 PE=1 SV=4
MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLSPLPSQAMDDLMLSPDDIEQWFTEDPGP
DEAPRMPEAAPPVAPAPAAPTPAAPAPAPSWPLSSSVPSQKTYQGSYGFR LGFLHSGTAK
SVTCTYSPALNKMFCQLAKTCPVQLWVDSTPPPGRTRVRAMAIYKQSQHMTEVVRRCPHHE
RCSDS DGLAPPQH LIRVEGNLRVEYLDDRNTFRHSVVVPYEPPEVGS DCTTIHNYMCNS
SCMGGMNRRPILTIITLEDSSGNLLGRNSFEVRVCACPGRRRTEENLRKKGEPHHELP
PGSTKRALPNNTSSSPQPKKKPLDGEYFTLQIRGRERFEMFRELNEALELKDAQAGKEPG
GSAHSSHLKSKKGQSTSRHKKLMFKTEGPDSD
```



# Precursor and fragment tolerance

Spectral lines will never have the exact same value

The problem is solved by defining a tolerance: when a value is within the tolerance window of another value, algorithms consider them equal

Since in MSMS there are two spectra, two tolerances are needed:

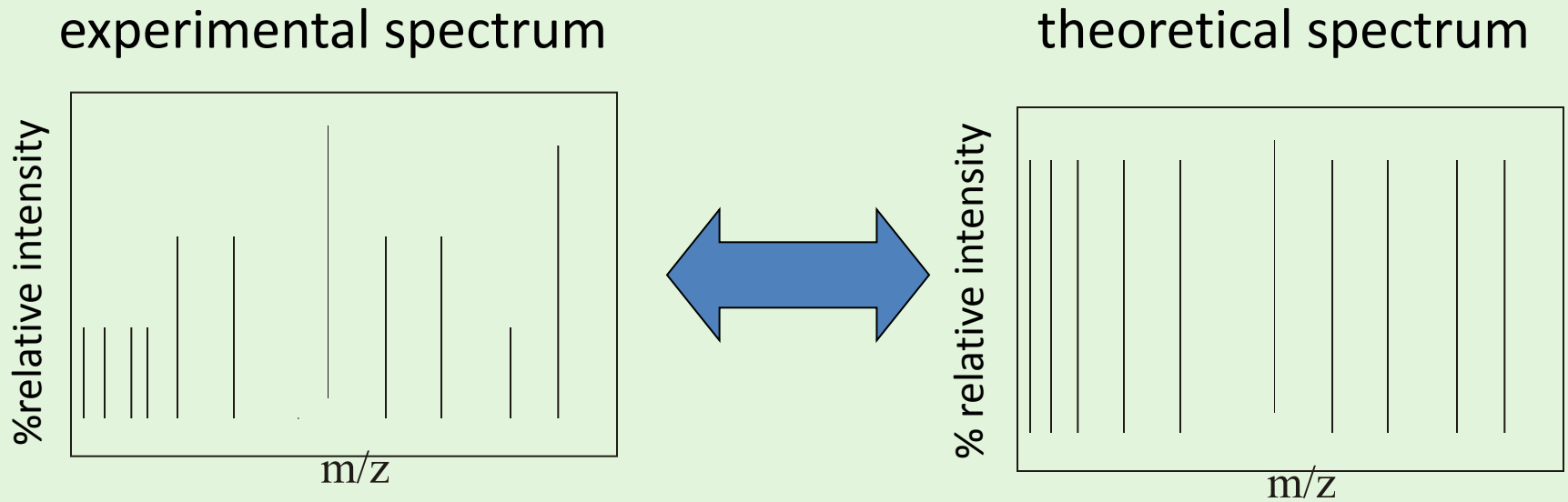
**the tolerance for the precursor** ion (which limits the database to the peptides having the same mass)

**the tolerance of the fragments** (which determines which lines in the spectrum are used)



# SEQUEST score

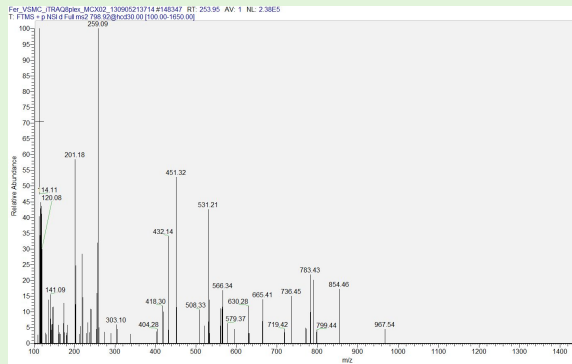
## *The XCorr*



SEQUEST measures the degree of correlation

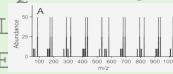
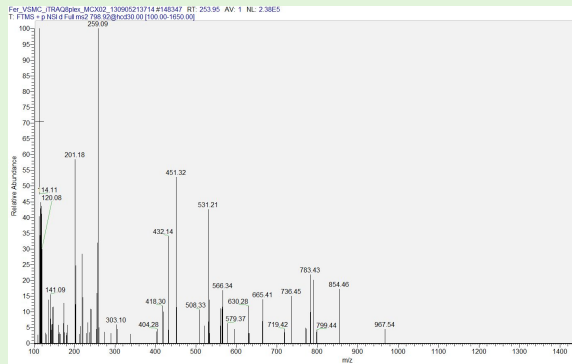
# Interpreting fragmentation spectra

```
>sp|P04637|P53_HUMAN Cellular tumor antigen p53 OS=Homo sapiens GN=TP53 PE=1 SV=4
MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLSPLPSQAMDDLMLSPDDIEQWFTEDPGP
DEAPRMPEAAPPVAPAPAAPTPAAPAPAPSWPLSSSVPSQKTYQGSYGFR LGFLHSGTAK
SVTCTYSPALNKMFCQLAKTCPVQLWVDSTPPPGRVVRAMAIYKQSQHMTEVVRRCPHHE
RCSDS DGLAPPQH LIRVEGNLRVEYLDDRNTFRHSVVVPYEPPEVGSDCTTIHNYMCNS
SCMGMNRRPILTIITLEDSSGNLLGRNSFEVRVCACPGRRRTEENLRKKGEPHHELP
PGSTKRALPNNTSSSPQPKKKPLDGEYFTLQIRGRERFEMFRELNEALELKDAQAGKEPG
GSAHSSHLKSKKGQSTSRHKKLMFKTEGPDSD
```

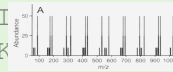


# Interpreting fragmentation spectra

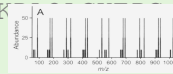
```
>sp|P04637|P53_HUMAN Cellular tumor antigen p53 OS=Homo sapiens GN=TP53 PE=1 SV=4
MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLSPLPSQAMDDLMLSPDDIEQWFTEDEPGP
DEAPRMPEAAPVAPAPAAPTPAAPAPAPSWPLSSSVPSQKTYQGSYGFRIR
SVTCTYSPALNKMFCQLAKTCTPVQLWVDSTPPPGRTRVRAMAIYKQSQHMTE
RCSDSDDLAPPQHLIRVEGNLRVEYLDDRNTRFHSVVVPYEPPEVGSDCTT
SCMGMNRRPILTIITLEDSSGNLLGRNSFEVRVCACPGRRRTEENLRK
PGSTKRALPNNTSSSPQPKKKPLDGEYFTLQIRGRERFEMFRELNEALELKR
GSRAHSSHLKSKKGQSTSRHKKLMFKTEGPDSD
```



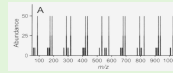
**EQLLPPK**



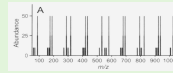
**ANDTLVR**



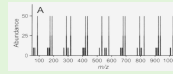
**LHVETVK**



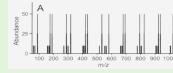
**TYTCDAK**



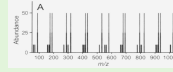
**AMAIYK**



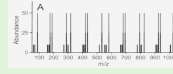
**EAAVLYK**



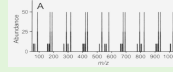
**EADVHGK**



**SILWTK**



**AEPANAGK**



**ELHSGAAR**

# Interpreting fragmentation spectra

```
>sp|P04637|P53_HUMAN Cellular tumor antigen p53 OS=Homo sapiens GN=TP53 PE=1 SV=4
MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLSPLPSQAMDDLMLSPDDIEQWFTEDPGP
DEAPRMPEAAPVAPAPAAPTPAAPAPAPSWPLSSVPEKSTVYSGYGFR
SVTCTYSPALNKMFCQLAKTCPVQLWVDSTPPPGTRVRFAMAIYKQSQHMTE
RCSDSDGLAPPQHLIRVEGNLRVEYLDNRNTRFRHSVVVPYERPEVGSDCTT
SCMGMNRRPILTIITLEDSSGNLLGRNSFEVRVCGEPIDDKLEENLRK
PGSTKRALPNNTSSSPQPKKKPLDGEYFTLQIRGRERFEMFRELNEALELK
GSRAHSSHLKSKKGQSTSRHKKLMFKTEGPDSD
```

1.6383

AMAIYK

1.2232

0.1504

2.5447

5.6032

2.0500

2.7363

1.4303

0.3376

1.9451

**EQLLPPK**

**ANDTLVR**

**LHVETVK**

**TYTCDAK**

**AMAIYK**

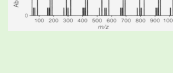
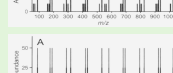
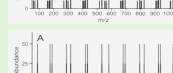
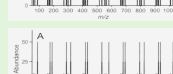
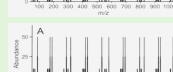
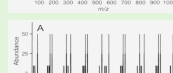
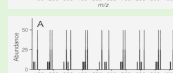
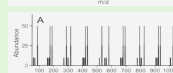
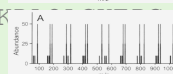
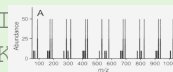
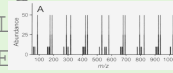
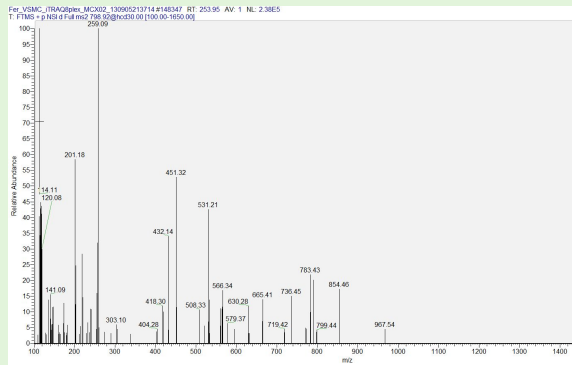
**EAAVLYK**

**EADVHGK**

**SILWTK**

**AEPANAGK**

**ELHSGAAR**



# Interpreting fragmentation spectra

>sp|P04637|P53\_HUMAN Cellular tumor antigen p53 OS=Homo sapiens GN=TP53 PE=1 SV=4  
 MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLSPLPSQAMDDLMLSPDDIEQWFTEDPGP  
 DEAPRMPEAAPVPAPAPAAPTPAAPAPAPSWPLSSVPEASTYSGSYGFRI  
 SVTCTYSPALNKMFCQLAKTCPVQLWVDSTPPPGTRVRA**AMAIYK**QSQHMTE  
 RCSDSDGLAPPQHLIRVEGNLRVEYLDNRNTRFHSVVVPYERPEVGSDCTT  
 SCMGMNRRPILTIITLEDSSGNLLGRNSFEVRVCGEPIDDKLEENLRK  
 PGSTKRALPNNTSSSPQPKKKPLDGEYFTLQIRGRERFEMFRELNEALELK  
 GSRAHSSHLKSKKGQSTSRHKKLMFKTEGPDSD

1.6383

1.2232

0.1504

2.5447

5.6032

2.0500

2.7363

1.4303

0.3376

1.9451

**EQLLPPK**

**ANDTLVR**

**LHVETVK**

**TYTCDAK**

**AMAIYK**

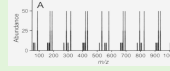
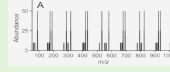
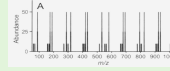
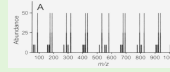
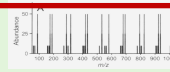
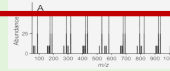
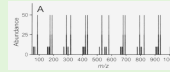
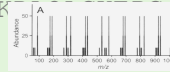
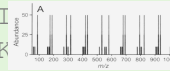
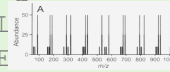
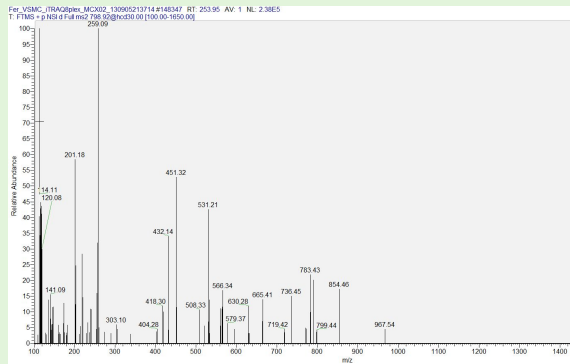
**EA AVL YK**

**EADVHGK**

**SILWTK**

**AE PANAGK**

**ELHSGAAR**



# Interpreting fragmentation spectra

```
>sp|P04637|P53_HUMAN Cellular tumor antigen p53 OS=Homo sapiens GN=TP53 PE=1 SV=4
MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLSPLPSQAMDDLMLSPDDIEQWFTEDEPGP
DEAPRMPEAAPVPAPAPAAPTPAAPAPAPSWPLSSVPEASTYSGSYGFR
SVTCTYSPALNKMFCQLAKTCVPQLWVDSTPPPGTRVRAAMAIYKQSQHMTE
RCSDSDGLAPPQHLIRVEGNLRVEYLDNRNTRFHSVVVPYERPEVGSDCIT
SCMGGMNRRPILTIITLEDSSGNLLGRNSFEVRVCGEPEDKKEENLRK
PGSTKRALPNNTSSSPQPKKKPLDGEYFTLQIRGRERFEMFRELNEALELK
GSRAHSSHLKSKKGQSTSRHKKLMFKTEGPDSD
```

**1.6383**

**1.2232**

**0.1504**

**2.5447**

**5.6032**

**2.0500**

**2.7363**

**1.4303**

**0.3376**

**1.9451**

**EQLLPPK**

**ANDTLVR**

**LHVETVK**

**TYTCDAK**

**AMAIYK**

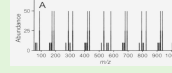
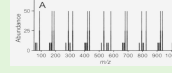
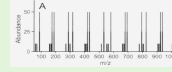
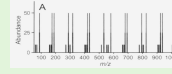
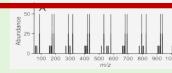
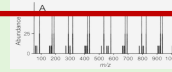
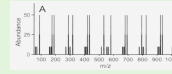
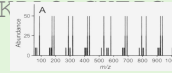
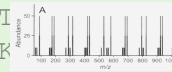
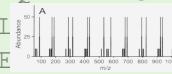
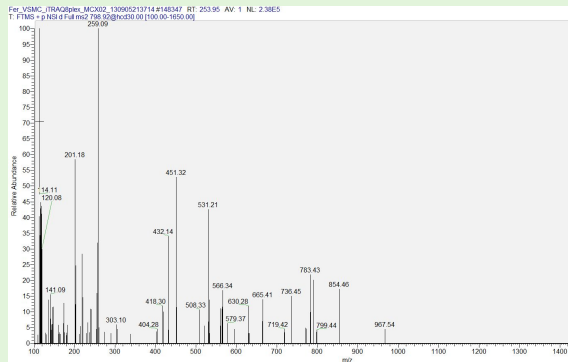
**EAAVLYK**

**EADVHGK**

**SILWTK**

**AEPANAGK**

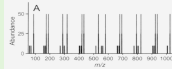
**ELHSGAAR**



the best XCorr is 5.6032,  
but is this good or bad?

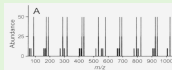
# XCorr

1.6383



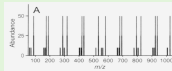
**EQLLPPK**

1.2232



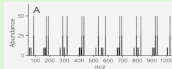
**ANDTLVR**

0.1504



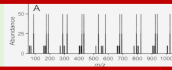
**LHVETVK**

2.5447



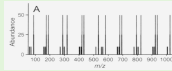
**TYTCDAK**

5.6032



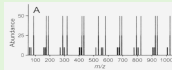
**AMAIYK**

2.0500



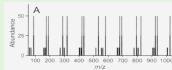
**EAAVLYK**

2.7363



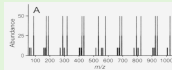
**EADVHGK**

1.4303



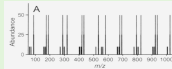
**SILWTK**

0.3376



**AEPANAGK**

1.9451

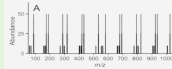


**ELHSGAAR**



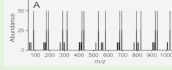
# XCorr

1.6383



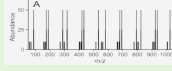
EQLLPPK

1.2232



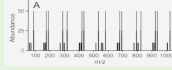
ANDTLVR

0.1504



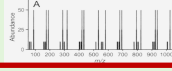
LHVETVK

2.5447



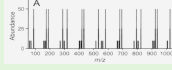
TYTCDAK

5.6032



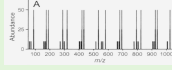
AMAIYK

2.0500



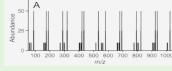
EAAVLYK

2.7363



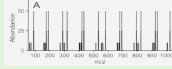
EADVHGK

1.4303



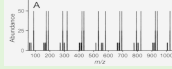
SILWTK

0.3376



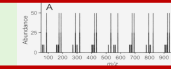
AEPANAGK

1.9451



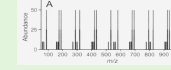
ELHSGAAR

5.6032



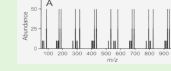
AMAIYK

2.7363



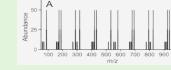
EADVHGK

2.5447



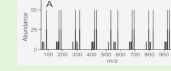
TYTCDAK

2.0500



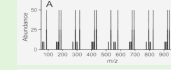
EAAVLYK

1.9451



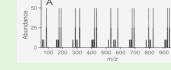
ELHSGAAR

1.6383



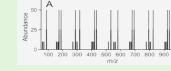
EQLLPPK

1.4303



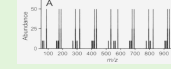
SILWTK

1.2232



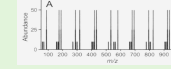
ANDTLVR

0.3376



AEPANAGK

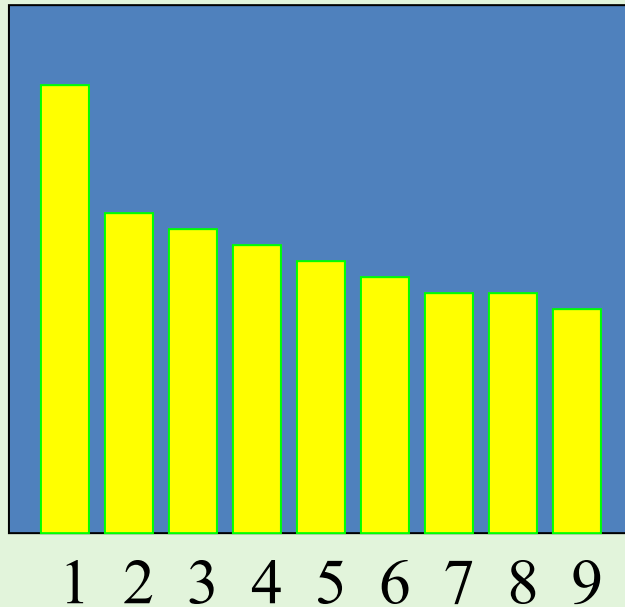
0.1504



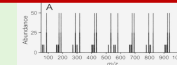
LHVETVK

# XCorr

Score

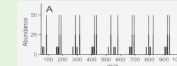


5.6032



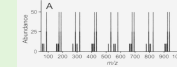
**AMAIYK**

2.7363



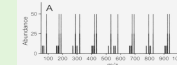
**EADVHGK**

2.5447



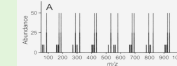
**TYTCDAK**

2.0500



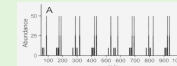
**EAAVLYK**

1.9451



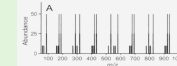
**ELHSGAAR**

1.6383



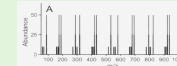
**EQLLPPK**

1.4303



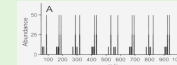
**SILWTK**

1.2232



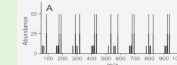
**ANDTLVR**

0.3376



**AEPANAGK**

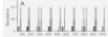
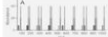
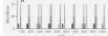
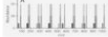
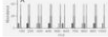
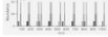
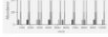
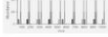
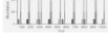
0.1504



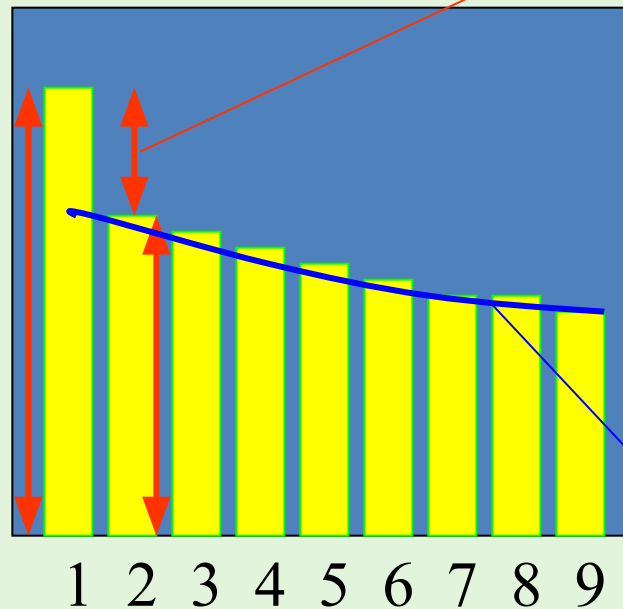
**LHVETVK**

# XCorr is not enough...

## *DeltaCn*

|        |                                                                                   |          |
|--------|-----------------------------------------------------------------------------------|----------|
| 5.6032 |  | AMAIYK   |
| 2.7363 |  | EADVHGK  |
| 2.5447 |  | TYTCDAK  |
| 2.0500 |  | EAAVLYK  |
| 1.9451 |  | ELHSGAAR |
| 1.6383 |  | EQLLPPK  |
| 1.4303 |  | SILWTK   |
| 1.2232 |  | ANDTLVR  |
| 0.3376 |  | AEPANAGK |
| 0.1504 |  | LHVETVK  |

Score



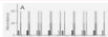
$$\Delta C_n = \frac{x_1 - x_2}{x_1}$$

$$\Delta C_n = \frac{5.6032 - 2.7363}{5.6032} = 0.51$$

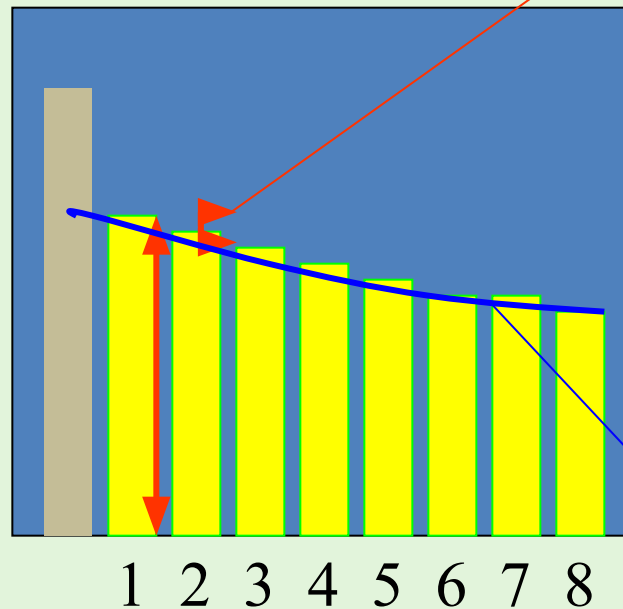
random  
behaviour

# XCorr is not enough...

## *DeltaCn*

|        |                                                                                   |          |
|--------|-----------------------------------------------------------------------------------|----------|
| 5.6032 |  | AMATYK   |
| 2.7363 |  | EADVHGK  |
| 2.5447 |  | TYTCDAK  |
| 2.0500 |  | EAAVLYK  |
| 1.9451 |  | ELHSGAAR |
| 1.6383 |  | EQLLPPK  |
| 1.4303 |  | SILWTK   |
| 1.2232 |  | ANDTLVR  |
| 0.3376 |  | AEPANAGK |
| 0.1504 |  | LHVETVK  |

Score



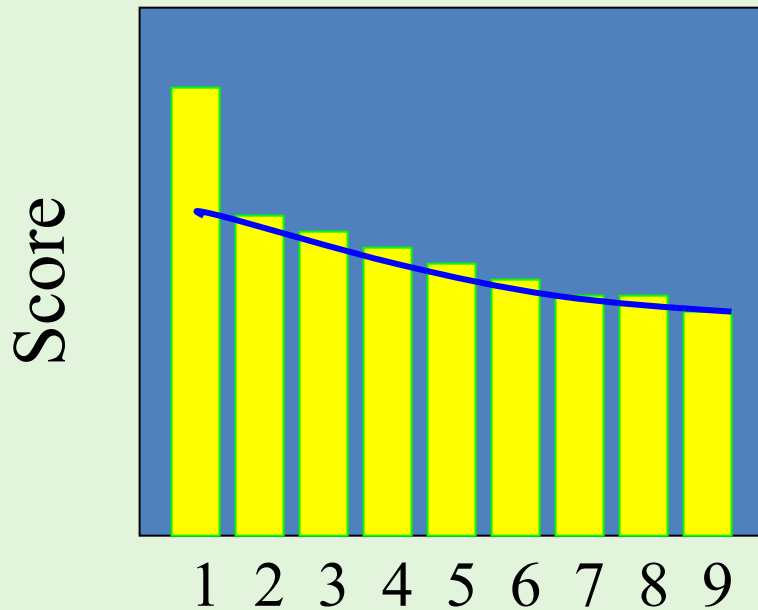
$$\Delta C_n = \frac{x_1 - x_2}{x_1}$$

$$\Delta C_n = \frac{2.7363 - 2.5447}{2.7363} = 0.07$$

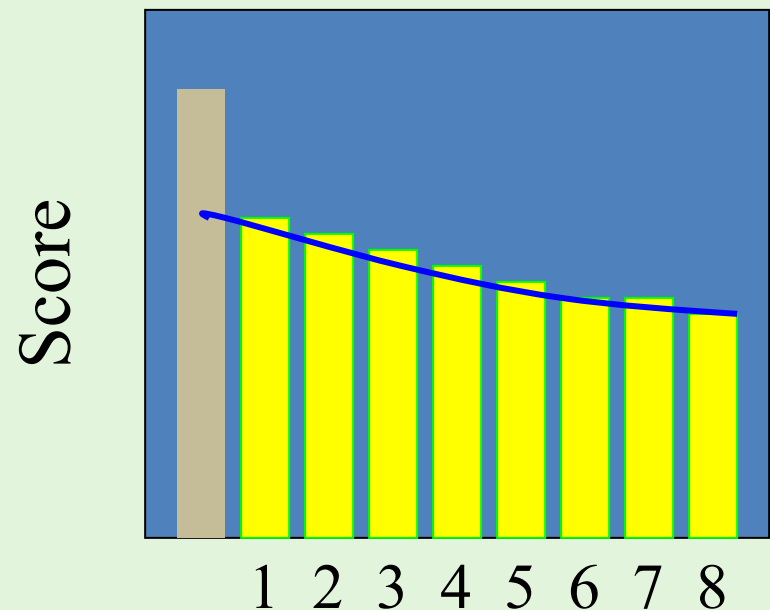
random  
behaviour

# XCorr is not enough...

## *DeltaCn*



$$\Delta C_n = 0.51$$



$$\Delta C_n = 0.07$$

# What does this SEQUEST search tell you?

```
010609_SILAC_alicPru_Sach.3843.3843.2.out
TurboSEQUEST v.27 (rev. 12), (c) 1999-2005
Molecular Biotechnology, Univ. of Washington, J.Eng/S.Morgan/J.Yates
Licensed to ThermoFinnigan Corp.
01/06/2010, 06:07 PM, 0.2 sec on PEDROBW
(M+H)+ mass = 1031.5554 ~ 2.0000 (+2), fragment tol = 1.2000, MONO/MONO
total inten = 4691.5, lowest Sp = 352.5, # matched peptides = 29680
# amino acids = 209702, # proteins = 26885,
E:\databases\quixotPlusHY\uniprot_sprot_may2009_R57_3_HumYeast_tryp_FC57_VM16K6R6.fasta.hdr
ion series nABY ABCDVWXYZ: 0 1 1 0.0 1.0 0.0 0.0 0.0 0.0 0.0 1.0 0.0
display top 10/0, ion % = 0.0, CODE = 101040
(M* +15.99490) (K# +6.02010) (R@ +6.02010) C=160.03018 Enzyme:None selected
```

| #   | Rank/Sp | Id#   | (M+H)+                | deltCn            | XCorr  | Sp     | Ions  | Reference                | Peptide          |
|-----|---------|-------|-----------------------|-------------------|--------|--------|-------|--------------------------|------------------|
| 1.  | 1 / 1   | 6443  | 1031.63031            | 0.0000            | 3.2703 | 1251.4 | 18/20 | sp P68104 EF1A1_HUMAN +3 | K.IGGIGTVPVGR@.V |
|     |         | 6444  | sp Q05639 EF1A2_HUMAN | Elongation facto  |        |        |       |                          |                  |
|     |         | 6445  | sp Q5VTE0 EF1A3_HUMAN | Putative elongat  |        |        |       |                          |                  |
|     |         | 6446  | sp P02994 EF1A_YEAST  | Elongation factor |        |        |       |                          |                  |
| 2.  | 2 / 22  | 515   | 1031.59558            | 0.5268            | 1.5474 | 568.9  | 12/16 | sp Q08828 ADCY1_HUMAN    | R.RALRTASEK.L    |
| 3.  | 3 / 232 | 2063  | 1031.55368            | 0.5560            | 1.4519 | 358.0  | 10/14 | sp P12644 BMP4_HUMAN     | R.INIYEVN*K#.P   |
| 4.  | 4 / 91  | 7421  | 1032.67317            | 0.5571            | 1.4484 | 437.9  | 12/18 | sp Q9HB96 FANCE_HUMAN +1 | R.GLGLGGRR@LK.S  |
|     |         | 7421  | sp Q9HB96 FANCE_HUMAN | Fanconi anemia g  |        |        |       |                          |                  |
| 5.  | 5 / 124 | 7421  | 1032.67317            | 0.5574            | 1.4474 | 411.4  | 12/18 | sp Q9HB96 FANCE_HUMAN +1 | R.GLGLGGRR@LK#.S |
|     |         | 7421  | sp Q9HB96 FANCE_HUMAN | Fanconi anemia g  |        |        |       |                          |                  |
| 6.  | 6 / 72  | 18533 | 1031.67792            | 0.5574            | 1.4473 | 465.4  | 11/16 | sp P05750 RS3_YEAST      | K.LLNLGAIR@.A    |
| 7.  | 7 / 13  | 17300 | 1032.57141            | 0.5674            | 1.4147 | 619.2  | 13/14 | sp P04049 RAF1_HUMAN +1  | K.NIIHRDMK#.S    |
|     |         | 17300 | sp P04049 RAF1_HUMAN  | RAF proto-oncogen |        |        |       |                          |                  |
| 8.  | 8 / 128 | 3174  | 1033.55371            | 0.5831            | 1.3635 | 410.4  | 11/14 | sp P29016 CD1B_HUMAN     | R.RRSYQNIP       |
| 9.  | 9 / 5   | 21975 | 1031.55181            | 0.5916            | 1.3355 | 675.8  | 16/18 | sp P21580 TNAP3_HUMAN    | R.TPGDR@TGTSK#.C |
| 10. | 10 / 94 | 18533 | 1031.67792            | 0.5991            | 1.3110 | 433.4  | 10/16 | sp P05750 RS3_YEAST      | K.LLNLGAIR@.A    |

I. interpret results from

# What does this SEQUEST search tell you?

```
010609_SILAC_alicPru_Sach.3843.3843.2.out
TurboSEQUEST v.27 (rev. 12), (c) 1999-2005
Molecular Biotechnology, Univ. of Washington, J.Eng/S.Morgan/J.Yates
Licensed to ThermoFinnigan Corp.
01/06/2010, 06:07 PM, 0.2 sec on PEDROBW
(M+H)+ mass = 1031.5554 ~ 2.0000 (+2), fragment tol = 1.2000, MONO/MONO
total inten = 4691.5, lowest Sp = 352.5, # matched peptides = 29680
# amino acids = 209702, # proteins = 26885,
E:\databases\quixotPlusHY\uniprot_sprot_may2009_R57_3_HumYeast_tryp_FC57_VM16K6R6.fasta.hdr
ion series nABY ABCDVWXYZ: 0 1 1 0.0 1.0 0.0 0.0 0.0 0.0 0.0 1.0 0.0
display top 10/0, ion % = 0.0, CODE = 101040
(M* +15.99490) (K# +6.02010) (R@ +6.02010) C=160.03018 Enzyme:None selected
```

| #   | Rank/Sp | Id#   | (M+H)+                | deltCn            | XCorr         | Sp     | Ions  | Reference                | Peptide          |
|-----|---------|-------|-----------------------|-------------------|---------------|--------|-------|--------------------------|------------------|
| 1.  | 1 / 1   | 6443  | 1031.63031            | 0.0000            | <b>3.2703</b> | 1251.4 | 18/20 | sp P68104 EF1A1_HUMAN +3 | K.IGGIGTVPVGR@.V |
|     |         | 6444  | sp Q05639 EF1A2_HUMAN | Elongation facto  |               |        |       |                          |                  |
|     |         | 6445  | sp Q5VTE0 EF1A3_HUMAN | Putative elongat  |               |        |       |                          |                  |
|     |         | 6446  | sp P02994 EF1A_YEAST  | Elongation factor |               |        |       |                          |                  |
| 2.  | 2 / 22  | 515   | 1031.59558            | <b>0.5268</b>     | 1.5474        | 568.9  | 12/16 | sp Q08828 ADCY1_HUMAN    | R.RALRTASEK.L    |
| 3.  | 3 / 232 | 2063  | 1031.55368            | 0.5560            | 1.4519        | 358.0  | 10/14 | sp P12644 BMP4_HUMAN     | R.INIYEVN*K#.P   |
| 4.  | 4 / 91  | 7421  | 1032.67317            | 0.5571            | 1.4484        | 437.9  | 12/18 | sp Q9HB96 FANCE_HUMAN +1 | R.GLGLGGRR@LK.S  |
|     |         | 7421  | sp Q9HB96 FANCE_HUMAN | Fanconi anemia g  |               |        |       |                          |                  |
| 5.  | 5 / 124 | 7421  | 1032.67317            | 0.5574            | 1.4474        | 411.4  | 12/18 | sp Q9HB96 FANCE_HUMAN +1 | R.GLGLGGRR@LK#.S |
|     |         | 7421  | sp Q9HB96 FANCE_HUMAN | Fanconi anemia g  |               |        |       |                          |                  |
| 6.  | 6 / 72  | 18533 | 1031.67792            | 0.5574            | 1.4473        | 465.4  | 11/16 | sp P05750 RS3_YEAST      | K.LLNLGAIR@.A    |
| 7.  | 7 / 13  | 17300 | 1032.57141            | 0.5674            | 1.4147        | 619.2  | 13/14 | sp P04049 RAF1_HUMAN +1  | K.NIIHRDMK#.S    |
|     |         | 17300 | sp P04049 RAF1_HUMAN  | RAF proto-oncogen |               |        |       |                          |                  |
| 8.  | 8 / 128 | 3174  | 1033.55371            | 0.5831            | 1.3635        | 410.4  | 11/14 | sp P29016 CD1B_HUMAN     | R.RRSYQNIP       |
| 9.  | 9 / 5   | 21975 | 1031.55181            | 0.5916            | 1.3355        | 675.8  | 16/18 | sp P21580 TNAP3_HUMAN    | R.TPGDR@TGTSK#.C |
| 10. | 10 / 94 | 18533 | 1031.67792            | 0.5991            | 1.3110        | 433.4  | 10/16 | sp P05750 RS3_YEAST      | K.LLNLGAIR@.A    |

I. interpret results from



# What about this one?

```
010609_SILAC_alicPru_Sach.3912.3912.2.out
TurboSEQUEST v.27 (rev. 12), (c) 1999-2005
Molecular Biotechnology, Univ. of Washington, J.Eng/S.Morgan/J.Yates
Licensed to ThermoFinnigan Corp.
01/06/2010, 06:07 PM, 0.3 sec on PEDROBW
(M+H)+ mass = 1199.4766 ~ 2.0000 (+2), fragment tol = 1.2000, MONO/MONO
total inten = 5777.8, lowest Sp = 346.4, # matched peptides = 31534
# amino acids = 207992, # proteins = 26885,
E:\databases\quixotPlusHY\uniprot_sprot_may2009_R57_3_HumYeast_tryp_FC57_VM16K6R6.fasta.hdr
ion series nABY ABCDVWXYZ: 0 1 1 0.0 1.0 0.0 0.0 0.0 0.0 0.0 1.0 0.0
display top 10/0, ion % = 0.0, CODE = 101040
(M* +15.99490) (K# +6.02010) (R@ +6.02010) C=160.03018 Enzyme:None selected
```

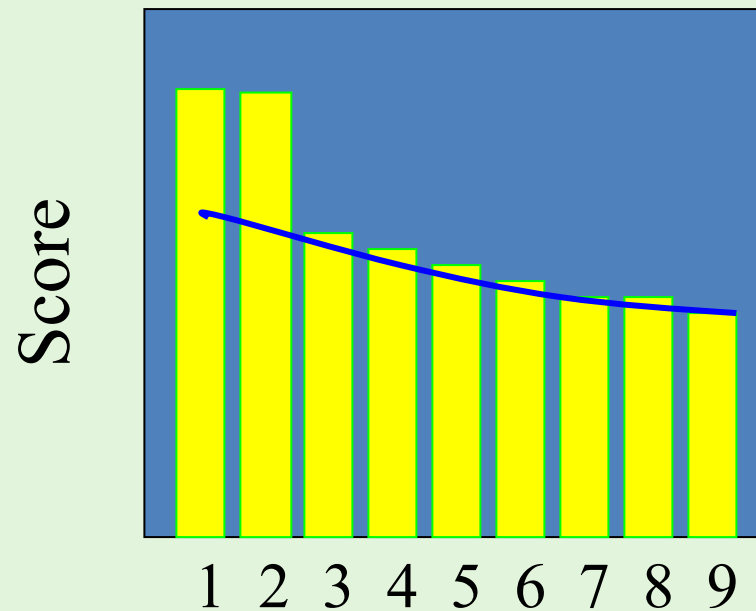
| #   | Rank/Sp | Id#   | (M+H)+                | deltCn           | XCorr  | Sp    | Ions  | Reference             | Peptide             |
|-----|---------|-------|-----------------------|------------------|--------|-------|-------|-----------------------|---------------------|
| 1.  | 1 /202  | 22517 | 1199.65292            | 0.0000           | 1.3928 | 358.1 | 12/18 | sp Q96AY4 TTC28_HUMAN | +1 K.DLGNK#R@EEAR.A |
|     |         | 22517 | sp Q96AY4 TTC28_HUMAN | Tetratricopeptid |        |       |       |                       |                     |
| 2.  | 2 / 40  | 6536  | 1198.64734            | 0.0294           | 1.3518 | 470.3 | 15/24 | sp Q8N6I1 EID2_HUMAN  | R.MAAARAAPAAAAR.G   |
| 3.  | 3 /135  | 12917 | 1198.64661            | 0.0413           | 1.3354 | 389.6 | 13/18 | sp Q96EY8 MMAB_HUMAN  | R.LSDYLFTLAR.Y      |
| 4.  | 4 / 10  | 25687 | 1197.72009            | 0.0737           | 1.2902 | 550.8 | 15/20 | sp Q08748 YO296_YEAST | K.SLVANIVKEPK.E     |
| 5.  | 5 / 79  | 7789  | 1199.61842            | 0.0741           | 1.2897 | 414.9 | 14/20 | sp P32785 FMT_YEAST   | R.LDNGSKPGMFK#.Y    |
| 6.  | 6 / 58  | 9666  | 1199.70323            | 0.0957           | 1.2596 | 439.5 | 13/20 | sp Q04432 HSP31_YEAST | K.NLATVEDVAK#K#.Y   |
| 7.  | 7 /181  | 5876  | 1199.73584            | 0.1027           | 1.2498 | 367.0 | 13/22 | sp Q9UPY3 DICER_HUMAN | R.ILGLTASILNGK.C    |
| 8.  | 8 /221  | 23009 | 1197.71005            | 0.1042           | 1.2478 | 352.5 | 12/18 | sp Q8NB66 UN13C_HUMAN | +1 K.SLDR@TVR@NPK.T |
|     |         | 23009 | sp Q8NB66 UN13C_HUMAN | Protein unc-13 h |        |       |       |                       |                     |
| 9.  | 9 /215  | 6942  | 1198.59230            | 0.1363           | 1.2030 | 353.8 | 11/16 | sp P57679 EVC_HUMAN   | R.IMEDHEER@K#.L     |
| 10. | 10 / 44 | 6357  | 1198.74060            | 0.1436           | 1.1928 | 459.4 | 14/20 | sp P39995 EAF5_YEAST  | K.LGINDILTIVK.N     |

# What about this one?

```
010609_SILAC_alicPru_Sach.3912.3912.2.out
TurboSEQUEST v.27 (rev. 12), (c) 1999-2005
Molecular Biotechnology, Univ. of Washington, J.Eng/S.Morgan/J.Yates
Licensed to ThermoFinnigan Corp.
01/06/2010, 06:07 PM, 0.3 sec on PEDROBW
(M+H)+ mass = 1199.4766 ~ 2.0000 (+2), fragment tol = 1.2000, MONO/MONO
total inten = 5777.8, lowest Sp = 346.4, # matched peptides = 31534
# amino acids = 207992, # proteins = 26885,
E:\databases\quixotPlusHY\uniprot_sprot_may2009_R57_3_HumYeast_tryp_FC57_VM16K6R6.fasta.hdr
ion series nABY ABCDVWXYZ: 0 1 1 0.0 1.0 0.0 0.0 0.0 0.0 0.0 1.0 0.0
display top 10/0, ion % = 0.0, CODE = 101040
(M* +15.99490) (K# +6.02010) (R@ +6.02010) C=160.03018 Enzyme:None selected
```

| #   | Rank/Sp | Id#   | (M+H)+                | deltCn           | XCorr         | Sp    | Ions  | Reference             | Peptide             |
|-----|---------|-------|-----------------------|------------------|---------------|-------|-------|-----------------------|---------------------|
| 1.  | 1 /202  | 22517 | 1199.65292            | 0.0000           | <b>1.3928</b> | 358.1 | 12/18 | sp Q96AY4 TTC28_HUMAN | +1 K.DLGNK#R@EEAR.A |
|     |         | 22517 | sp Q96AY4 TTC28_HUMAN | Tetratricopeptid |               |       |       |                       |                     |
| 2.  | 2 / 40  | 6536  | 1198.64734            | <b>0.0294</b>    | 1.3518        | 470.3 | 15/24 | sp Q8N6I1 EID2_HUMAN  | R.MAAARAAPAAAAR.G   |
| 3.  | 3 /135  | 12917 | 1198.64661            | 0.0413           | 1.3354        | 389.6 | 13/18 | sp Q96EY8 MMAB_HUMAN  | R.LSDYLFTLAR.Y      |
| 4.  | 4 / 10  | 25687 | 1197.72009            | 0.0737           | 1.2902        | 550.8 | 15/20 | sp Q08748 YO296_YEAST | K.SLVANIVKEPK.E     |
| 5.  | 5 / 79  | 7789  | 1199.61842            | 0.0741           | 1.2897        | 414.9 | 14/20 | sp P32785 FMT_YEAST   | R.LDNGSKPGMFK#.Y    |
| 6.  | 6 / 58  | 9666  | 1199.70323            | 0.0957           | 1.2596        | 439.5 | 13/20 | sp Q04432 HSP31_YEAST | K.NLATVEDVAK#K#.Y   |
| 7.  | 7 /181  | 5876  | 1199.73584            | 0.1027           | 1.2498        | 367.0 | 13/22 | sp Q9UPY3 DICER_HUMAN | R.ILGLTASILNGK.C    |
| 8.  | 8 /221  | 23009 | 1197.71005            | 0.1042           | 1.2478        | 352.5 | 12/18 | sp Q8NB66 UN13C_HUMAN | +1 K.SLDR@TVR@NPK.T |
|     |         | 23009 | sp Q8NB66 UN13C_HUMAN | Protein unc-13 h |               |       |       |                       |                     |
| 9.  | 9 /215  | 6942  | 1198.59230            | 0.1363           | 1.2030        | 353.8 | 11/16 | sp P57679 EVC_HUMAN   | R.IMEDHEER@K#.L     |
| 10. | 10 / 44 | 6357  | 1198.74060            | 0.1436           | 1.1928        | 459.4 | 14/20 | sp P39995 EAF5_YEAST  | K.LGINDILTIVK.N     |

# DetlaCn looks nice, but what if...?



$$\Delta C_n = 0.03$$

# converting scores to (un)certainties: p-value

«Probability for a spectrum to have a score at least as extreme as the one obtained, when assigning a random peptide sequence»

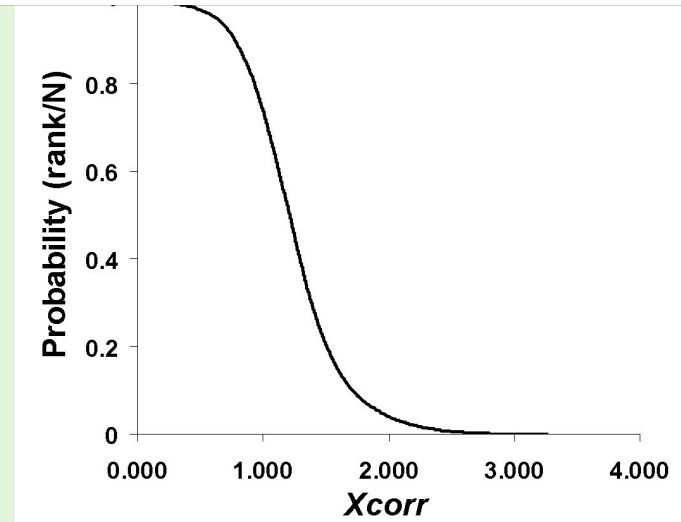
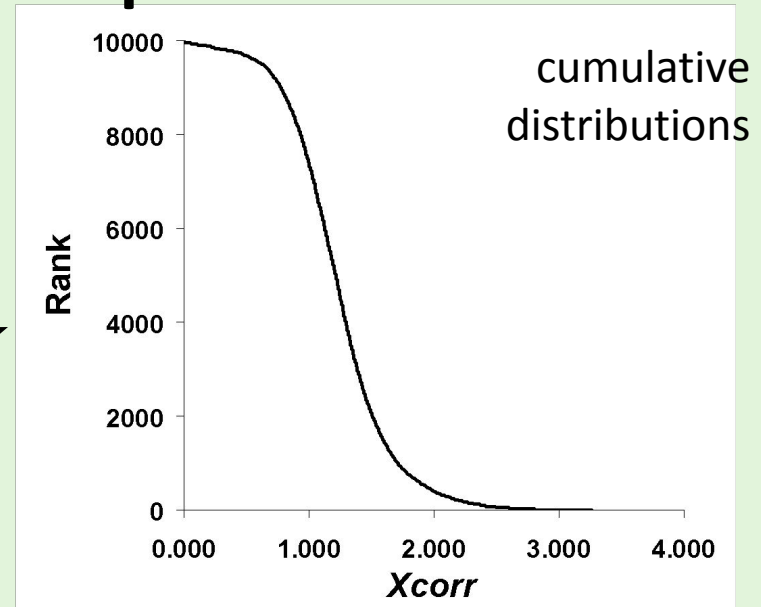
# converting scores to (un)certainities: p-value

- p-value of the event «flip a coin *three* times, and get *three* tails»?  $\rightarrow (1/2)^3 = 0.125$
- p-value of the event «flip a coin *five* times, and get **at least three** tails»?  $\rightarrow \dots = 0.5$
- p-value of the event «flip a coin *ten* times, and get nine or ten tails»  $\rightarrow \dots \approx \mathbf{0.011}$

# converting scores to (un)certainities: p-value

- \* take all the scores from a search against a random database
- \* sort the scores from best to worst (for XCorr this means from highest to lowest)
- \* get the normalised rank

| posición  | puntuación | posición/N |
|-----------|------------|------------|
| 1         | 6.71       | 0.0001     |
| 2         | 6.01       | 0.0002     |
| 3         | 5.64       | 0.0003     |
| 4         | 5.31       | 0.0004     |
| 5         | 3.2        | 0.0005     |
| 6         | 3.18       | 0.0006     |
| 7         | 3.13       | 0.0007     |
| ...       | ...        | ...        |
| N= 10,000 | 1.35       | 1          |



# too much data? e-value or expected value

- So, you have got 1000 experiments, and in one of them there is one that has a 1/1000 chance to appear (p-value = 0.001)

is that significant?



# too much data? e-value or expected value

- So, you have got 1000 experiments, and in one of them there is one that has a 1/1000 chance to appear (p-value = 0.001)

NO!! It is not!

In fact, that is what you would expect

# too much data? e-value or expected value

«The number of assignments you expect to get with a given p-value (or lower), when using N candidates in your search»

$$e = N \cdot p$$

used very often in algorithms like BLAST or Mascot

# e-value and p-value

Do not forget:

the p-value is a probability

the e-value is an expected value

what is the range of the p-values?

what is the range of e-values?

if an identification is good,

how will be the p-value?

how will be the e-value?

# e-value and p-value

example: a kid has to wear

- a t-shirt
- trousers
- pants
- and shoes

1) There is a probability of  $1/2$  of wearing any of these on the wrong side. **What is the p-value of the event «he will wear *everything* on the wrong side»?**

2) If during one month this happened once, could this have happened just by chance, unintentionally? Is he trying to get late to school to get the day off? **Calculate the probability of this happening at least once in a month.**

3) **Calculate the e-value.**

II. calculate de p-value and e-value in everyday's life.

# Part C

## Key concepts

- \* the confusion matrix
- \* FDR, sensitivity, specificity
- \* ROC curve
- \* target and decoy databases
- \* how to make decoy databases
- \* narrow precursor tolerances

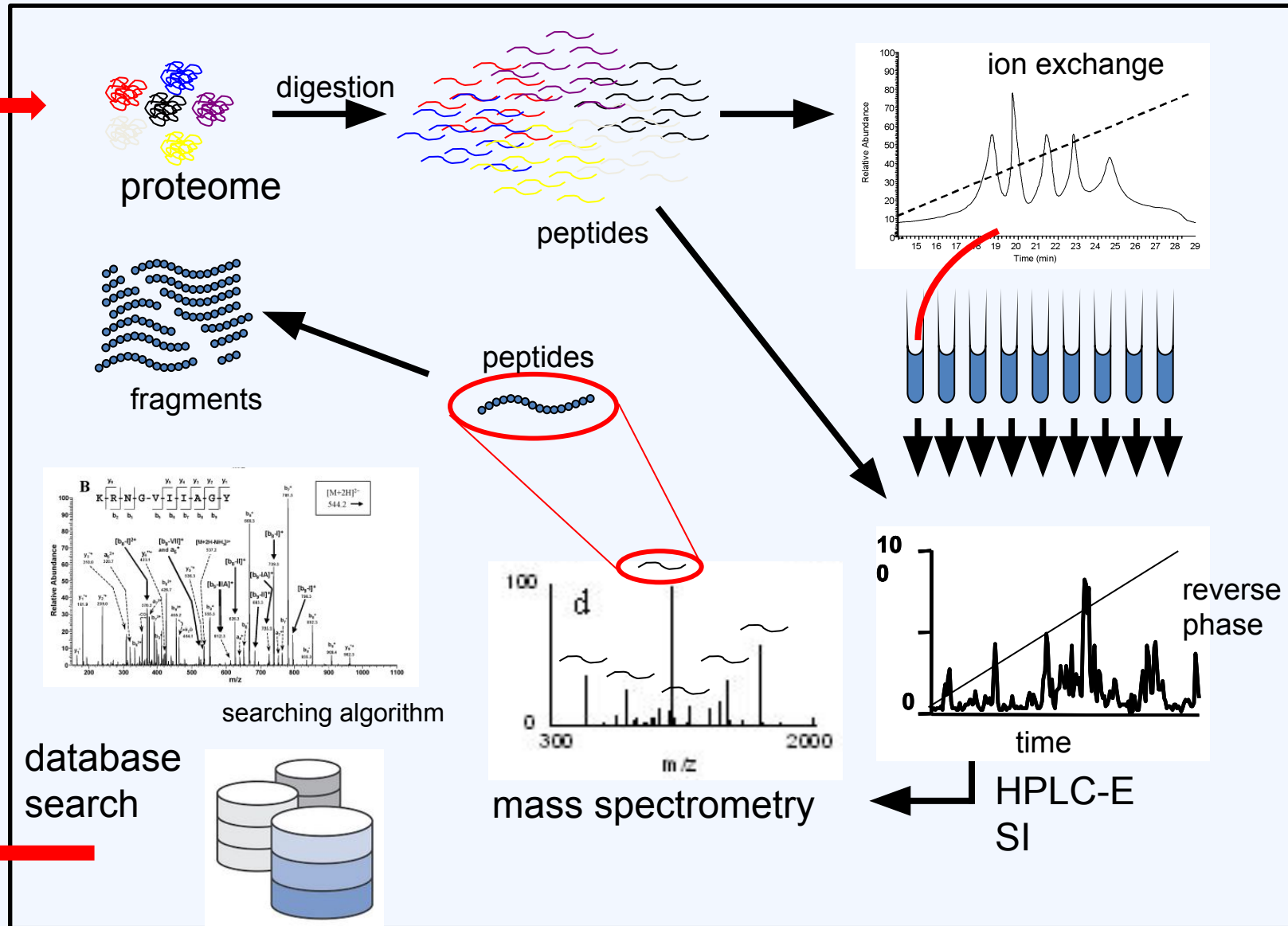
## End of Part II

In Part III we will apply these things to identify  
our own spectrum

## Part III

Search your own spectrum!

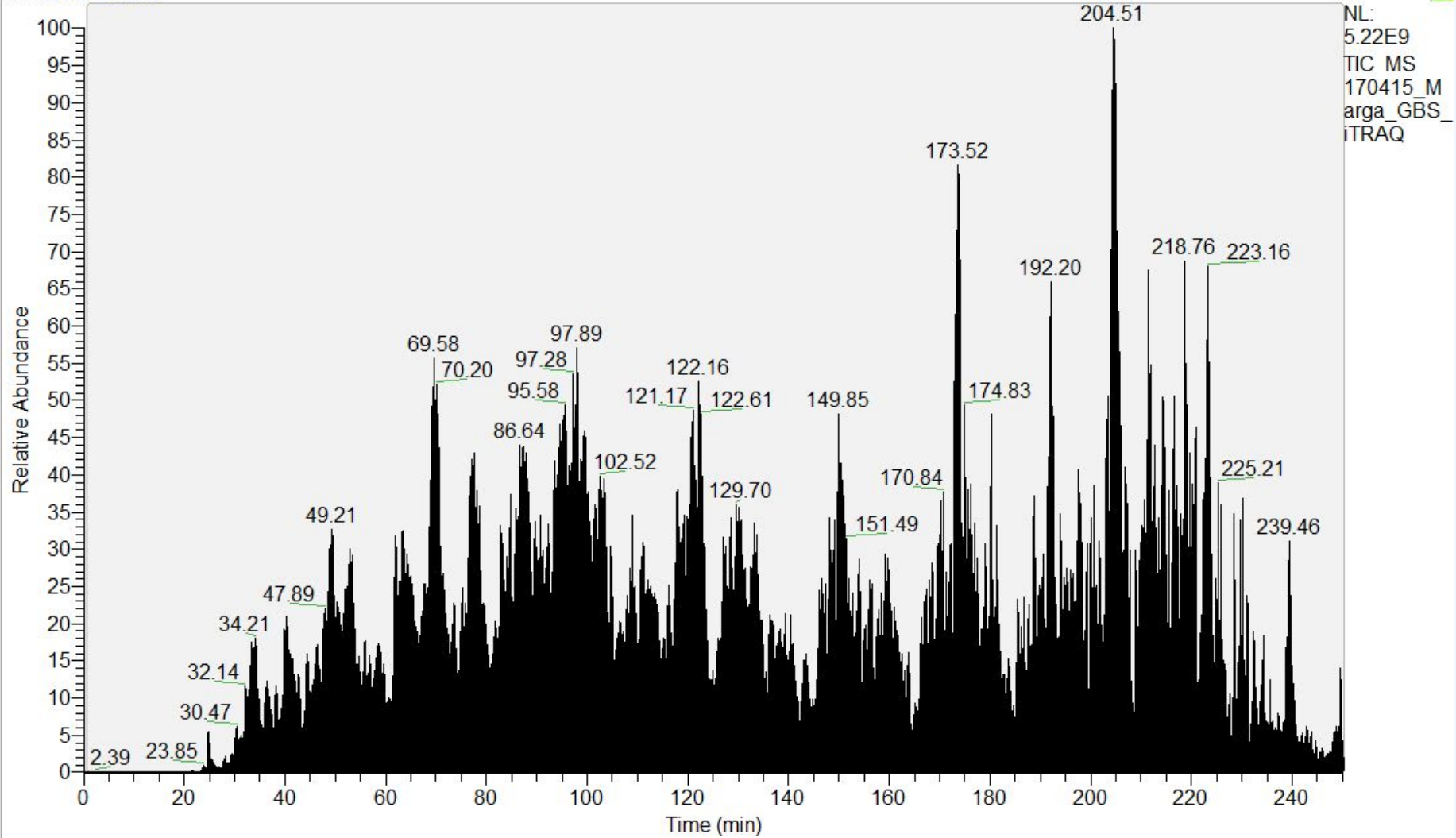
# Overview of protein identification



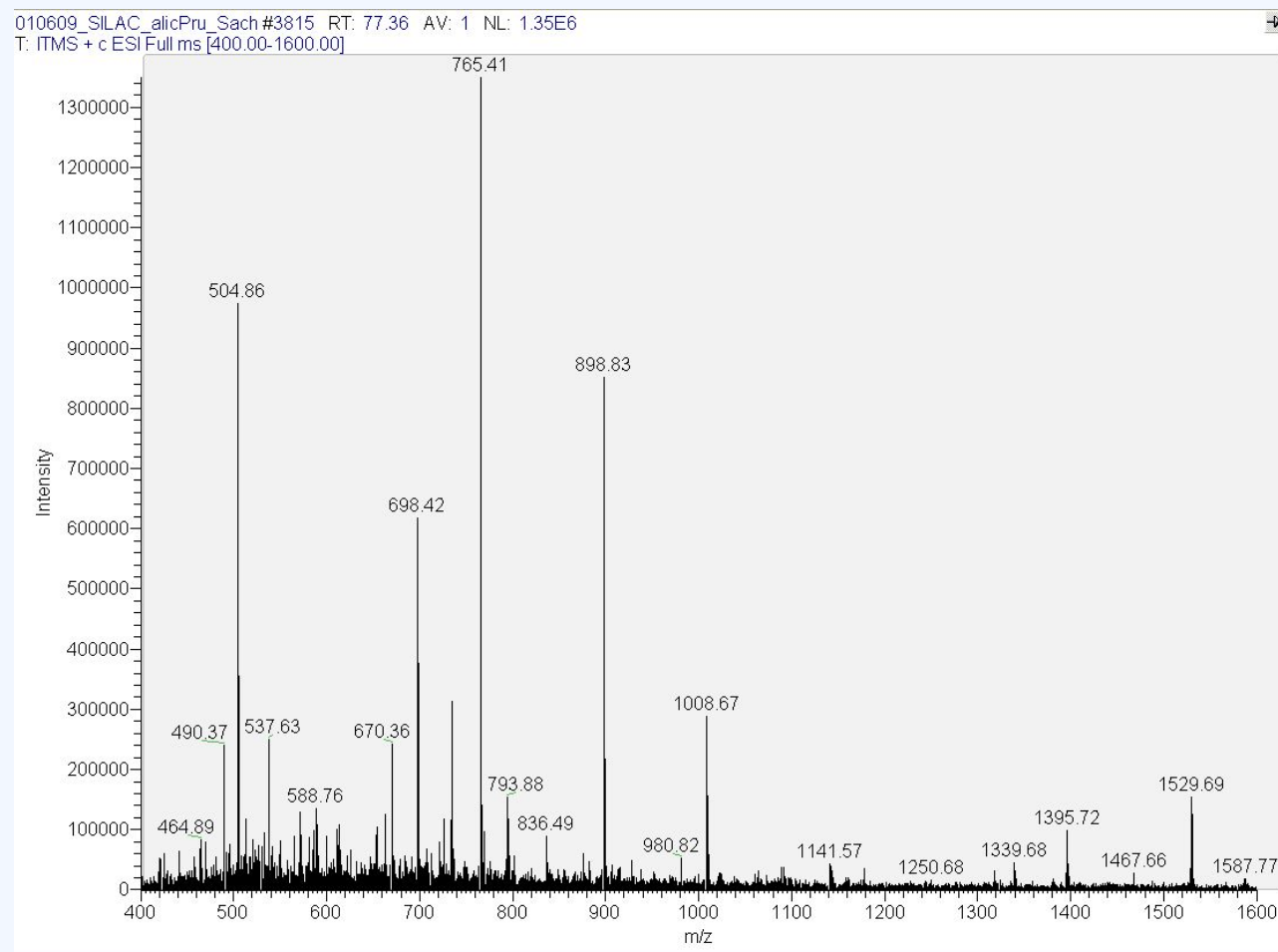


# Chromatogram

RT: 0.00 - 250.17

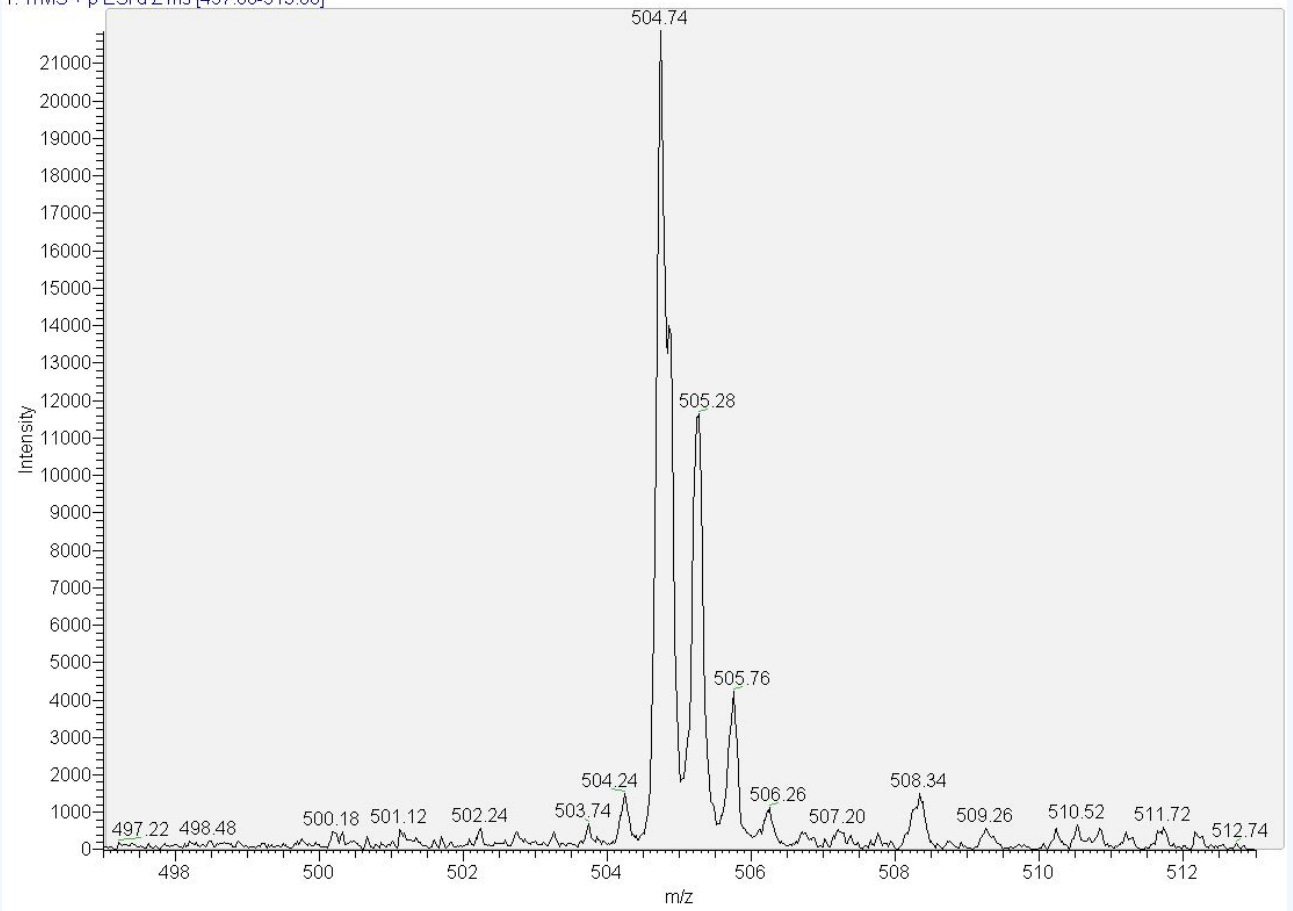


# Full scan (MS1)

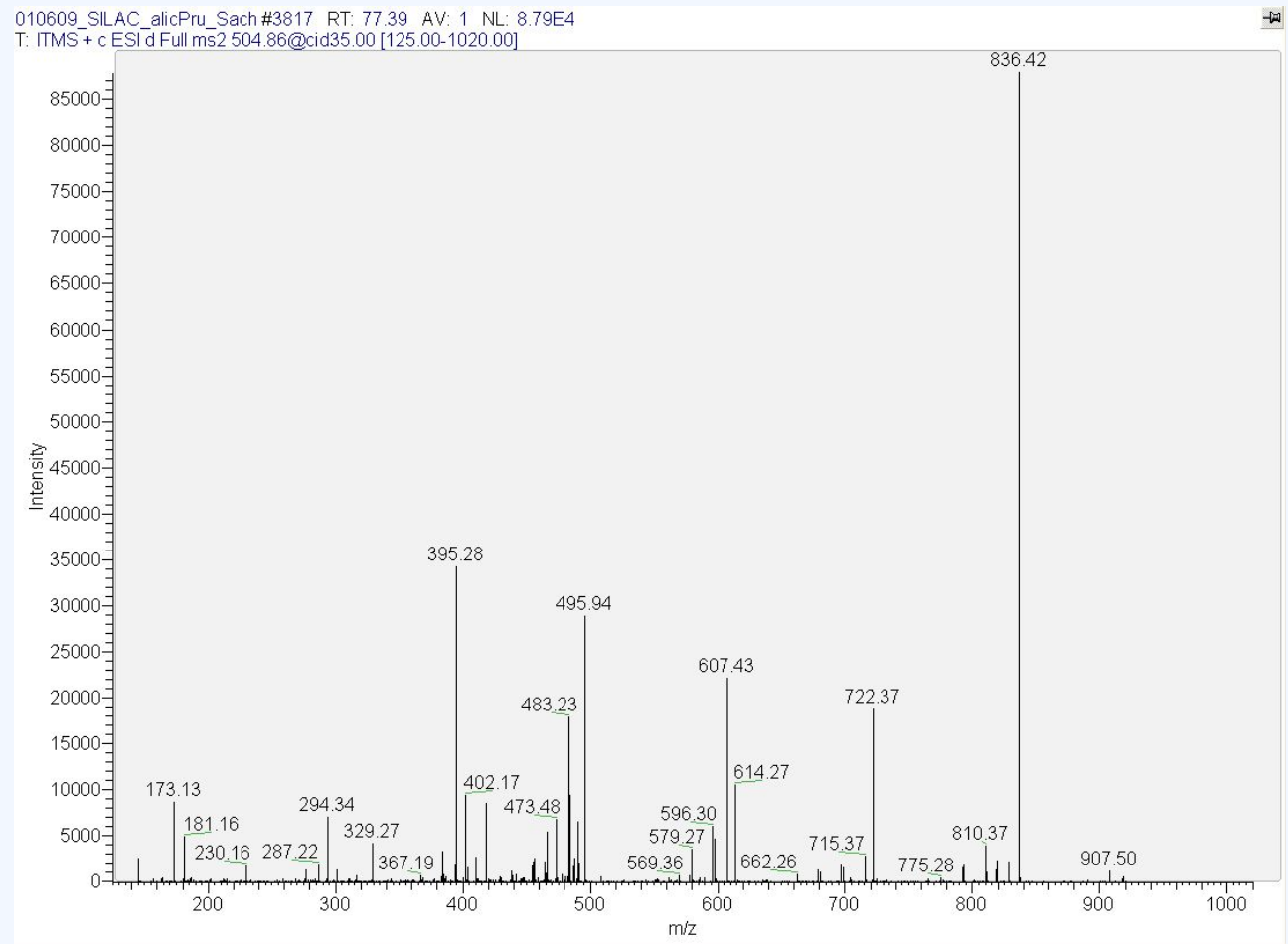


# Zoom scan (MS1)

010609\_SILAC\_aliPrü\_Sach#3816 RT: 77.37 AV: 1 NL: 2.19E4  
T: ITMS + p ESI d Z ms [497.00-513.00]



# Fragmentation spectrum (MS2)



# structure of an MGF file

```
BEGIN IONS  
TITLE=example.mgf  
PEPMASS=549.261138  
CHARGE=2+  
159.204 2870  
161.172 4580  
162.202 2930  
163.268 2790  
165.347 9800  
167.002 690  
170.214 640  
171.096 1040  
175.187 5470  
178.158 620  
181.163 970  
183.106 2660  
185.179 2900  
187.245 1020  
189.227 7130  
END IONS
```

# structure of an MGF file

mass of the precursor *ion* (MH+) →

charge of precursor ion →

mass/charge ratios

intensities

```
BEGIN IONS
TITLE=example.mgf
PEPMASS=549.261138
CHARGE=2+
159.204 2870
161.172 4580
162.202 2930
163.268 2790
165.347 9800
167.002 690
170.214 640
171.096 1040
175.187 5470
178.158 620
181.163 970
183.106 2660
185.179 2900
187.245 1020
189.227 7130
END IONS
```

The diagram illustrates the structure of an MGF file. It shows a list of mass-to-charge ratios and their corresponding intensities. Annotations include: 'mass of the precursor ion (MH+)' pointing to the PEPMASS value; 'charge of precursor ion' pointing to the CHARGE value; 'mass/charge ratios' pointing to the list of m/z values; and 'intensities' pointing to the list of intensity values.

# a handmade MGF file

mass of the precursor *ion*  
(MH<sup>+</sup>)

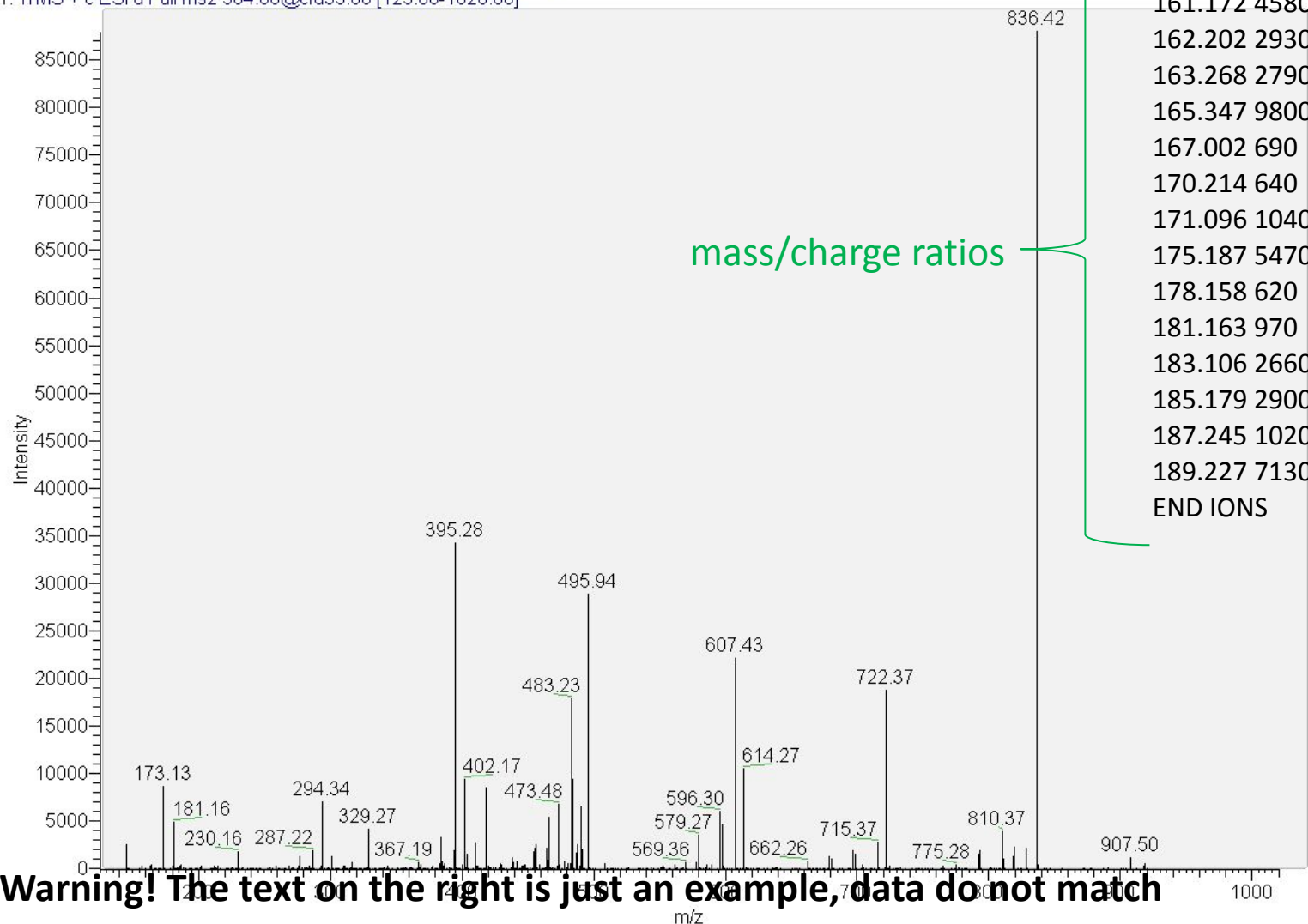
BEGIN IONS  
TITLE=example.mgf  
PEPMASS=549.261138  
CHARGE=2+

charge of  
precursor  
ion

mass/charge ratios

intensity

010609\_SILAC\_alicPru\_Sach#3817 RT: 77.39 AV: 1 NL: 8.79E4  
T: ITMS + c ESI d Full ms2 504.86@cid35.00 [125.00-1020.00]



# identifying MS2 spectra with Mascot



**MATRIX SCIENCE**

Search this site

Mascot: The trusted reference standard for protein identification by mass spectrometry for 25 years

Home | Free trial | Solutions | Products | Help | Support | Training | Monthly newsletter | About

**Access Mascot Server** | Evaluate Mascot Distiller

Proteomics solutions you can trust

Matrix Science develops software solutions for mass spectrometry proteomics. Mascot is a great choice for **researchers, core labs, industry and educators.**

Many third-party systems integrate directly with Mascot:

**Blog**

September 30, 2025

Mascot Daemon includes a protein level quantitation report for both precursor based methods (LFQ, SILAC MS/MS quantitation (iTRAQ, TMT)). The exported format is suitable for importing into [...]

Latest products

**What's new in Mascot Server 3.1**

Protein identification

Mascot Server is the trusted reference standard for protein

[www.matrixscience.com](http://www.matrixscience.com)

...o just google

*matrix science*



# identifying MS2 spectra with Mascot



Search this site 

Mascot: The trusted reference standard for protein identification by mass spectrometry for 25 years

[Home](#) [Free trial](#) [Solutions](#) [Products](#) [Help](#) [Support](#) [Training](#) [Monthly newsletter](#) [About](#)

[Access Mascot Server](#) | [Evaluate Mascot Distiller](#)

[Free trial](#) > [Access Mascot Server](#)

## Free trial: Mascot Server

You are welcome to submit searches to this free Mascot Server, subject to these restrictions:

- Searches of MS/MS data are limited in size.
- Some functions, such as no-enzyme searches, are unavailable.
- Automated searching of batches of files is not permitted.

If you want to automate search submission, perform large searches, search additional sequence databases, or customise the modifications, quantitation methods, etc., you'll need to license your own, in-house copy of Mascot Server. [Get a quote.](#)

Please [contact us](#) if you would like to trial Mascot Server in the cloud.

### More info

- > [Mascot overview](#)
- > [Search parameter reference](#)
- > [Data file format](#)
- > [Results report overview](#)



## Peptide Mass Fingerprint

The experimental data are a list of peptide mass values from the digestion of a protein by a specific enzyme such as trypsin.

[Perform search](#) | [Example of results report](#) | [Tutorial](#)

## Sequence Query

One or more peptide mass values associated with information such as partial or ambiguous sequence strings, amino acid composition information, MS/MS fragment ion masses, etc. A super-set of a sequence tag query.

[Perform search](#) | [Example of results report](#) | [More information](#)

## MS/MS Ions Search

Identification based on raw MS/MS data from one or more peptides.

[Perform search](#) | [Example of results report](#) | [Tutorial](#)

# identifying MS2 spectra with Mascot

[Home](#)
[Free trial](#)
[Products](#)
[Help](#)
[Support](#)
[Training](#)
[Monthly newsletter](#)
[About](#)

[Access Mascot Server](#)
[Evaluate Mascot Distiller](#)

[Access Mascot Server > MS/MS Sine Search](#)

## MASCOT MS/MS Ions Search

Your name
Email

Search title

Database(s)

contaminants (AA)
SwissProt (AA)

>
<

Amino acid (AA)
cRAP
UP186698\_X\_tactis
UP19116\_T\_aestivum
UP1940\_C\_elegans
UP2195\_D\_discoideum
UP219602\_F\_oxysporum
UP2311\_S\_cerevisiae
UP241690\_T\_harizianum
UP2485\_S\_pombe

Taxonomy
All entries

Enzyme
Trypsin
Allow up to
1 missed cleavages

Quantitation
SILAC K+6 R+6 multiplex

Crosslinking
None

Fixed modifications

Carbamidomethyl (C)

>
<

6C-CysPAT (C)
6C-CysPAT (N-term)
Acetyl (K)
Acetyl (N-term)
Acetyl (Protein N-term)
Amidated (C-term)
Amidated (Protein C-term)
Ammonia-loss (N-term C)
Carbamidomethyl (N-term)
Carbamyl (K)
Carbamyl (N-term)

☐ Display all modifications

Variable modifications

Oxidation (M)

>
<

Error tolerant
☐ Automatic second pass search of selected modification classes

Peptide tol. ±
0.6 Da # 13c 0 MS/MS tol. ±
2 Da

Data file
Explore... 3817\_manual.mgf

Data format
Mascot generic

Instrument
Default

Target FDI
5%

Machine learning
☐ Refine results with machine learning (Percolator)

Start Search ...
Reset Form

# identifying MS2 spectra with Mascot

LEAVE DEFAULT BY PAY  
ATTENTION

WE WILL MODIFY  
THIS HERE

Home Free trial Products Help Support Training Monthly newsletter About

Access Mascot Server | Evaluate Mascot Distiller

Access Mascot Server > MS/MS Ions Search

### MASCOT MS/MS Ions Search

Your name:  Email:

Search title:

Database(s):

Amino acid (AA):

Taxonomy:

Enzyme:  Allow up to:  missed cleavages

Quantitation:

Crosslinking:

Fixed modifications:

Variable modifications:

Display all modifications: ☐

Error tolerant: ☐ Automatic second pass search of selected modification classes

Peptide tol.  $\pm$   Da #  C  MS/MS tol.  $\pm$   Da

Data file:

Data format:

Instrument:

Target FDR:

Machine learning: ☐ Refine results with machine learning (Percolator)

# identifying MS2 spectra with Mascot



Mascot training materials are available for free: [Training materials](#). All Mascot Server documentation is [available online for free](#). Mascot Distiller ships with reference documentation and tutorials, available from the Distiller GUI, Help, Mascot Distiller Help.

**Search running...** 

**Time now** Sun Oct 19 20:18:58 2025

Your Mascot search is running. This page will refresh at the end of the search to show the results report.

If you close your browser window before the search ends, the search will finish in the background.

# identifying MS2 spectra with Mascot

## MASCOT Search Results

The Mascot software suite is a complete solution for [industrial and pharmaceutical research](#).

**User** : Marco  
**E-mail** : marco.trevisan@newcastle.ac.uk  
**Search title** : 3817  
**MS data file** : 3817\_manual.mgf  
**Databases** : 1: contaminants 20160129 (247 sequences; 128,130 residues)  
2: SwissProt 2025\_04 (573,661 sequences; 207,922,125 residues)  
**Timestamp** : 19 Oct 2025 at 19:18:59 GMT

[Re-search](#) ☒ All ☐ Non-significant ☐ Unassigned [\[help\]](#)

[Export](#) As XML ▼

► **Search parameters**

► **Score distribution**

► **Modification statistics for all protein families**

► **Legend**

### Protein Family Summary

|                                       |                                     |                                      |                                   |                        |
|---------------------------------------|-------------------------------------|--------------------------------------|-----------------------------------|------------------------|
| Significance threshold p<             | <input type="text" value="0.0100"/> | Max. number of families              | <input type="text" value="AUTO"/> | <a href="#">[help]</a> |
| Target FDR (overrides sig. threshold) | <span>5%</span> ▼                   | FDR type                             | <span>Sequence</span> ▼           |                        |
| Display non-sig. matches              | <input type="checkbox"/>            | Min. number of sig. unique sequences | <span>1</span> ▼                  |                        |
|                                       |                                     | Dendrograms cut at                   | <input type="text" value="0"/>    |                        |
| Preferred taxonomy                    | <span>All entries</span> ▼          |                                      |                                   |                        |

# identifying MS2 spectra with Mascot

Report peptide ratios ☒ Report protein ratios ☒

▼Sensitivity and FDR (reversed protein sequences)

|                        | Target | Decoy | FDR   |                                                                |
|------------------------|--------|-------|-------|----------------------------------------------------------------|
| Protein family members | 1      | 0     | 0.00% | Not enough decoy protein hits for a meaningful FDR calculation |
| Sequences              | 1      | 0     | 0.00% | Not enough decoy matches above significance threshold          |

Decoy results are available in [the decoy report](#).

Proteins (1) [Report Builder](#)

**Protein family 1 (out of 1)**

10 per page 1

Accession contains

►1 **2::KPYK1\_YEAST** 69 Pyruvate kinase 1 OS=Saccharomyces cerevisiae (strain ATCC 204508 / S288c) OX=559292 GN=CDC19 PE=1 SV=2

10 per page 1

If everything went fine, the protein that generated the spectrum has been generated!

It looks like it was a **pyruvate kinase** in **yeast**.

Congrats!

Let's check now what other information we have here...

# identifying MS2 spectra with Mascot

## **MASCOT Search Results**

The Mascot software suite is a complete solution for [industrial and pharmaceutical research](#).

User : Marco  
E-mail : marco.trevisan@newcastle.ac.uk  
Search title : 3817  
MS data file : 3817\_manual.mgf  
Databases : 1: contaminants 20160129 (247 sequences; 128,130 residues)  
2: SwissProt 2025\_04 (573,661 sequences; 207,922,125 residues)  
Timestamp : 19 Oct 2025 at 19:18:59 GMT

☒ All ☐ Non-significant ☐ Unassigned [\[help\]](#)

As

### ▼ Search parameters

Type of search : MS/MS Ion Search  
Target FDR : 5%  
Quantitation : [SILAC K+6 R+6 multiplex](#) (Zhang and Neubert, MCP 5 401-411 (2006))  
Enzyme : Trypsin  
Fixed modifications : [Carbamidomethyl \(C\)](#)  
Variable modifications : [Oxidation \(M\)](#), [13C6\\_NL\\_K \(C-term K\)](#), [13C6\\_NL\\_R \(C-term R\)](#)  
Mass values : Monoisotopic  
Protein mass : Unrestricted  
Peptide mass tolerance :  $\pm 0.6$  Da  
Fragment mass tolerance :  $\pm 2$  Da  
Max missed cleavages : 1  
Instrument type : Default  
Number of queries : 1

We can retrieve the search parameters we had introduced by unfolding the *search parameters* option.



# identifying MS2 spectra with Mascot



## MASCOT Search Results

The Mascot software suite is a complete solution for [industrial and pharmaceutical research](#).

User : Marco  
E-mail : marco.trevisan@newcastle.ac.uk  
Search title : 3817  
MS data file : 3817\_manual.mgf  
Databases : 1: contaminants 20160129 (247 sequences; 128,130 residues)  
2: SwissProt 2025\_04 (573,661 sequences; 207,922,125 residues)  
Timestamp : 19 Oct 2025 at 19:18:59 GMT

Re-search

☒ All ☐ Non-significant ☐ Unassigned

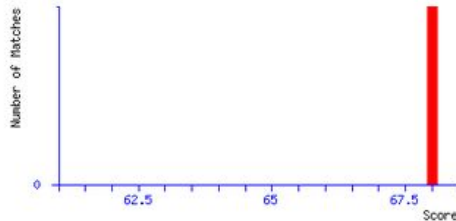
[\[help\]](#)

Export

As XML

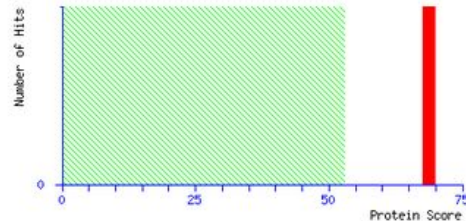
Search parameters

▼ Score distribution



**Peptide score distribution.** Ions score is  $-10\log(P)$ , where  $P$  is the probability that the observed match is a random event.

There is **1** peptide match above identity threshold and **1** match above homology threshold for **1** queries. On average, individual ions scores **> 53** (beyond green shading) indicate **identity or extensive homology** ( $p < 0.01001$ ).



**[Deprecated]** Protein score distribution. Score distribution for family members in the first 50 proteins. Protein scores are derived from ions scores as a non-probabilistic basis for ranking protein families.

We have just unfolded here the information of *score distribution* to understand how good or bad is the identification.



# identifying MS2 spectra with Mascot

Proteins (1) [Report Builder](#)

**Protein family 1 (out of 1)**

10 per page 1 [Expand all](#) [Collapse all](#)

Accession contains  [Find](#) [Clear](#)

▼1 **2::KPYK1\_YEAST** 69 Pyruvate kinase 1 OS=Saccharomyces cerevisiae (strain ATCC 204508 / S288c) OX=559292 GN=CDC19 PE=1 SV=2

|     |                                | Score | Mass  | Matches | Sequences | Heavy/<br>Light |                                                                               |
|-----|--------------------------------|-------|-------|---------|-----------|-----------------|-------------------------------------------------------------------------------|
| 1.1 | <a href="#">2::KPYK1_YEAST</a> | 69    | 54909 | 1 (1)   | 1 (1)     | ---             | Pyruvate kinase 1 OS=Saccharomyces cerevisiae (strain ATCC 204508 / S288c) O: |

▼1 peptide matches (1 non-duplicate, 0 duplicate)

☒ Auto-fit to window

| Query             | Dupes | Observed | Mr (expt) | Mr (calc) | Delta M | Score | Expect | Rank   | U  | Heavy/<br>Light | Peptide |                                      |
|-------------------|-------|----------|-----------|-----------|---------|-------|--------|--------|----|-----------------|---------|--------------------------------------|
| <a href="#">1</a> |       | 504.8636 | 1007.7127 | 1007.5707 | 0.1420  | 0     | 69     | 0.0003 | ►1 | U               | ---     | R.TANDVLTIR.E + 13C6_NL_R (C-term R) |

Here we have unfolded the details about the identification to see which peptide had been "seen" by the search algorithm.

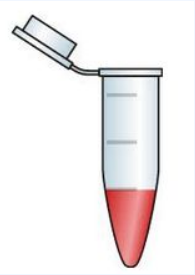
# #7) What happens if a peptide is not in the database?

- A) We get a new sequence that is not in the database, so we can improve the database we have
- B) The system warns us that the peptide is not in the database
- C) We get the peptide from the database that matches the experimental spectrum with the best score
- D) None of the above

<https://tinyurl.com/omics2025-q7>

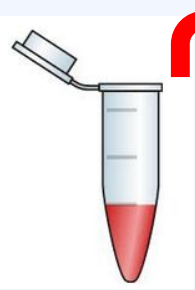
# Part III

## Summary



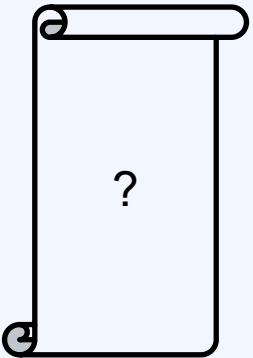
**what are the  
cells doing?**

# Overview of protein identification

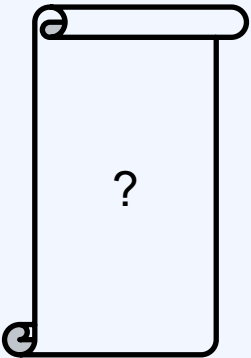
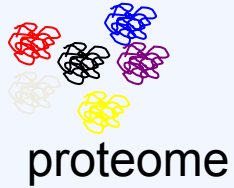
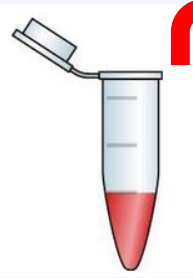


**what are the  
cells doing?**

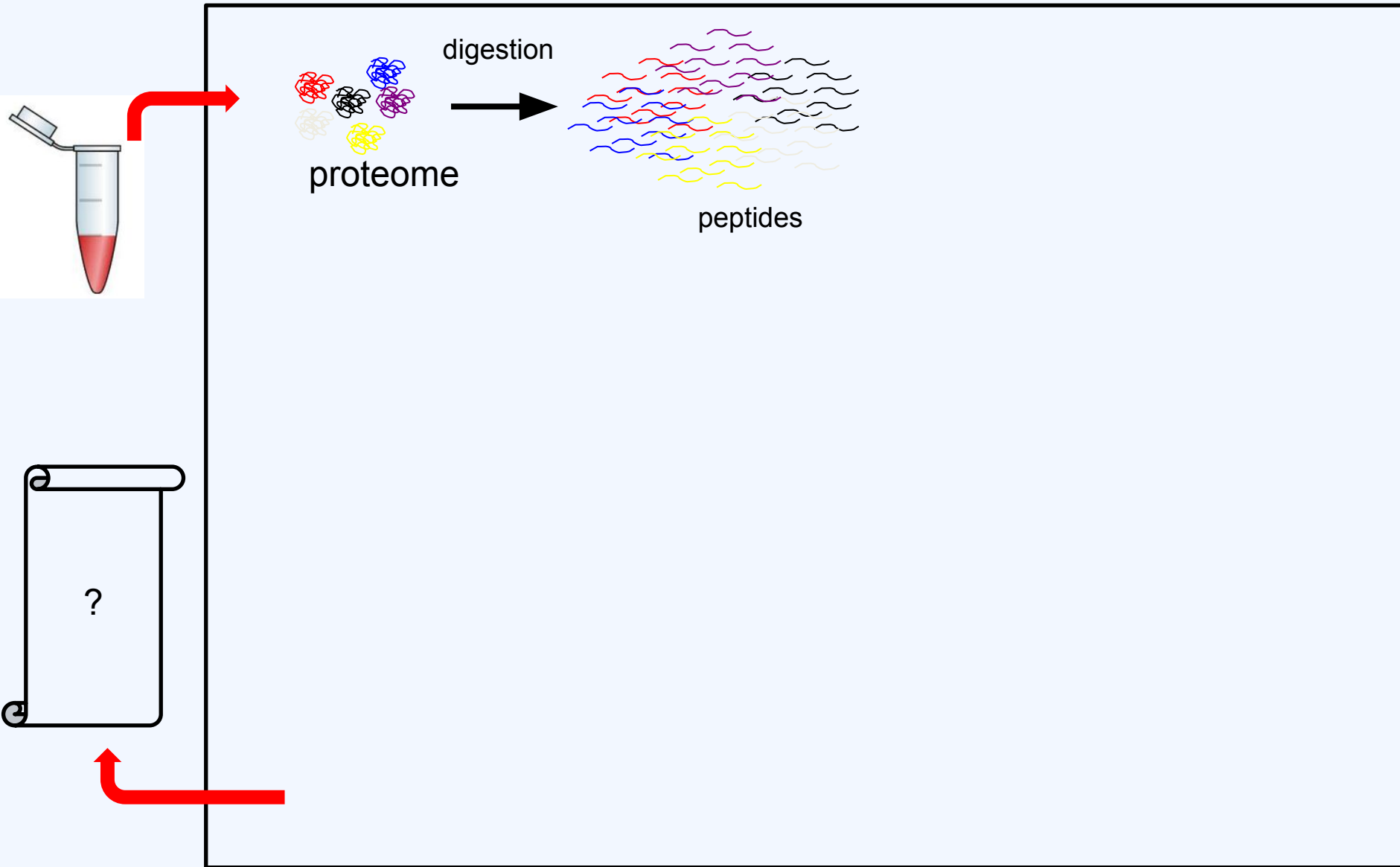
which proteins are  
in that sample?



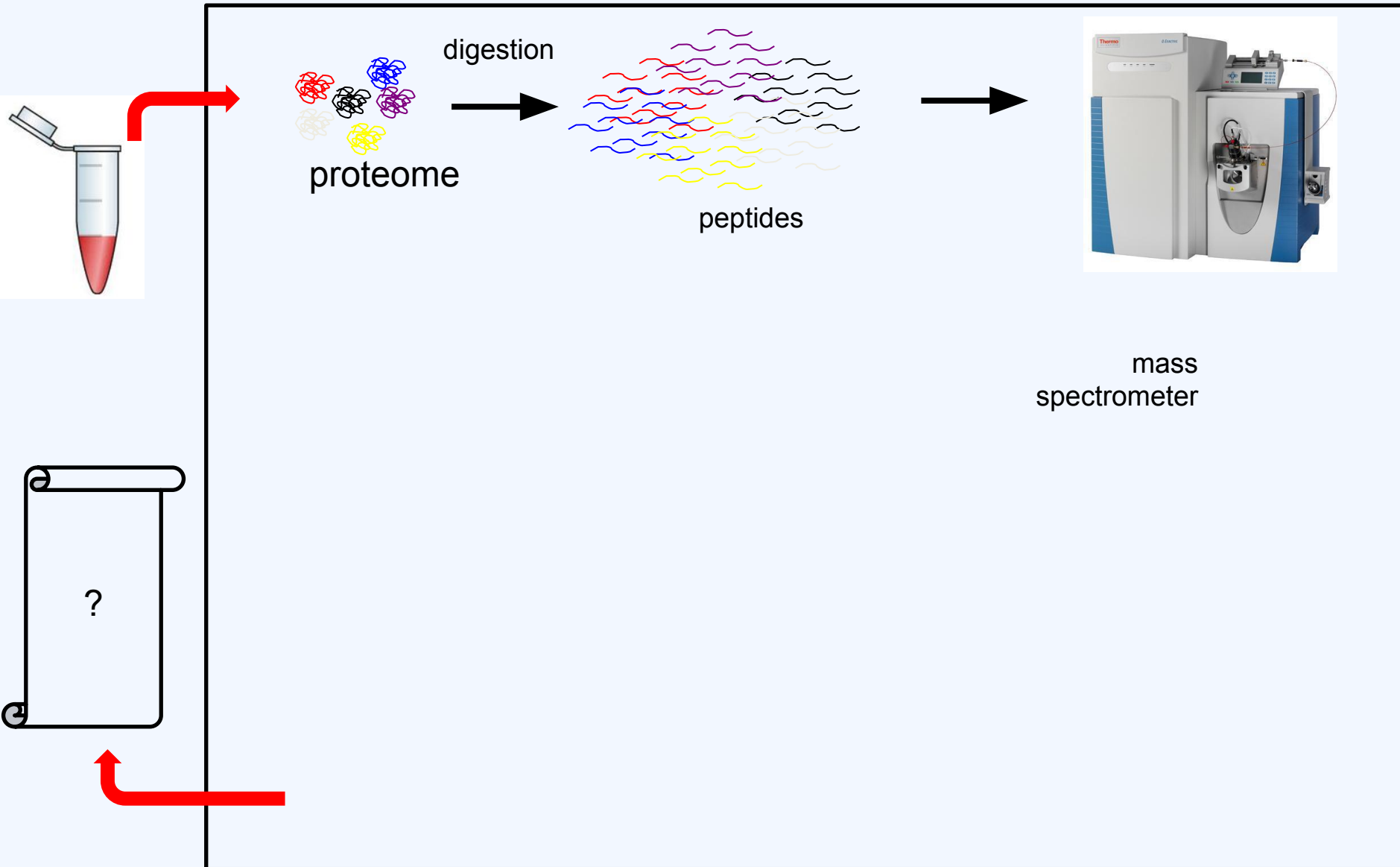
# Overview of protein identification



# Overview of protein identification

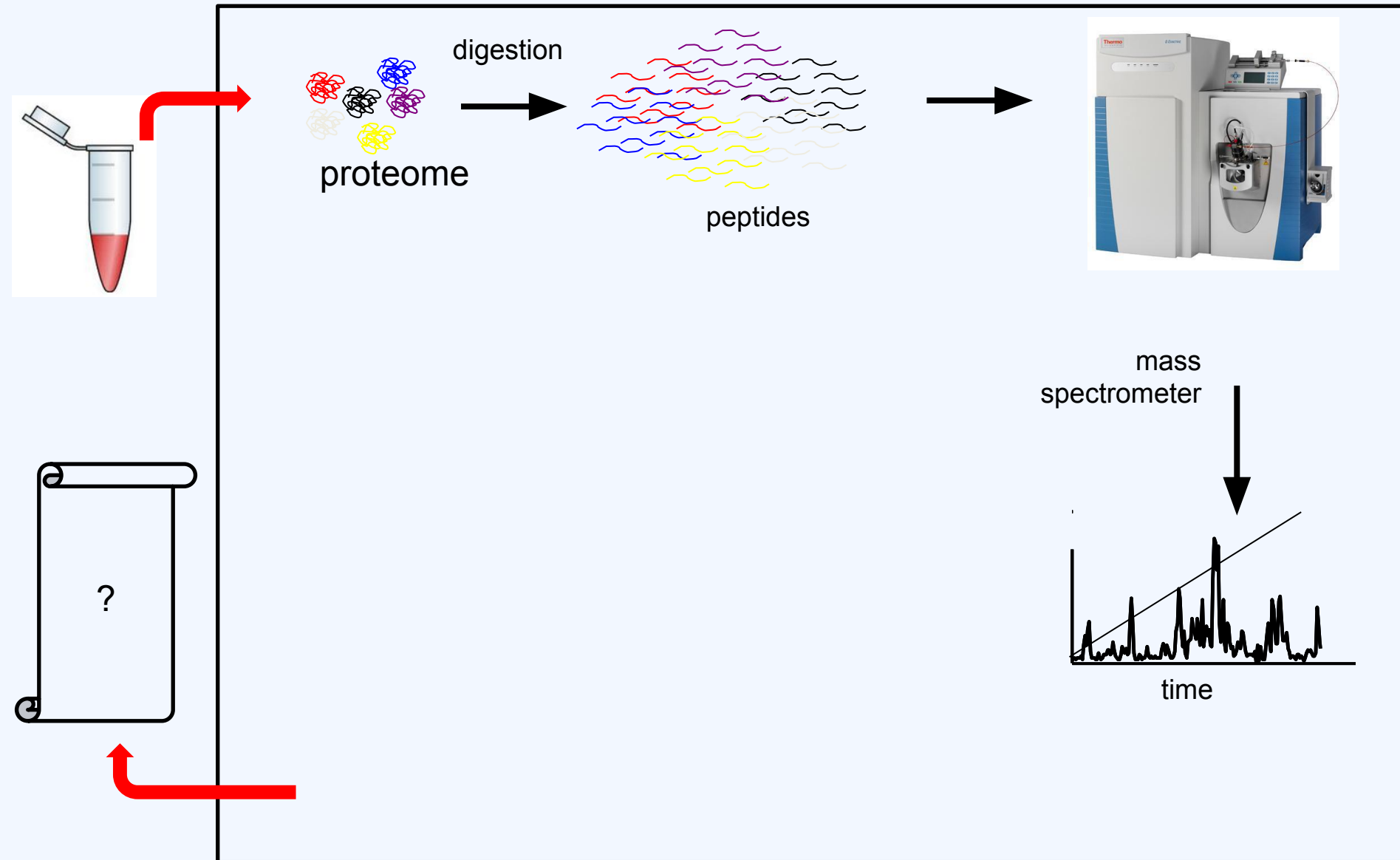


# Overview of protein identification

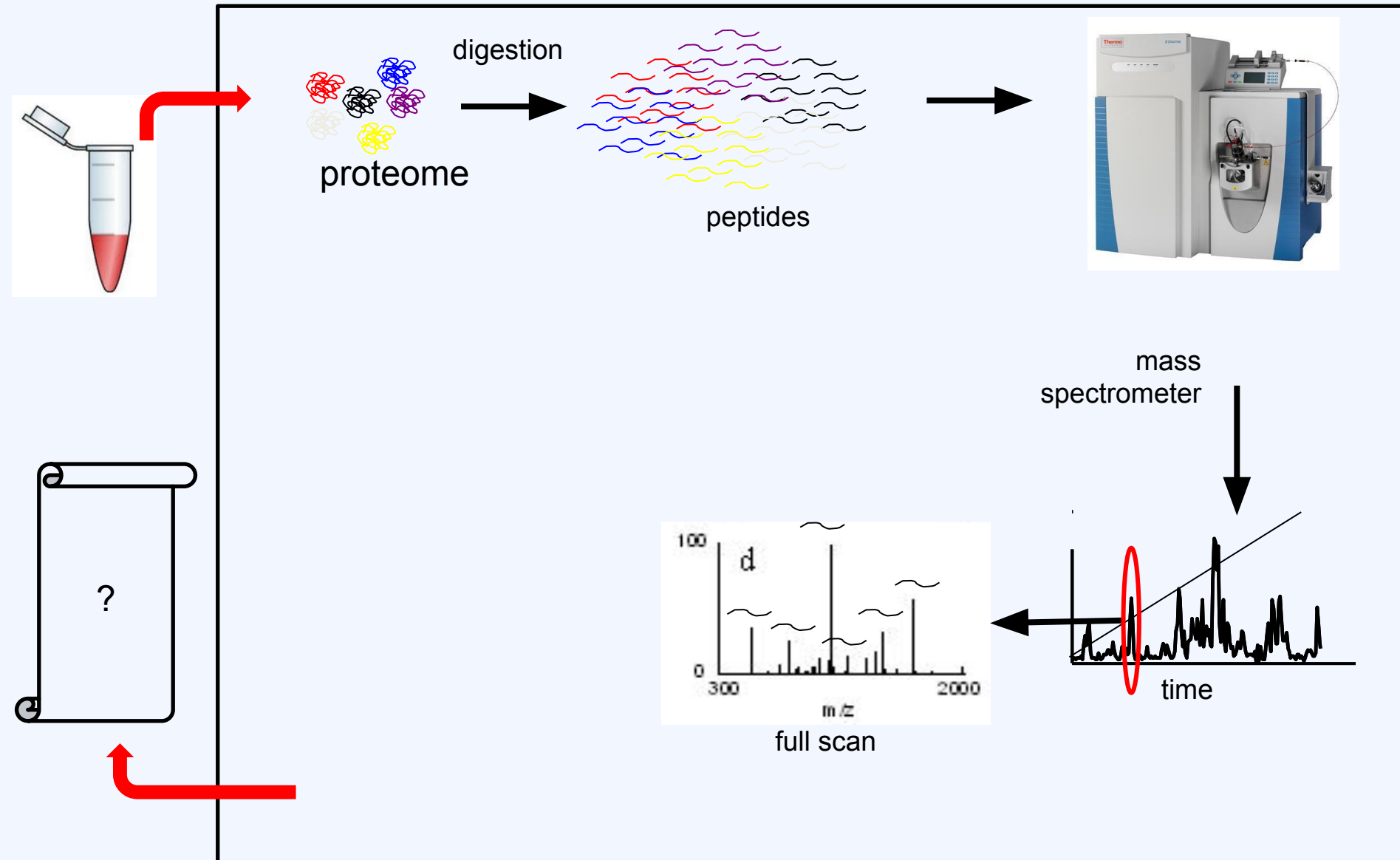




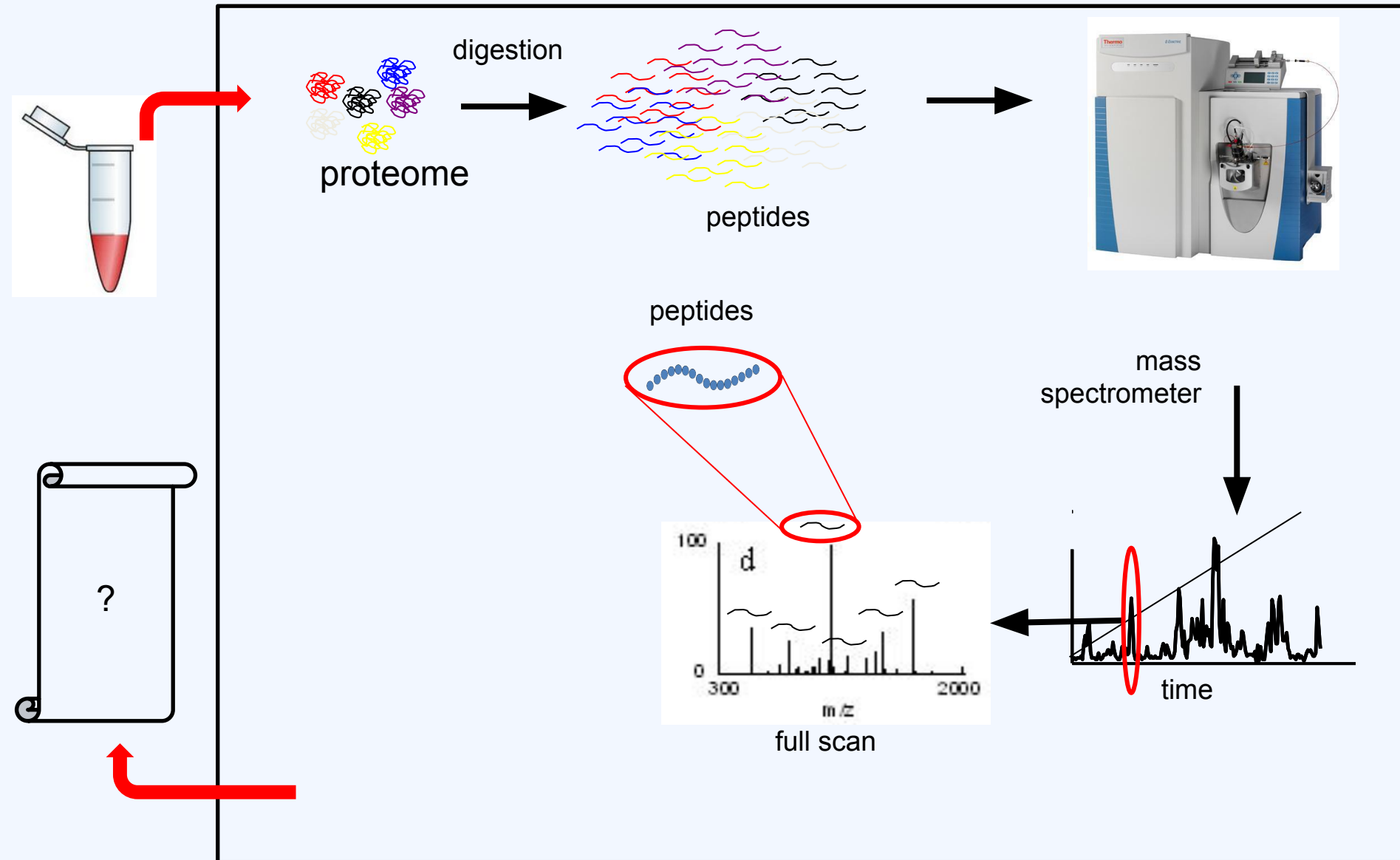
# Overview of protein identification



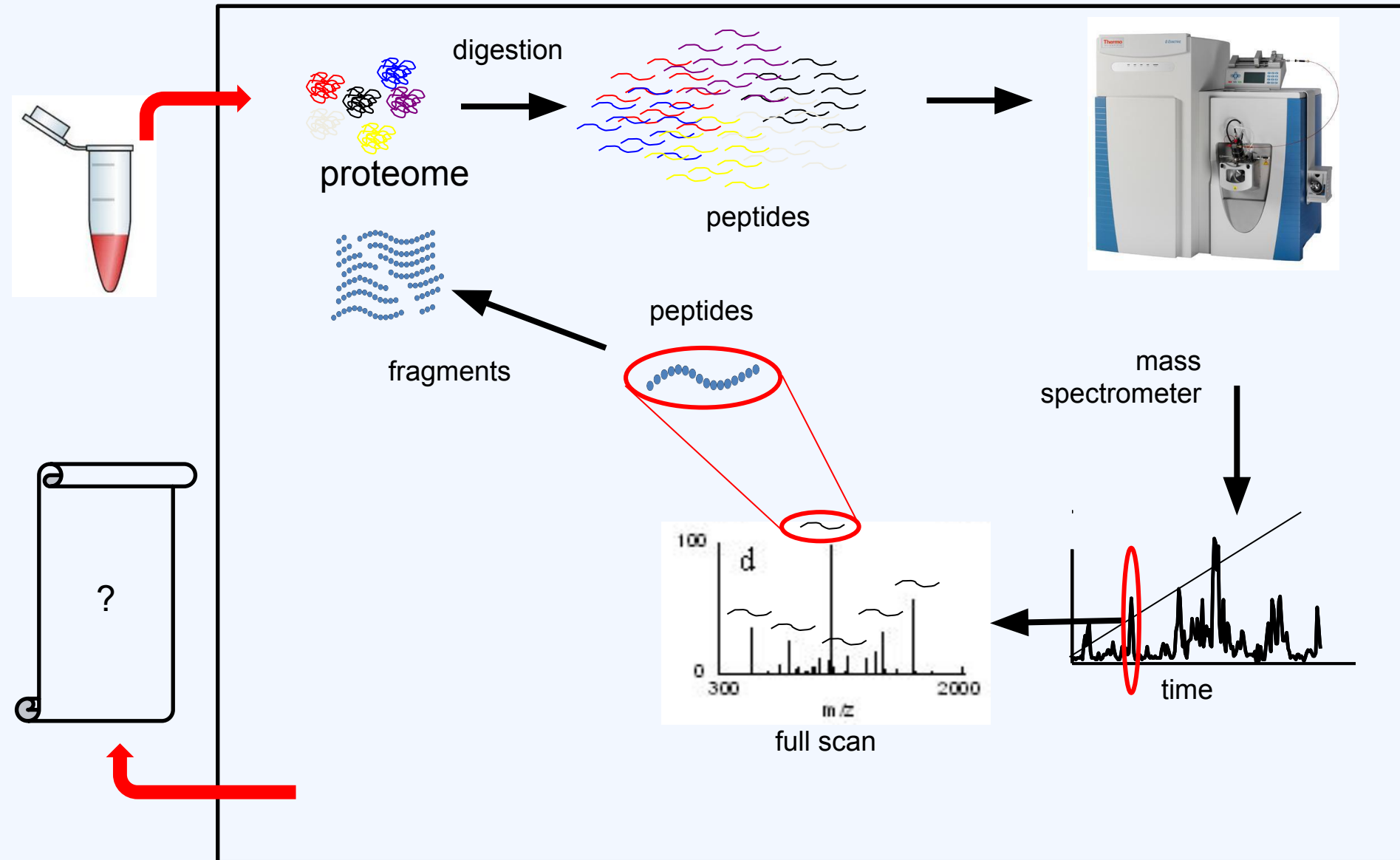
# Overview of protein identification



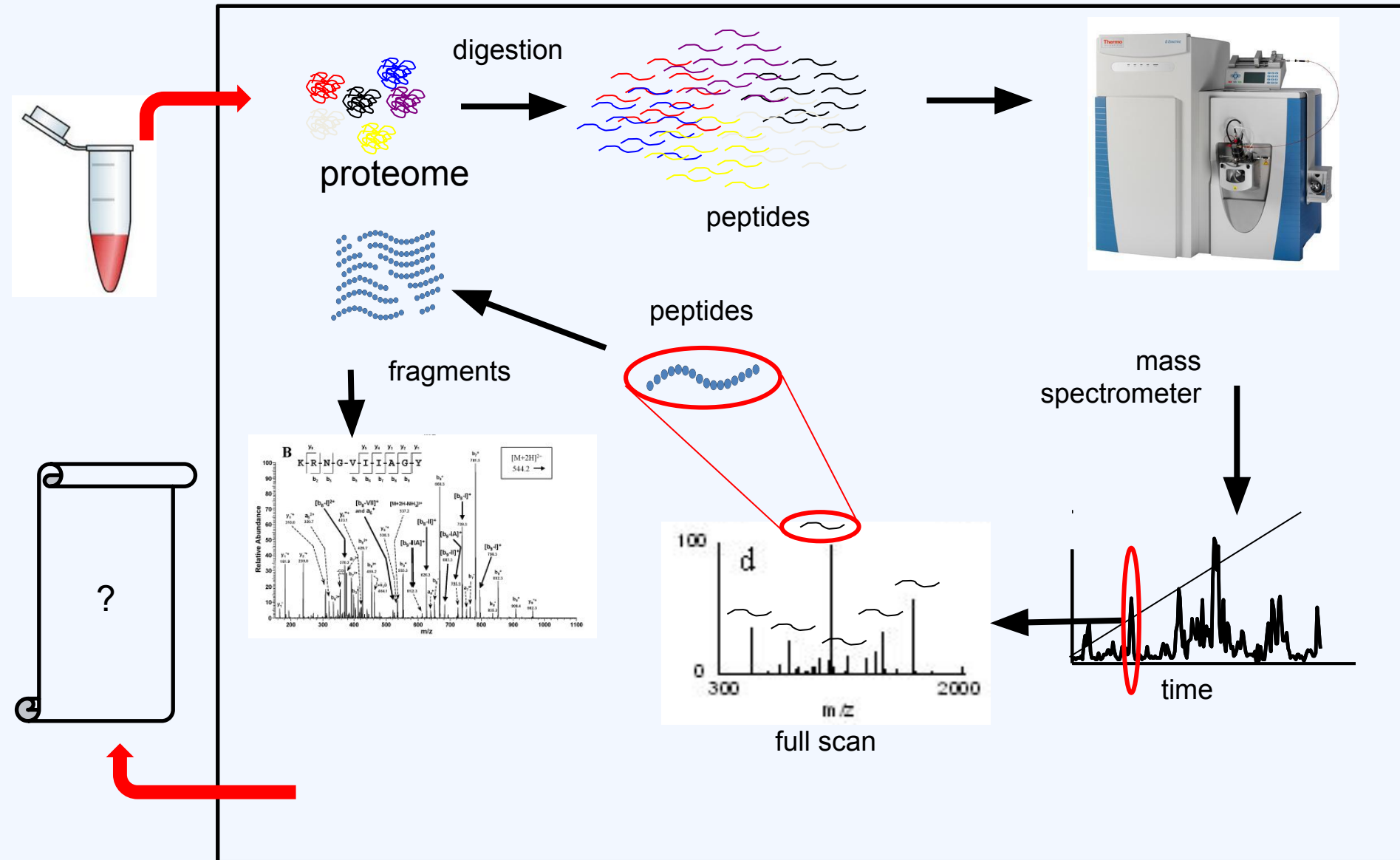
# Overview of protein identification



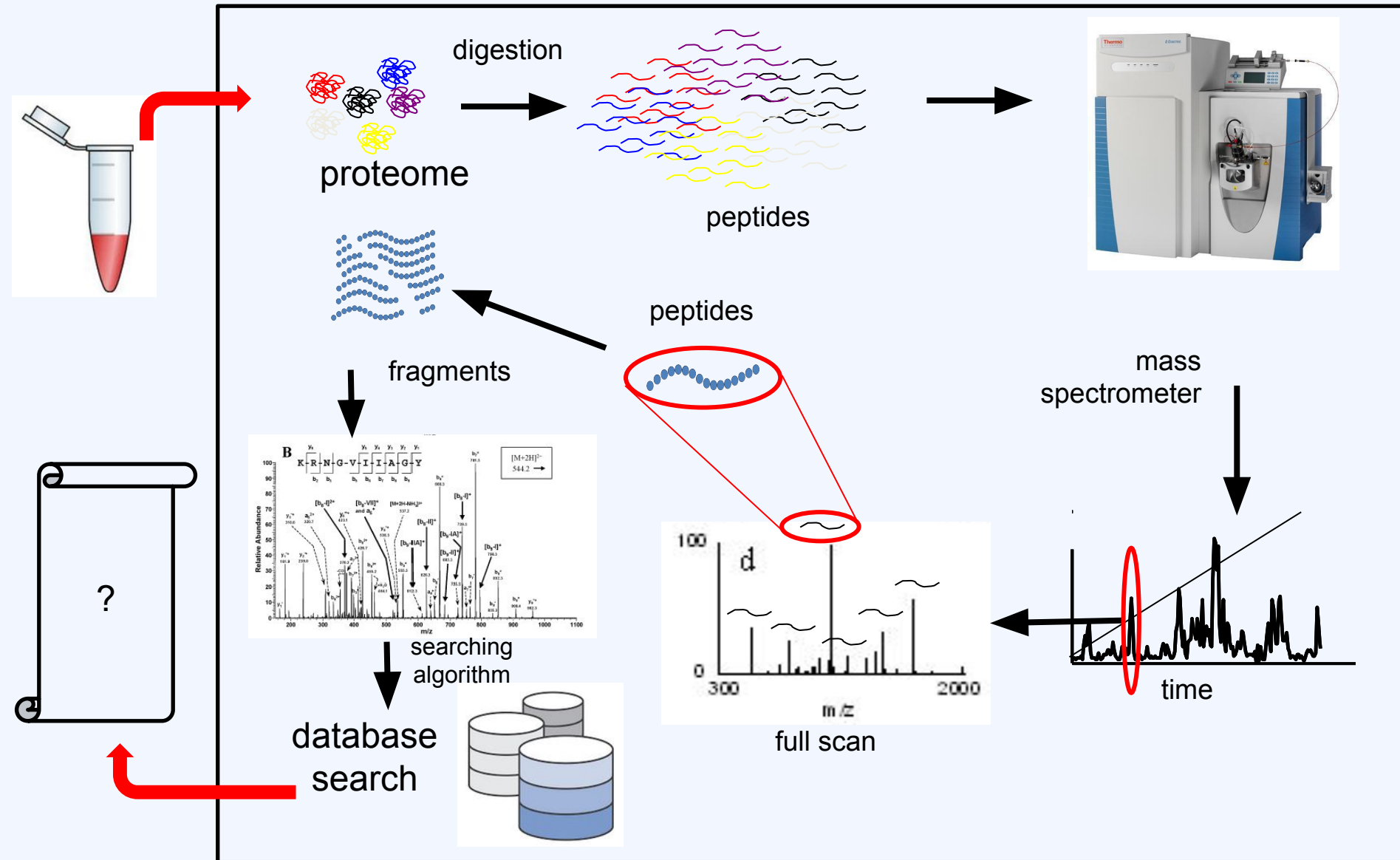
# Overview of protein identification



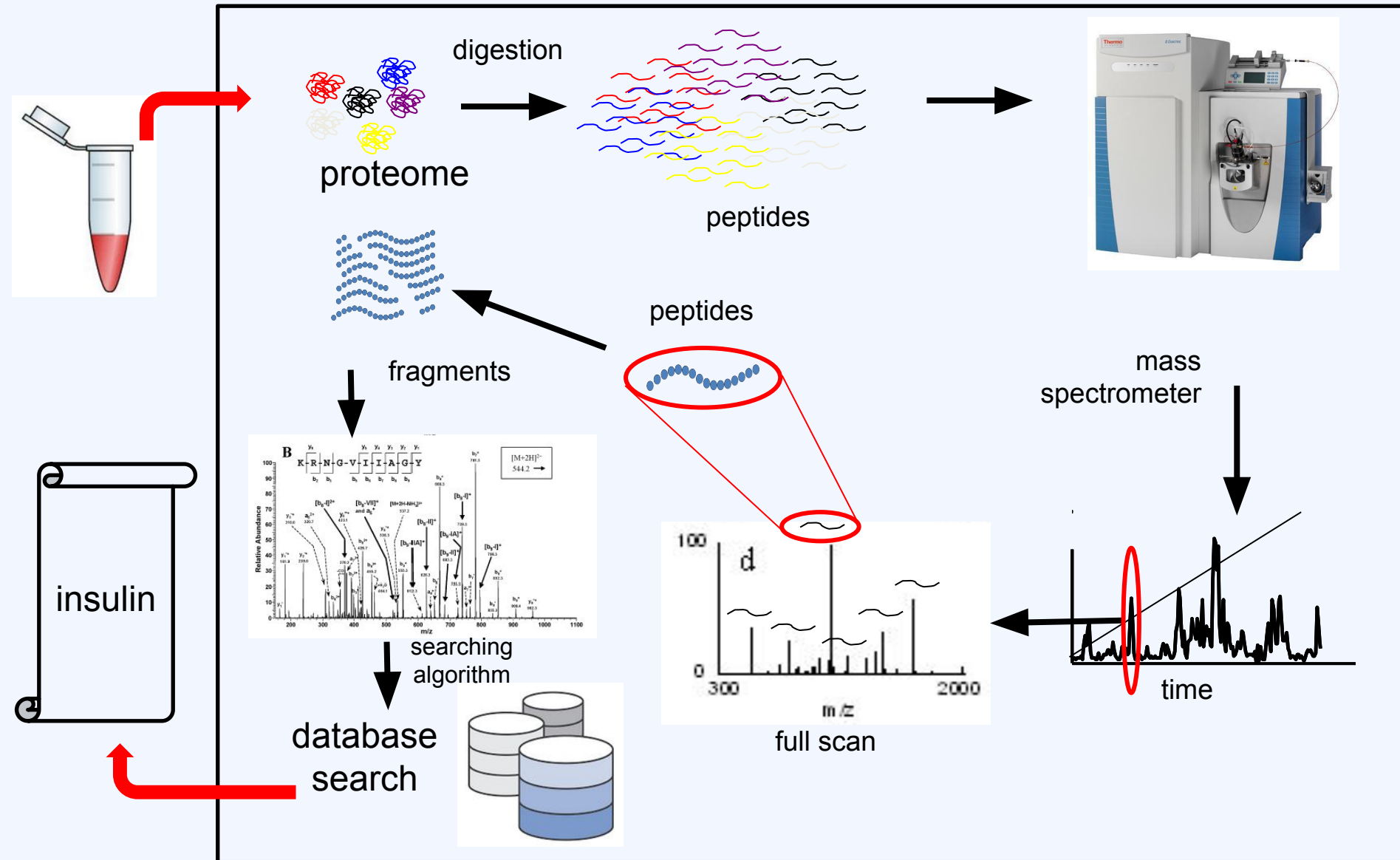
# Overview of protein identification



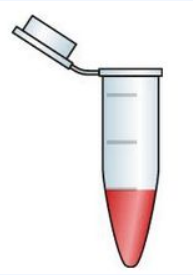
# Overview of protein identification



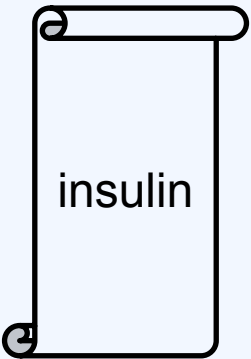
# Overview of protein identification



# what are the cells doing?



**“some cells are busy taking glucose  
from the blood, and storing it as  
glycogen via glycogenesis”**





## End of Part III

If there is time, in Part IV we will learn about how to compare the protein abundance in two samples

# Part IV

Quantitative proteomics

# Overview of methods to quantify proteins

## Data Dependent Acquisition (DDA)

- \* discovery-driven techniques
  - \* the mass spectrometre makes an MS1 (full or survey scan)
  - \* the most abundant precursor ions are isolated
  - \* isolated ions are fragmented, generating a number of MS2 spectra (fragmentation)
- \* target-driven techniques
  - \* data recorded in LC-time dimension at predefined m/z for precursor and fragments (pairs called *transitions*)
  - \* fragment intensities are monitored across chromatographic time
  - \* it is a target-driven technique

# Overview of methods to quantify proteins

## Data Independent Acquisition (DIA)

- \* the spectrometre fragments all the precursors within a defined (usually large)  $m/z$
- \* it does not explicitly target single precursors, but whole bands of the precursor ion

# Overview of methods to quantify proteins

## Data Dependent Acquisition (DDA)

- \* discovery-driven techniques
  - \* stable isotope labelling (SIL) techniques
    - \* SILAC
    - \*  $^{18}\text{O}$
    - \* iTRAQ
    - \* TMT (tandem mass tag)
  - \* label free / spectral counting
- \* target-driven techniques
  - \* SRM (selected reaction monitoring)
  - \* MRM (multiple reaction monitoring)

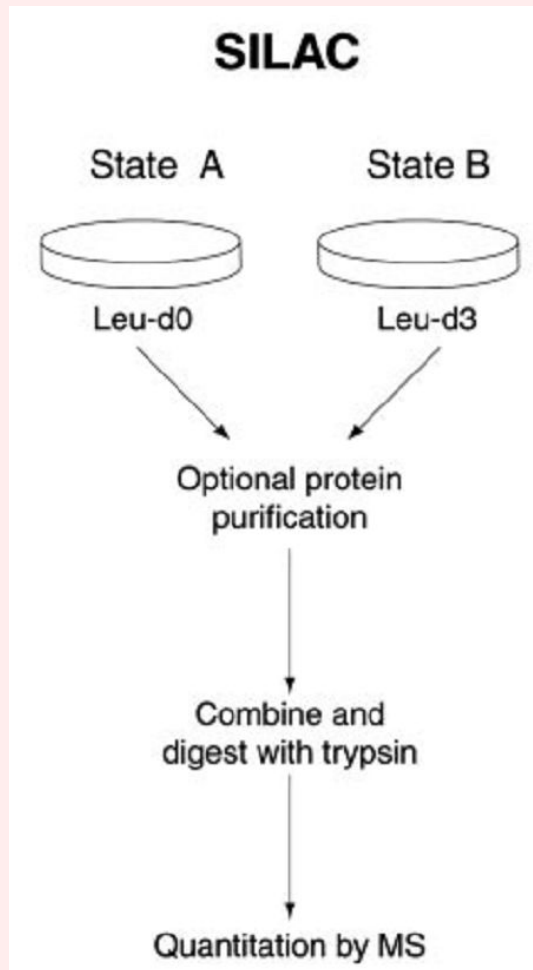
## Data Independent Acquisition (DIA)

- \* Parallel reaction monitoring (PRM)
- \* Precursor acquisition independent from ion count (PAcIFIC)
- \* Data-independent scanning (DiS)
- \* SWATH-MS
- \* MSX (multiplexed data-independent acquisition)

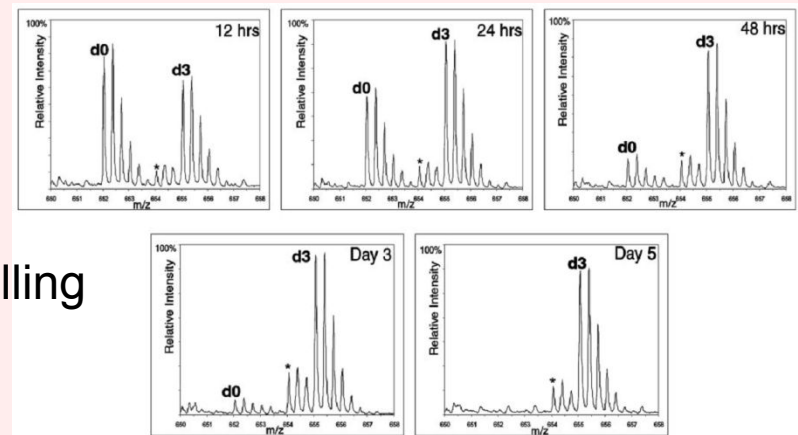
# Some common methods for Stable Isotope Labelling (SIL)

- SILAC
- $^{18}\text{O}$
- isobaric methods
  - iTRAQ
  - TMT

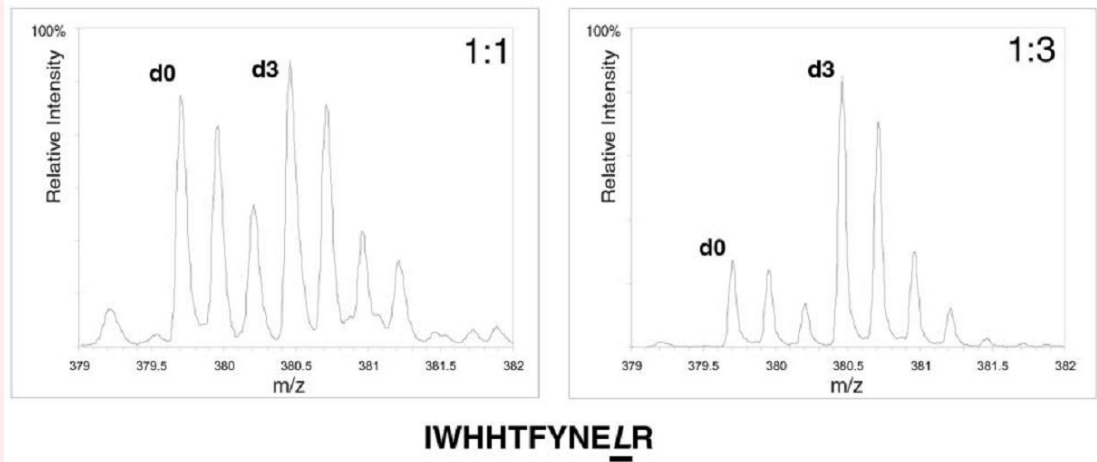
# Stable Isotope Labelling of Amino acids in Culture (SILAC)



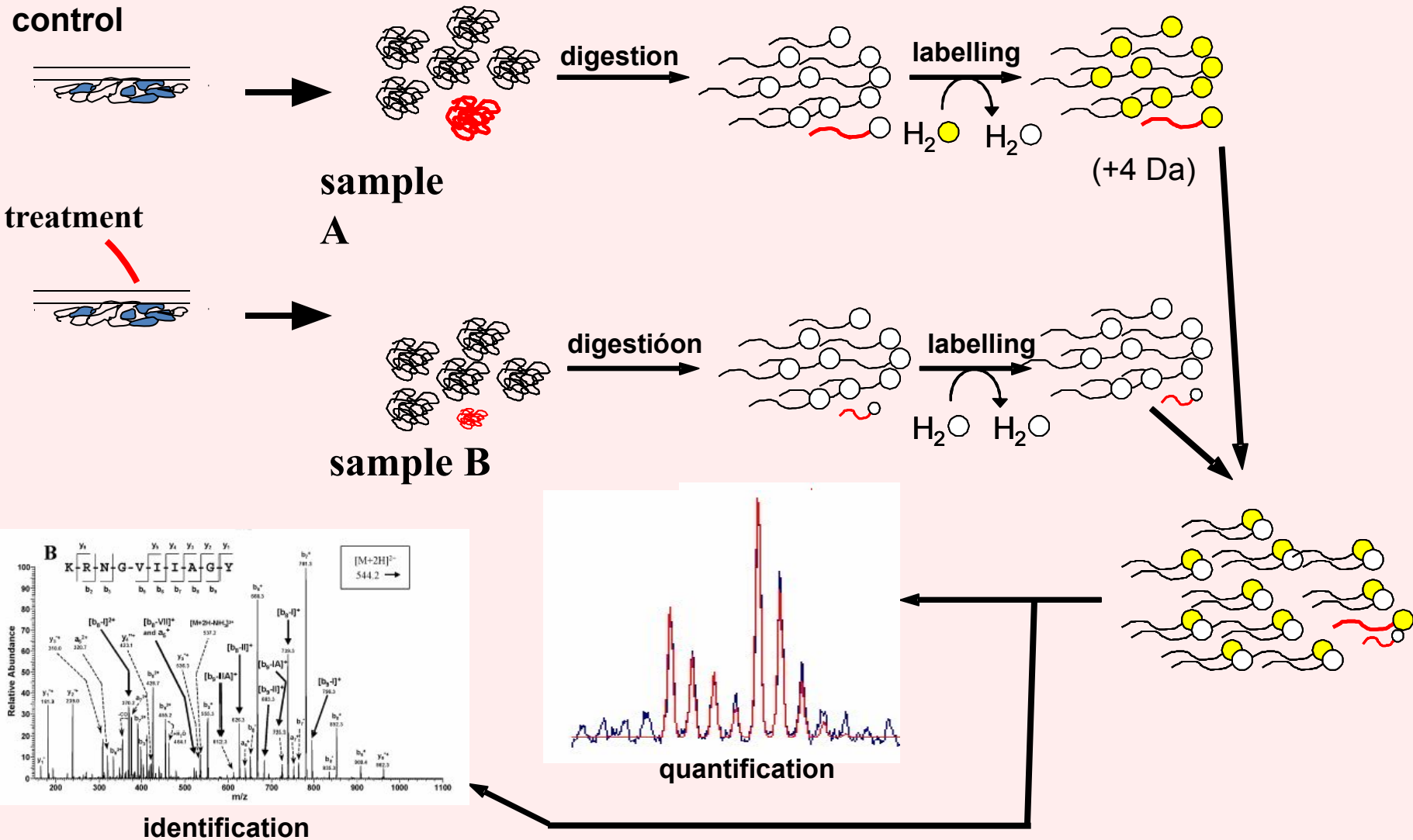
Labelling



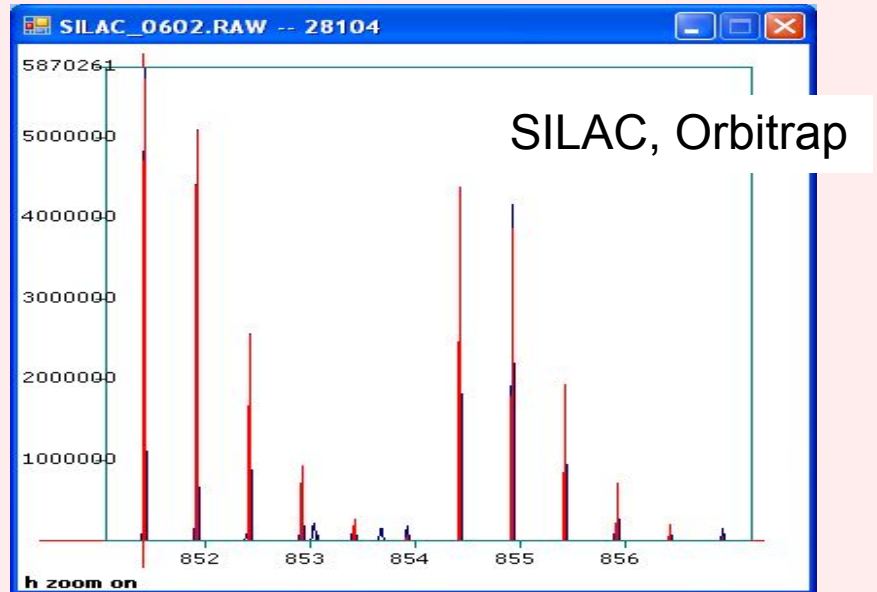
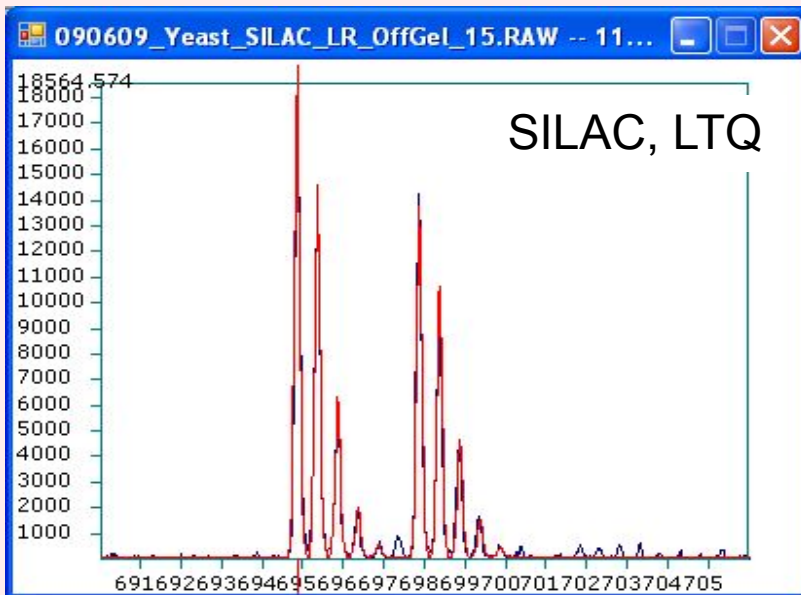
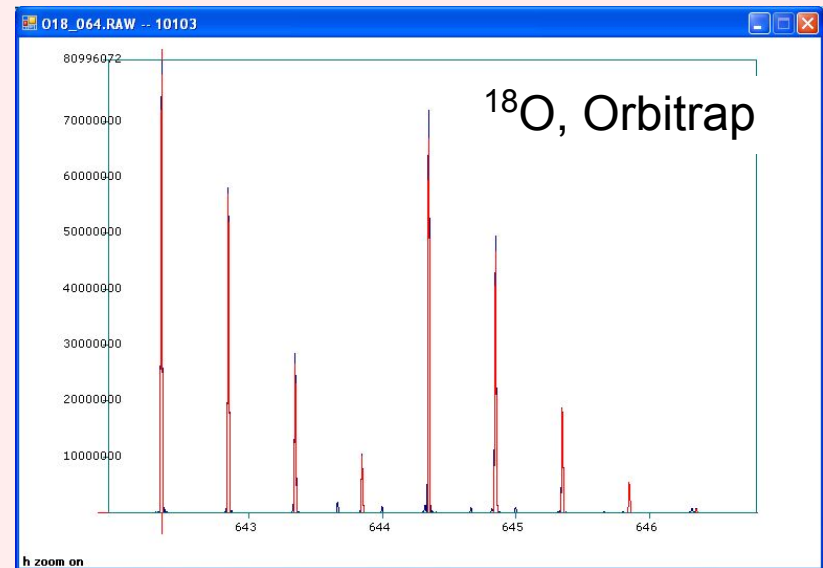
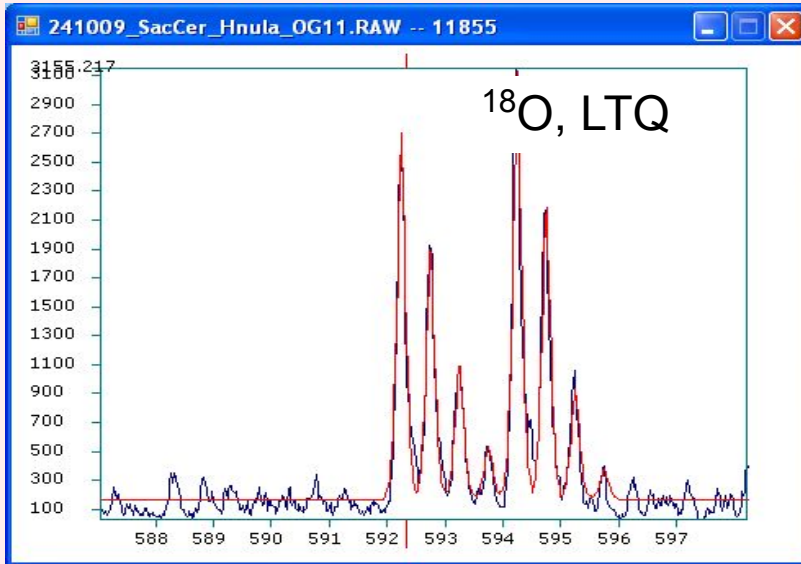
Quantification



# $^{16}\text{O}/^{18}\text{O}$ labelling



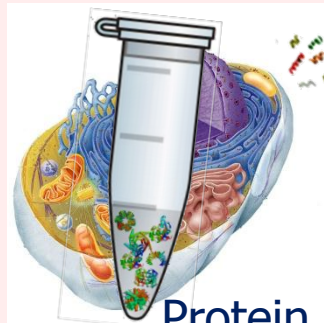




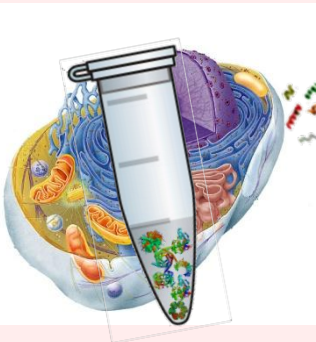
comparison of quantifications in high- and low-resolutions

# Workflow for the analysis of samples using a post-digestion labelling method

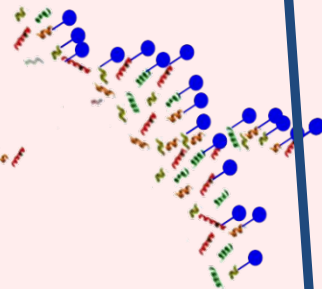
Samples    Extraction of proteins    Digestion    Labelling    Mixture    MS Analysis



Protein  
variability

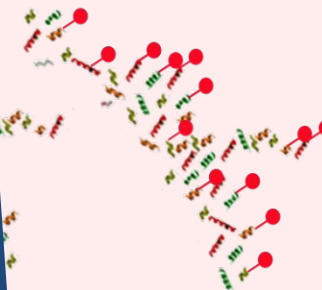


Digestion



Peptide  
variability

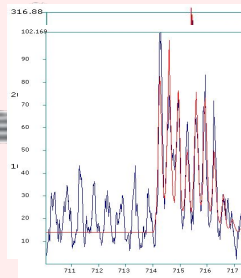
Labelling



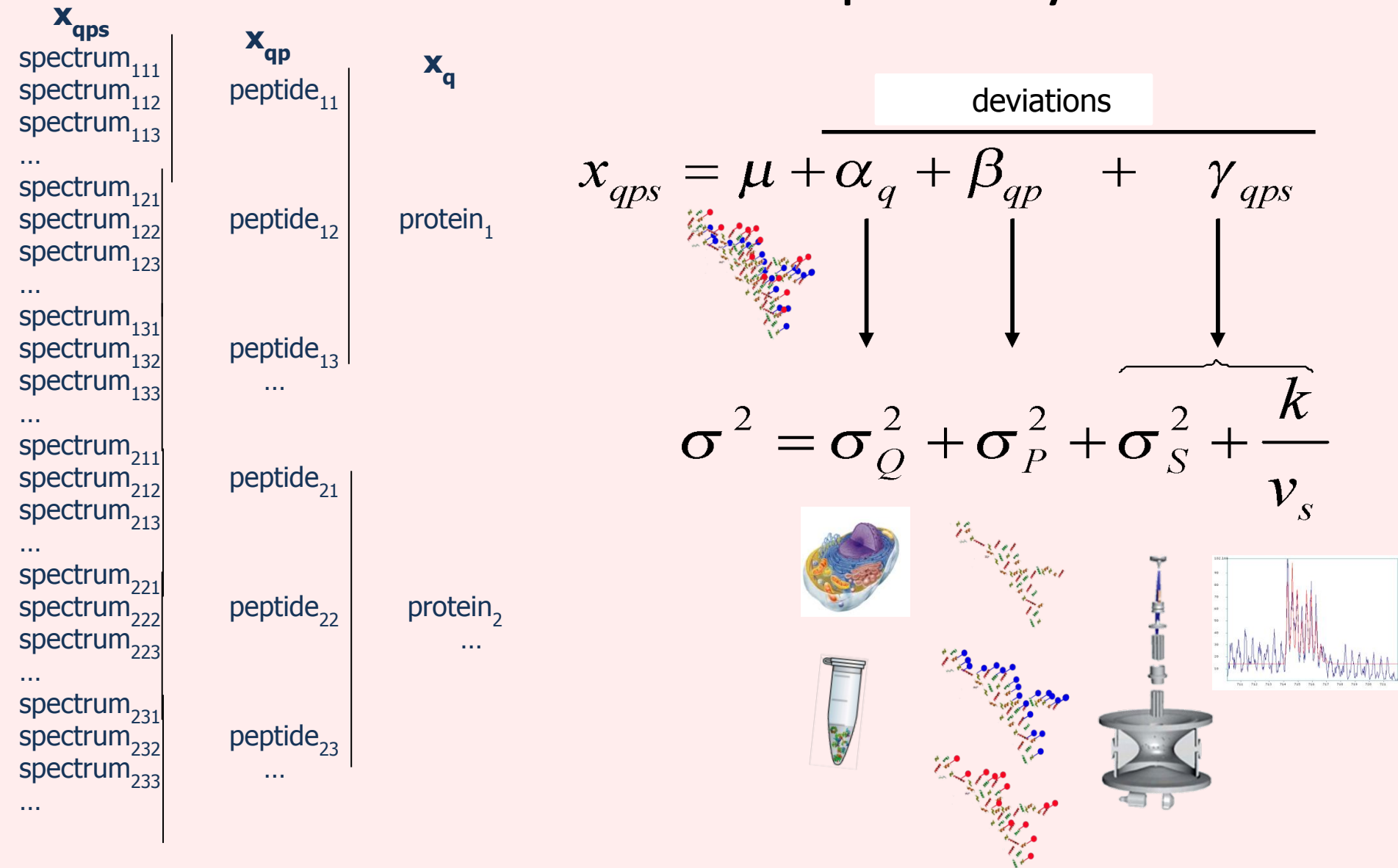
Mixture

MS Analysis

Spectrum  
variability



# Statistical model: the sources of error are considered separately



# Statistical model: integration of quantitative information

spectrum<sub>111</sub>  
 spectrum<sub>112</sub>  
 spectrum<sub>113</sub>  
 ...  
 spectrum<sub>121</sub>  
 spectrum<sub>122</sub>  
 spectrum<sub>123</sub>  
 ...  
 spectrum<sub>131</sub>  
 spectrum<sub>132</sub>  
 spectrum<sub>133</sub>  
 ...  
 spectrum<sub>211</sub>  
 spectrum<sub>212</sub>  
 spectrum<sub>213</sub>  
 ...  
 spectrum<sub>221</sub>  
 spectrum<sub>222</sub>  
 spectrum<sub>223</sub>  
 ...  
 spectrum<sub>231</sub>  
 spectrum<sub>232</sub>  
 spectrum<sub>233</sub>  
 ...

peptide<sub>11</sub>  
 ...  
 peptide<sub>13</sub>  
 ...

peptide<sub>21</sub>  
 ...  
 peptide<sub>22</sub>  
 ...  
 peptide<sub>23</sub>  
 ...

protein<sub>1</sub>  
 ...  
 protein<sub>2</sub>  
 ...

$$x_{qp} = \frac{\sum_s w_{qps} x_{qps}}{\sum_s w_{qps}}$$

the peptide is a weighted average of the spectra used to identify it

$$x_q = \frac{\sum_p w_{qp} x_{pq}}{\sum_p w_{qp}}$$

the protein is the weighted average of the peptides it includes

The weights are the inverse of the variances detected

$$w_{qps} = \frac{1}{\frac{k}{v_{qps}} + \sigma_s^2} \quad w_{qp} = \frac{1}{\frac{1}{\sum_s w_{qps}} + \sigma_p^2} \quad w_q = \frac{1}{\frac{1}{\sum_p w_{qp}} + \sigma_q^2}$$

# Some examples of software in proteomics

Software for identification of proteins:

- SEQUEST (Proteome Discoverer)

- Comet (an open source SEQUEST)

- Andromeda

- Mascot

- X!Tandem

Software for quantitative proteomics by SIL

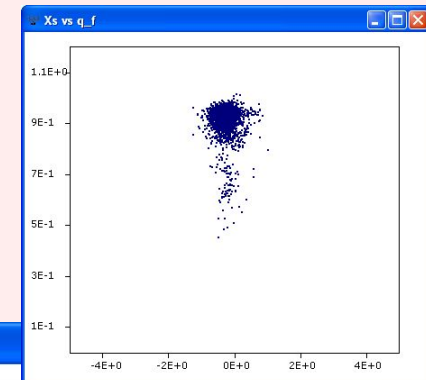
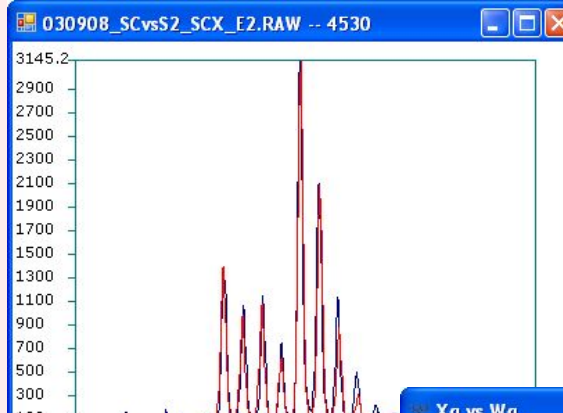
- SanXoT (and the previous, QuiXoT)

- MaxQuant

- Mascot Distiller

- Proteome Discoverer

- UNiquant



**QuiXoT**

Identifications XML file:

[FDRq], q\_index, p\_index

st\_excluded=" and vs>10

☒ show only data used in stats ☐ hide manually discarded

**Load status**

- ☒ Isotopes file
- ☒ Aminoacids file
- ☒ Parameters file
- ☒ Identifications XML file
- ☒ stack index

| FASTAProteinDescription                                               | Sequence                  | st_Meth | st_Cterm |
|-----------------------------------------------------------------------|---------------------------|---------|----------|
| P80723IBASP_HUMAN Brain acid soluble protein 1 - Homo sapiens (Human) | K.KTEAPAAPAAQETK.S        | 0       | 0        |
| P80723IBASP_HUMAN Brain acid soluble protein 1 - Homo sapiens (Human) | K.KTEAPAAPAAQETK.S        | 0       | 0        |
| P80723IBASP_HUMAN Brain acid soluble protein 1 - Homo sapiens (Human) | K.KTEAPAAPAAQETK.S        | 0       | 0        |
| P80723IBASP_HUMAN Brain acid soluble protein 1 - Homo sapiens (Human) | K.SDGAPASD SKPGSSEAPSSK.E | 0       | 0        |
| P49327FAS_HUMAN Fatty acid synthase - Homo sapiens (Human)            | K.ADEASELACPTPK#E         | 0       | 0        |
| P49327FAS_HUMAN Fatty acid synthase - Homo sapiens (Human)            | K.EQGVTFPSGDIQEQLIR@S     | 0       | 0        |
| P49327FAS_HUMAN Fatty acid synthase - Homo sapiens (Human)            | K.EQGVTFPSGDIQEQLIR@S     | 0       | 0        |
| P49327FAS_HUMAN Fatty acid synthase - Homo sapiens (Human)            | K.LPEDPLL SGLLDSPALK#A    | 0       | 0        |
| P49327FAS_HUMAN Fatty acid synthase - Homo sapiens (Human)            | K.VVQVLAEPEAVLK#G         | 0       | 0        |
| P49327FAS_HUMAN Fatty acid synthase - Homo sapiens (Human)            | R.FPQLDSTSFANSR.D         | 0       | 0        |
| P49006IMRP_HUMAN MARCKS-related protein - Homo sapiens (Human)        | K.AAATPESQEPQAK#G         | 0       | 0        |
| P49006IMRP_HUMAN MARCKS-related protein - Homo sapiens (Human)        | K.AAATPESQEPQAK#G         | 0       | 0        |
| P49006IMRP_HUMAN MARCKS-related protein - Homo sapiens (Human)        | K.AAATPESQEPQAK.G         | 0       | 0        |
| P49006IMRP_HUMAN MARCKS-related protein - Homo sapiens (Human)        | K.AAATPESQEPQAK.G         | 0       | 0        |
| P49006IMRP_HUMAN MARCKS-related protein - Homo sapiens (Human)        | K.AAATPESQEPQAK.G         | 0       | 0        |
| P49006IMRP_HUMAN MARCKS-related protein - Homo sapiens (Human)        | K.AAATPESQEPQAK.G         | 0       | 0        |
| P49006IMRP_HUMAN MARCKS-related protein - Homo sapiens (Human)        | R.GDVTAEAAAGASPAK#A       | 0       | 0        |
| P49006IMRP_HUMAN MARCKS-related protein - Homo sapiens (Human)        | R.GDVTAEAAAGASPAK#A       | 0       | 0        |

**Stats**

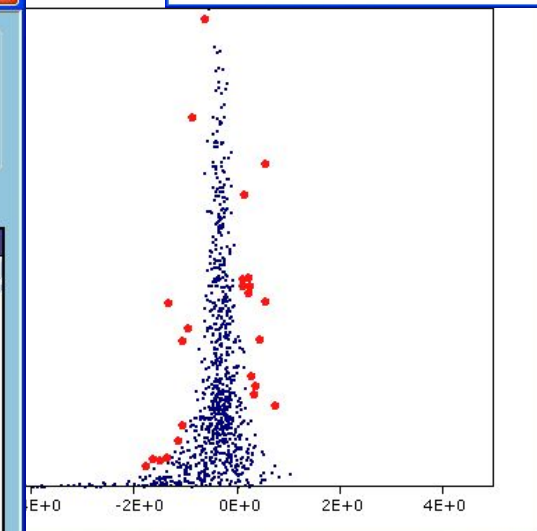
Xs : -1.28 Xp : -1.2006 Xq : -0.9268 X : -0.3416  
 Ws : 29.8651 Wp : 31.3808 Wq : 120.4592 Ns : 5911  
 Psp : -0.1776 Ppq : -0.9026 Pq : -9.8737 Np : 2334  
 FDRs : 1.7793 FDRp : 1.2812 FDRq : 0 Nq : 975

**Pep. Id.**

Xcorr : 3.8429  
 DCn : 0.4294  
 FDR : 0  
 Vs : 11.7371  
 f : 0.8577

**statistics constants** [change values](#)

K : .17  $\sigma^2s$  : .019  $\sigma^2p$  : .018  $\sigma^2q$  : .005



More information at:

<https://www.cnice.es/wiki/proteomica/index.php/QSP>



# The SanXoT workflow

statistical tools

systems biology tools

parsing tool

## SanXoT

Generic algorithm to perform the integration of experimental data to a higher level (such as integration from peptide data to protein data), while determining the variance between them.

Lower level  
 $\{x_p, v_p\}$



Higher level  
 $\{x_h, v_h\}$

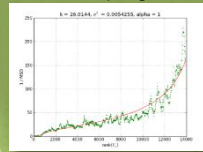
## Aljamia

Parser to convert and filter xml files or tab separated tables into a reduced tab separated values archive to be used either as relations or data files.

This makes possible using output of any identification software that produces text files as input for the SanXoT workflow.

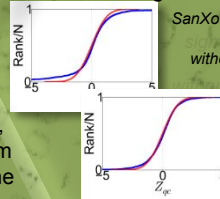
## Klibrate

Calibrates experimental data, as a first step to integrate them into higher levels along with the SanXoT program.



## SanXoTSieve

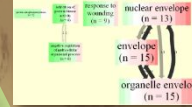
Program used to remove automatically lower level outliers in an integration.



SanXoT sigmoids, with and without outliers.

## Sanson

It creates the similarity graph: a graph that shows relationships between a set of categories according to the proteins they share, finding emerging clusters of related categories having proteins changing their expression.

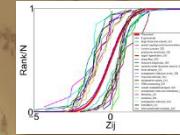


## SanXoTSqueezer

This program filters out relevant changing categories, allowing posterior representation using SanXoTGauss, Arbor or Sanson.

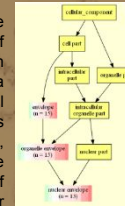
## SanXoTGauss

It depicts the sigmoids of proteins within their categories: each category is a colour sigmoid, and its dots are proteins.



## Arbor

Generates the tree graph of a set of categories (when they have a hierarchical structure, such as Gene Ontology), displaying the changes of their proteins



global variances are estimated by a robust, iterative method taking into account unbalanced structure

$$z_{gp} = \frac{x_{gp} - x_g}{\sqrt{\frac{1}{\sum w_{gp}} + \sigma_p^2}} \cdot \frac{n_{gp}}{n_{gp} - 1}, \quad n_{gp} > 1$$

$$z_i \sim N(0,1)$$

$$\text{median}(z^2) / \{\Phi^{-1}(3/4)\}^2 \approx 1$$

Calibrated Spectrum

Peptide

Protein

Category

All

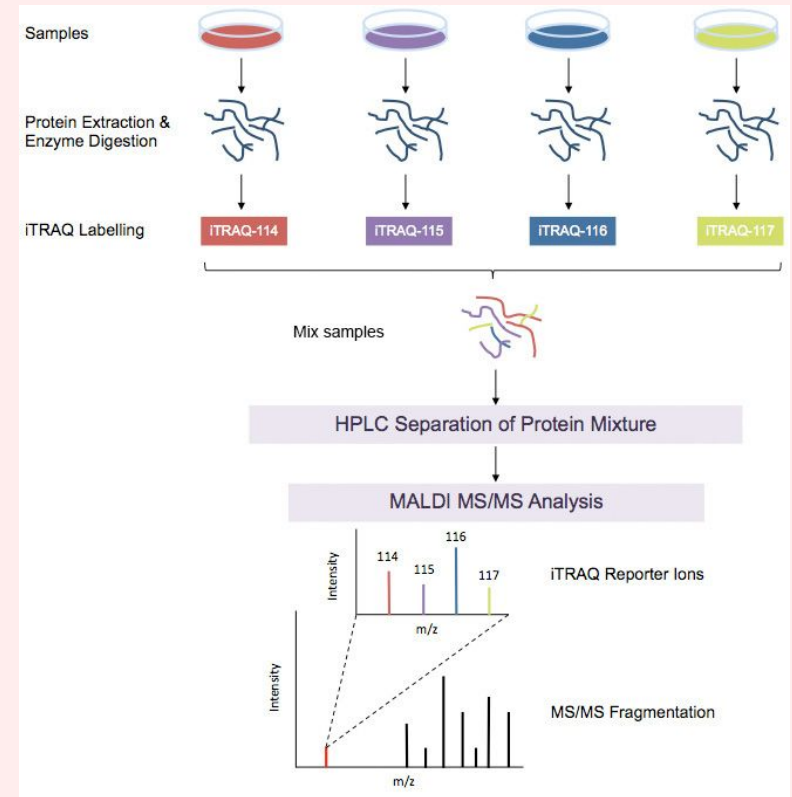
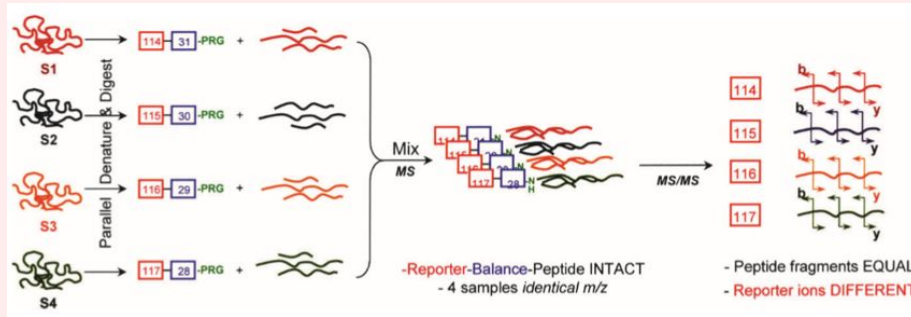
Random

- Protein quantitation
- The Systems Biology Triangle
- Randomisation to calculate coordination
- Application of variances

More information at:

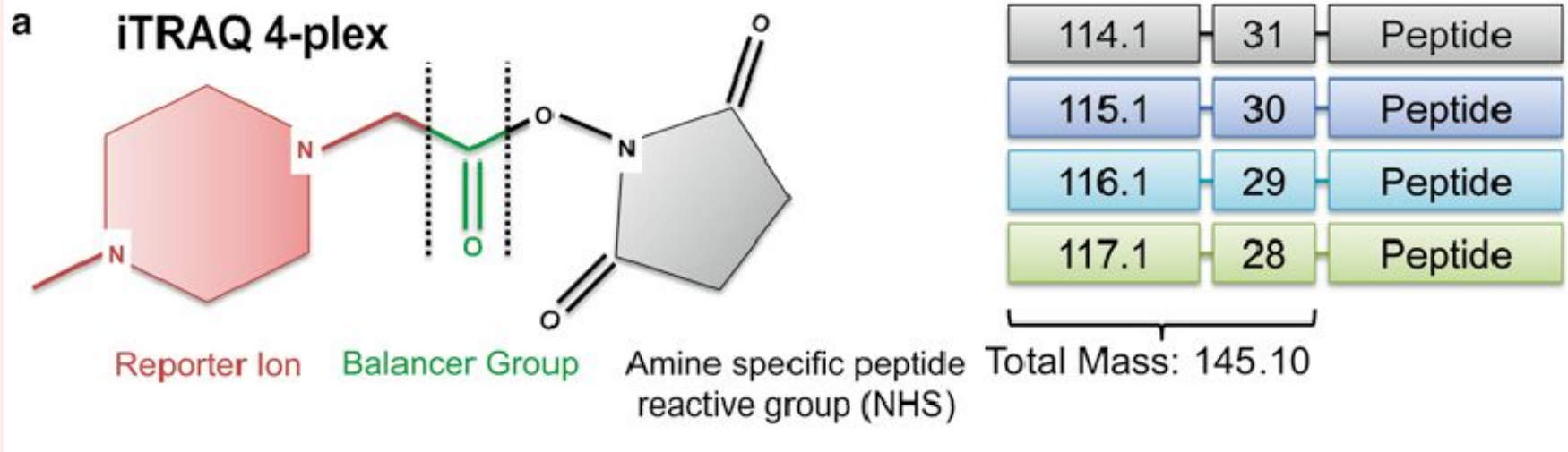
<https://www.cnice.es/wiki/proteomica/index.php/SSP>

# Isobaric tags for relative and absolute quantitation (iTRAQ)





# Isobaric tags for relative and absolute quantitation (iTRAQ)



# Isobaric methods

## iTRAQ vs TMT

|                                                               | iTRAQ | TMT |
|---------------------------------------------------------------|-------|-----|
| Isobaric labeling method for quantitative proteomics          | ✓     | ✓   |
| Labels peptide N-termini and lysine side chains               | ✓     | ✓   |
| Uses reporter ions in MS/MS for quantification                | ✓     | ✓   |
| Typical bottom-up workflow: digest first, then label peptides | ✓     | ✓   |
| Same precursor mass in MS1 across channels (isobaric)         | ✓     | ✓   |
| Needs isotopic impurity correction                            | ✓     | ✓   |
| 4-plex or 8-plex                                              | ✓     | ✗   |
| reporter ions between 113 and 121 Da                          | ✓     | ✗   |
| between 10-plex and 35-plex                                   | ✗     | ✓   |
| reporter ions between 126 and 135 Da                          | ✗     | ✓   |

# Let's play a bit with iTRAQ data



Research

## Addressing Accuracy and Precision Issues in iTRAQ Quantitation \*

Natasha A. Karp , Wolfgang Huber , Pawel G. Sadowski , Philip D. Charles , Svenja V. Hester   
, Kathryn S. Lilley  

Show more 

 Add to Mendeley  Share  Cite

<https://doi.org/10.1074/mcp.M900628-MCP200> 

[Get rights and content](#) 

Under a Creative Commons [license](#) 

 Open access

iTRAQ (isobaric tags for relative or absolute quantitation) is a mass spectrometry technology that allows quantitative comparison of protein abundance by measuring peak intensities of reporter ions released from iTRAQ-tagged peptides by fragmentation during MS/MS. However, current data analysis techniques for iTRAQ struggle to report reliable relative protein abundance estimates and suffer with problems of precision and accuracy.

# Let's play a bit with iTRAQ data

Research

## Addressing Accuracy in iTRAQ Quantitation

Natasha A. Karp <sup>‡</sup>, Wolfgang Huber <sup>§</sup>,  
Kathryn S. Lilley <sup>¶</sup>  

Show more 

 Add to Mendeley  Share  Cite

<https://doi.org/10.1074/mcp.M900628-MCP200>

Under a Creative Commons [license](#)

iTRAQ (isobaric tags for relative or absolute quantitation) is a mass spectrometry technology that allows quantitative comparison of protein abundances by measuring the intensities of reporter ions released from iTRAQ-tagged peptides by fragmentation during MS/MS. However, current data analysis techniques for iTRAQ struggle to report reliable relative protein abundance estimates and suffer with problems of precision and accuracy.



TABLE I

Summary of various data sets used within this study

| Study type           | Sample type (data set name)                         | iTRAQ system | LC-MS/MS system | Quantitation system |
|----------------------|-----------------------------------------------------|--------------|-----------------|---------------------|
| Same-sample          | <i>Erwinia</i> ( <i>Erwinia</i> B)                  | 4-Plex       | QSTAR           | i-Tracker           |
|                      | <i>Erwinia</i> ( <i>Erwinia</i> C)                  | 4-Plex       | QSTAR           | Mascot              |
|                      | <i>Erwinia</i>                                      | 4-Plex       | OrbitrapXL      | i-Tracker           |
|                      | Phosphorylase B                                     | 8-Plex       | QSTAR           | Mascot              |
|                      |                                                     |              | OrbitrapXL      | Mascot              |
| Known ratio          | Proteins at known ratios                            | 4-Plex       | QSTAR           | i-Tracker           |
| Biologically unknown | Yeast grown under nutritionally limiting conditions | 4-Plex       | Premier         |                     |
|                      |                                                     |              | QSTAR           | i-Tracker           |

TABLE II

Breakdown of proteins included in known ratio preparation and level of these proteins for each iTRAQ reporter tag

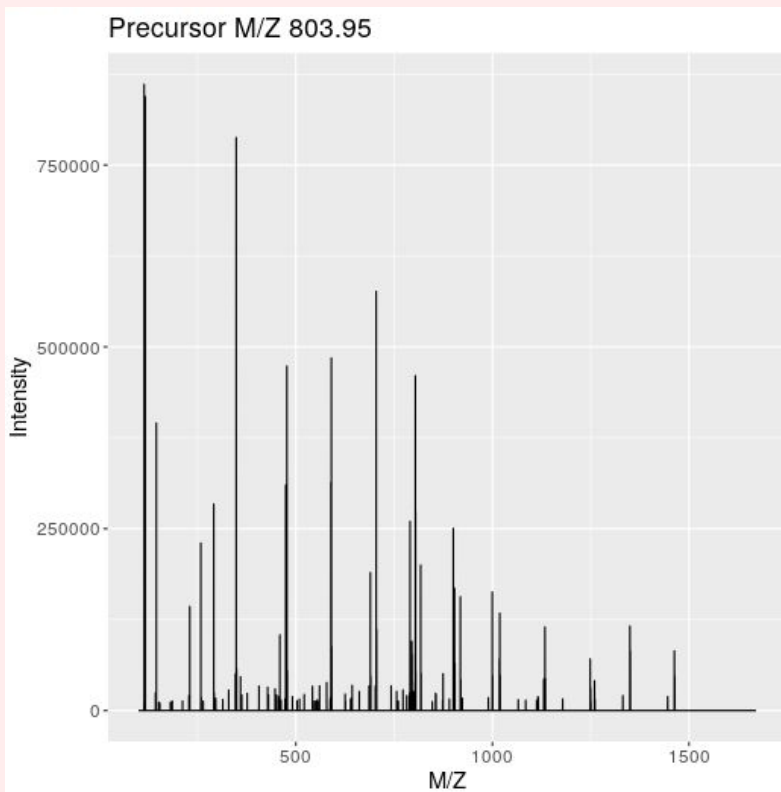
| Protein         | 114 | 115 | 116 | 117 |
|-----------------|-----|-----|-----|-----|
| BSA             | 1   | 2   | 3   | 4   |
| Cytochrome c    | 1   | 1   | 1   | 1   |
| Enolase         | 4   | 3   | 2   | 1   |
| Phosphorylase b | 1   | 1   | 1   | 1   |



# Let's play a bit with iTRAQ data

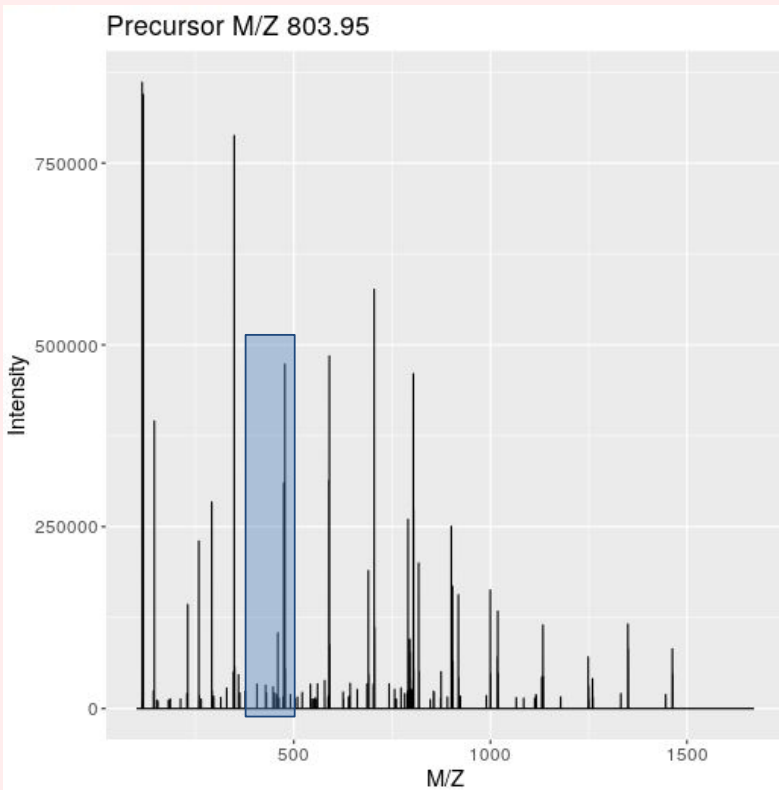
```
> library(MSnbase)
# Let's have a look (we need to have XMinG installed for X11 forwarding)
plot(sp)>
> # Let's load the test dataset of MSnbase
> data(itraqdata)
>
> # How many spectra do we have here?
> length(itraqdata)
[1] 55
>
> # Which are those spectra?
> featureNames(itraqdata)
 [1] "X1"  "X10" "X11" "X12" "X13" "X14" "X15" "X16" "X17" "X18" "X19" "X2"
[13] "X20" "X21" "X22" "X23" "X24" "X25" "X26" "X27" "X28" "X29" "X3"  "X30"
[25] "X31" "X32" "X33" "X34" "X35" "X36" "X37" "X38" "X39" "X4"  "X40" "X41"
[37] "X42" "X43" "X44" "X45" "X46" "X47" "X48" "X49" "X5"  "X50" "X51" "X52"
[49] "X53" "X54" "X55" "X6"  "X7"  "X8"  "X9"
>
> # Let's load spectrum X42
> sp = itraqdata[["X42"]]
> sp
Object of class "Spectrum2"
Precursor: 803.9468
Retention time: 33:38
Charge: 2
MSn level: 2
Peaks count: 2001
Total ion count: 49352468
>
> # Let's have a look (we need to have XMinG installed for X11 forwarding)
> plot(sp)
```

# Let's play a bit with iTRAQ data

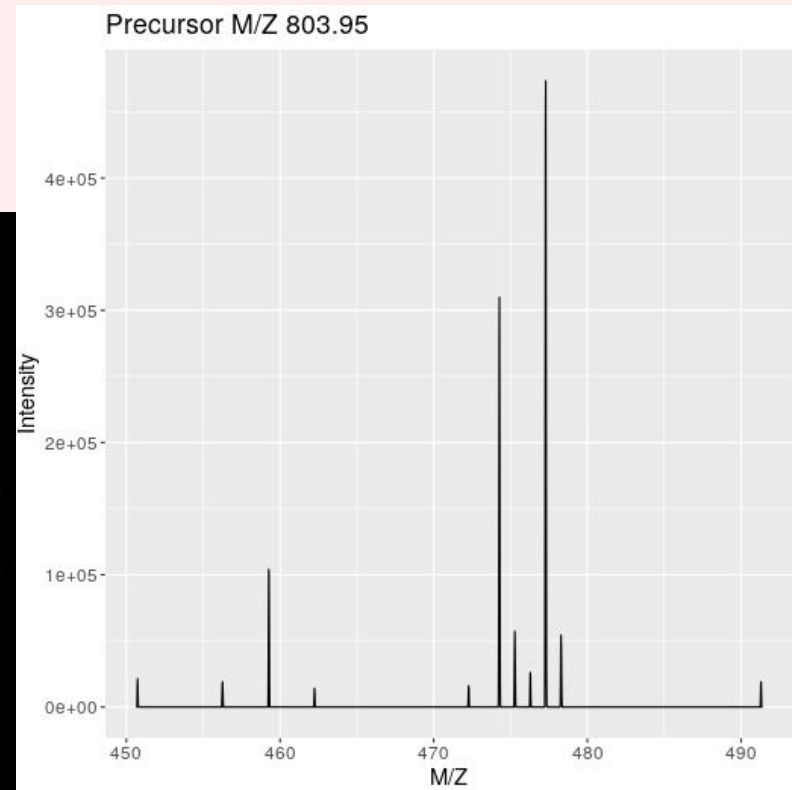


```
> sp
Object of class "Spectrum2"
Precursor: 803.9468
Retention time: 33:38
Charge: 2
MSn level: 2
Peaks count: 2001
Total ion count: 49352468
>
> # Let's have a look (we need to have Xming installed for X11 forwarding)
> plot(sp)
```

# Let's play a bit with iTRAQ data



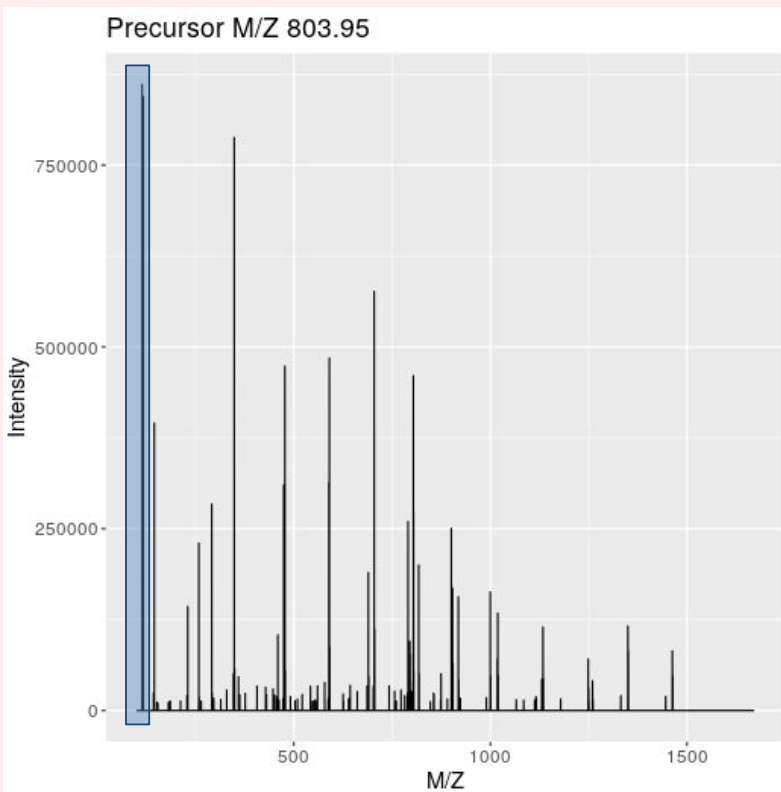
"X15"  
"X26"  
"X37"  
"X48"  
"X9"



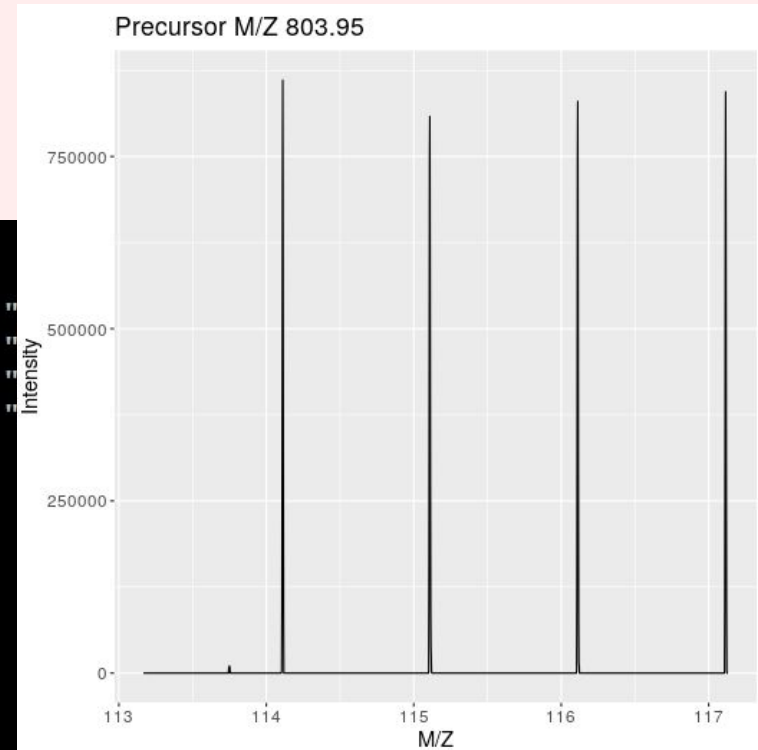
```
Precursor: 803.9468  
Retention time: 33:38  
Charge: 2  
MSn level: 2  
Peaks count: 2001  
Total ion count: 49352468
```

```
>  
> # Let's have a look (we need to have XMinG installed for X11 forwarding)  
> plot(sp)  
> # Let's zoom into a region  
> plot(trimMz(sp, mzlim = c(450, 500)))
```

# Let's play a bit with iTRAQ data



```
X15" "X16" "X17" "  
X26" "X27" "X28" "  
X37" "X38" "X39" "  
X48" "X49" "X5" "  
X9"
```



```
MSn level: 2
```

```
Peaks count: 2001
```

```
Total ion count: 49352468
```

```
>  
> # Let's have a look (we need to have Xming installed for X11 forwarding)  
> plot(sp)  
> # Let's zoom into a region  
> plot(trimMz(sp, mzlim = c(450, 500)))  
  
>  
> # Can we see the reporter ions?  
> plot(trimMz(sp, mzlim = c(110, 130)))
```



# Let's play a bit with iTRAQ data

```
>
> # Which are those spectra?
> featureNames(itraqdata)
 [1] "X1"  "X10" "X11" "X12" "X13" "X14" "X15" "X16" "X17" "X18" "X19" "X2"
[13] "X20" "X21" "X22" "X23" "X24" "X25" "X26" "X27" "X28" "X29" "X3"  "X30"
[25] "X31" "X32" "X33" "X34" "X35" "X36" "X37" "X38" "X39" "X4"  "X40" "X41"
[37] "X42" "X43" "X44" "X45" "X46" "X47" "X48" "X49" "X5"  "X50" "X51" "X52"
[49] "X53" "X54" "X55" "X6"  "X7"  "X8"  "X9"
>
> # Let's load spectrum X42
> sp = itraqdata[["X42"]]
> sp
Object of class "Spectrum2"
Precursor: 803.9468
Retention time: 33:38
Charge: 2
MSn level: 2
Peaks count: 2001
Total ion count: 49352468
>
> # Let's have a look (we need to have XMinG installed for X11 forwarding)
> plot(sp)
>
> # Let's zoom into a region
> plot(trimMz(sp, mzlim = c(450, 500)))
>
> # Can we see the reporter ions?
> plot(trimMz(sp, mzlim = c(110, 130)))
>
> # Since this has been identified already, we can check the identification data
> fData(itraqdata)[["X42", c("spectrum", "ProteinAccession", "ProteinDescription", "PeptideSequence")]]
      spectrum ProteinAccession      ProteinDescription PeptideSequence
X42          42      ECA0631 conserved hypothetical protein      LTDDDLTVIEGK
```

# Let's play a bit with iTRAQ data

```
>
> # Which are those spectra
> featureNames(itraqdata)
[1] "X1"  "X10" "X11" "X12"
[13] "X20" "X21" "X22" "X23"
[25] "X31" "X32" "X33" "X34"
[37] "X42" "X43" "X44" "X45"
[49] "X53" "X54" "X55" "X56"
>
> # Let's load spectrum X42
> sp = itraqdata[["X42"]]
> sp
Object of class "Spectrum2D"
Precursor: 803.9468
Retention time: 33:38
Charge: 2
MSn level: 2
Peaks count: 2001
Total ion count: 49352468
>
> # Let's have a look (we will use the plot function)
> plot(sp)
>
> # Let's zoom into a region of interest
> plot(trimMz(sp, mzlim = c(450, 500)))
>
> # Can we see the reporter ions?
> plot(trimMz(sp, mzlim = c(110, 130)))
>
> # Since this has been identified already, we can check the identification data
> fData(itraqdata)[["X42"], c("spectrum", "ProteinAccession", "ProteinDescription", "PeptideSequence")]
      spectrum ProteinAccession ProteinDescription PeptideSequence
X42         42          ECA0631 conserved hypothetical protein  LTDDDLTVIEGK
```

UniProt  BLAST Align Peptide search ID mapping SPARQL

Release 2026\_01 | Statistics    Help

## Find your protein

UniProtKB ▾ ECA0631 Advanced | List Search

Examples: Insulin, APP, Human, P05067, organism\_id:9606

UniProt is the world's leading high-quality, comprehensive and freely accessible resource of protein sequence and functional information. [Cite UniProt](#)

[Help](#)

# Let's play a bit with iTRAQ data

```
>
> # Which are those spectra?
> featureNames(itraqdata)
[1] "X1" "X10" "X11" "X12" "
[13] "X20" "X21" "X22" "X23" "
[25] "X31" "X32" "X33" "X34" "
[37] "X42" "X43" "X44" "X45" "
[49] "X53" "X54" "X55" "X6" "
>
> # Let's load spectrum X42
> sp = itraqdata[["X42"]]
> sp
Object of class "Spectrum2"
Precursor: 803.9468
Retention time: 33:38
Charge: 2
MSn level: 2
Peaks count: 2001
Total ion count: 49352468
>
> # Let's have a look (we need
> plot(sp)
>
> # Let's zoom into a region
> plot(trimMz(sp, mzlim = c(45
>
> # Can we see the reporter io
> plot(trimMz(sp, mzlim = c(110, 130)))
>
```

```
> # Since this has been identified already, we can check the identification data
> fData(itraqdata)[["X42", c("spectrum", "ProteinAccession", "ProteinDescription", "PeptideSequence")]
      spectrum ProteinAccession ProteinDescription PeptideSequence
X42         42      ECA0631 conserved hypothetical protein LTDDDLTVIEGK
```

UniProtKB BLAST Align Peptide search ID mapping SPARQL UniProtKB ECA0631 Advanced List Search

Status

- Reviewed (Swiss-Prot) (1)
- Unreviewed (TrEMBL) (1)

Taxonomy

Filter by taxonomy

Group by

- Taxonomy
- Keywords
- Gene Ontology
- Enzyme Class

Proteins with

- Chain (1)
- Domain (1)

Protein existence

- Homology (2)

Annotation score

- 1 (2)

Sequence length

- 1 - 200 (2)

The unreviewed UniProtKB/TrEMBL database will be reduced in size in release 2026\_02 (first half of 2026).

- Entries to be retained in UniProtKB:
  - Entries from reference proteomes
  - All reviewed (Swiss-Prot) entries
  - Selected unreviewed (TrEMBL) entries with experimental or biologically important data
- Entries to be removed: Unreviewed (TrEMBL) entries that are not part of a reference proteome

Entries removed from unreviewed UniProtKB/TrEMBL will remain accessible in the UniParc sequence archive. Please read our [help page](#), view [affected entries](#) and [proteomes](#), or contact us with any questions.

## UniProtKB 2 results

or search "ECA0631" as a Protein Name or Gene Name

Tools Download (2) Add View: Cards Table Customize columns Share

| Entry                           | Entry Name   | Protein Names           | Gene Names  | Organism                                                                                              | Length |
|---------------------------------|--------------|-------------------------|-------------|-------------------------------------------------------------------------------------------------------|--------|
| <input type="checkbox"/> Q6D9I4 | Y631_PECAS   | UPF0337 protein ECA0631 | ECA0631     | Pectobacterium atrosepticum (strain SCRI 1043 / ATCC BAA-672) (Erwinia carotovora subsp. atroseptica) | 69 AA  |
| <input type="checkbox"/> F4N205 | F4N205_YEREN | UPF0337 protein ECA0631 | YEW_BH06290 | Yersinia enterocolitica W22703                                                                        | 69 AA  |



# Let's play a bit with iTRAQ data

```
>
> # Which are those spectra?
> featureNames(itraqdata)
 [1] "X1"  "X10" "X11" "X12" "X13" "X14"
[13] "X20" "X21" "X22" "X23" "X24" "X25"
[25] "X31" "X32" "X33" "X34" "X35" "X36"
[37] "X42" "X43" "X44" "X45" "X46" "X47"
[49] "X53" "X54" "X55" "X6"  "X7"  "X8"
>
> # Let's load spectrum X42
> sp = itraqdata[["X42"]]
> sp
Object of class "Spectrum2"
Precursor: 803.9468
Retention time: 33:38
Charge: 2
MSn level: 2
Peaks count: 2001
Total ion count: 49352468
>
> # Let's have a look (we need to have XM
> plot(sp)
>
> # Let's zoom into a region
> plot(trimMz(sp, mzlim = c(450, 500)))
>
> # Can we see the reporter ions?
> plot(trimMz(sp, mzlim = c(110, 130)))
>
> # Since this has been identified already,
> fData(itraqdata)[["X42", c("spectrum", "ProteinAccession", "ProteinDescription", "PeptideSequence")]]
      spectrum ProteinAccession ProteinDescription PeptideSequence
X42      42      ECA0631 conserved hypothetical protein LTDDDLTVIEGK
```

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB Advanced | List Search

## Q6D9I4 · Y631\_PECAS

|                       |                                                                                                       |                                |                        |
|-----------------------|-------------------------------------------------------------------------------------------------------|--------------------------------|------------------------|
| Protein <sup>i</sup>  | UPF0337 protein ECA0631                                                                               | Amino acids                    | 69 (go to sequence)    |
| Status <sup>i</sup>   | UniProtKB reviewed (Swiss-Prot)                                                                       | Protein existence <sup>i</sup> | Inferred from homology |
| Organism <sup>i</sup> | Pectobacterium atrosepticum (strain SCRI 1043 / ATCC BAA-672) (Erwinia carotovora subsp. atroseptica) | Annotation score <sup>i</sup>  | 1/3                    |

Entry Variant viewer Feature viewer Genomic coordinates Publications External links History

Tools Download Add Add a publication Entry feedback

### Function<sup>i</sup>

### Names & Taxonomy<sup>i</sup>

#### Protein names<sup>i</sup>

|                  |                         |
|------------------|-------------------------|
| Recommended name | UPF0337 protein ECA0631 |
|------------------|-------------------------|

#### Gene names<sup>i</sup>

|                     |         |
|---------------------|---------|
| Ordered locus names | ECA0631 |
|---------------------|---------|

#### Organism names

|                                   |                                                                                                       |
|-----------------------------------|-------------------------------------------------------------------------------------------------------|
| Taxonomic identifier <sup>i</sup> | 218491 (NCBI)                                                                                         |
| Organism <sup>i</sup>             | Pectobacterium atrosepticum (strain SCRI 1043 / ATCC BAA-672) (Erwinia carotovora subsp. atroseptica) |
| Strain                            | SCRI 1043 / ATCC BAA-672                                                                              |
| Taxonomic                         | cellular organisms > Bacteria (bacteria) > Pseudomonadati > Pseudomonadota                            |

# Let's play a bit with iTRAQ data

```
>
> # Which are those spectra?
> featureNames(itraqdata)
[1] "X1" "X10" "X11" "X12" "X13" "X14"
[13] "X20" "X21" "X22" "X23" "X24" "X25"
[25] "X31" "X32" "X33" "X34" "X35" "X36"
[37] "X42" "X43" "X44" "X45" "X46" "X47"
[49] "X53" "X54" "X55" "X6" "X7" "X8"
>
> # Let's load spectrum X42
> sp = itraqdata[["X42"]]
> sp
```

UniProtKB • BLAST Align Peptide search ID mapping SPARQL UniProtKB • Advanced | List Search

## Q6D9I4 · Y631\_PECAS

**Function**

**Names & Taxonomy** Protein<sup>i</sup> UPF0337 protein ECA0631 Amino acids 69 (go to sequence)

**Subcellular Location** Status<sup>i</sup> UniProtKB reviewed (Swiss-Prot) Protein existence<sup>i</sup> Inferred from homology

**Phenotypes & Variants** Organism<sup>i</sup> Pectobacterium atrosepticum (strain SCRI 1043 / ATCC BAA-672) (Erwinia carotovora subsp. atroseptica) Annotation score<sup>i</sup> 1/5

**PTM/Processing**

**Expression**

**Interaction**

**Structure**

**Family & Domains**

**Sequence**

Entry Variant viewer Feature viewer Genomic coordinates Publications External links Hist

Tools Download Add Add a publication Entry feedback

**Function<sup>i</sup>**

TABLE I

Summary of various data sets used within this study

| Study type           | Sample type (data set name)                         | iTRAQ system | LC-MS/MS system | Quantitation system |
|----------------------|-----------------------------------------------------|--------------|-----------------|---------------------|
| Same-same            | Erwinia (Erwinia B)                                 | 4-Plex       | QSTAR           | i-Tracker           |
|                      | Erwinia (Erwinia C)                                 | 4-Plex       | QSTAR           | Mascot              |
|                      | Erwinia                                             | 4-Plex       | OrbitrapXL      | Mascot              |
|                      | Phosphorylase B                                     | 8-Plex       | QSTAR           | Mascot              |
| Known ratio          | Proteins at known ratios                            | 4-Plex       | OrbitrapXL      | Mascot              |
|                      |                                                     |              | QSTAR           | i-Tracker           |
| Biologically unknown | Yeast grown under nutritionally limiting conditions | 4-Plex       | Premier         | i-Tracker           |
|                      |                                                     |              | QSTAR           |                     |

```
>
> # Can we see the reporter ions?
> plot(trimMz(sp, mzlim = c(110, 130)))
>
> # Since this has been identified already,
> fData(itraqdata)[["X42", c("spectrum", "ProteinAccession", "ProteinDescription", "PeptideSequence")]]
spectrum ProteinAccession ProteinDescription PeptideSequence
X42 42 ECA0631 conserved hypothetical protein LTDDDLTVIEGK
```

Identifier<sup>i</sup>

Organism<sup>i</sup> Pectobacterium atrosepticum (strain SCRI 1043 / ATCC BAA-672) (Erwinia carotovora subsp. atroseptica)

Strain SCRI 1043 / ATCC BAA-672

Taxonomic cellular organisms > Bacteria (bacteria) > Pseudomonadati > Pseudomonadota

# Let's play a bit with iTRAQ data

```
> # Let's zoom into a region
> plot(trimMz(sp, mzlim = c(450, 500)))
>
> # Can we see the reporter ions?
> plot(trimMz(sp, mzlim = c(110, 130)))
>
> # Since this has been identified already, we can check the identification data
> fData(itraqdata)["X42", c("spectrum", "ProteinAccession", "ProteinDescription", "PeptideSequence")]
      spectrum ProteinAccession      ProteinDescription PeptideSequence
X42         42      ECA0631 conserved hypothetical protein  LTDDDLTVIEGK
> # Let's confirm whether we get the same identification with Mascot
> # First we create a function to filter a bit the lot intensity peaks
>
> showMgfData <- function(sp, t = 10000) {
+   tf <- tempfile(fileext = ".mgf")
+   on.exit(unlink(tf), add = TRUE)
+
+   sp2 <- pickPeaks(sp)
+   sp2 <- clean(removePeaks(sp2, t = t))
+
+   writeMgfData(sp2, con = tf)
+   cat(readLines(tf), sep = "\n")
+   invisible(NULL)
+ }
>
```

# Let's play a bit with iTRAQ data

```
> # Let's confirm whether we get the same identification with Mascot
> # First we create a function to filter a bit the lot intensity peaks
>
> showMgfData <- function(sp, t = 10000) {
+   tf <- tempfile(fileext = ".mgf")
+   on.exit(unlink(tf), add = TRUE)
+
+   sp2 <- pickPeaks(sp)
+   sp2 <- clean(removePeaks(sp2, t = t))
+
+   writeMgfData(sp2, con = tf)
+   cat(readLines(tf), sep = "\n")
+   invisible(NULL)
+ }
>
>
> # Then, we obtain the MGF text
> showMgfData(sp)
COM=Spectrumexported by MSnbase on Wed Mar 18 14:21:59 2026
BEGIN IONS
SCANS=44
TITLE=msLevel 2; retentionTime 2018.28; scanNum 44; precMz 803.9468; precCharge 2
RTINSECONDS=2018.28
PEPMASS=803.9468
CHARGE=2+
113.749305725098 10169.232421875
114.111358642578 860852.875
115.108963012695 808308.5
116.112144470215 830120.875
117.115089416504 844034.5
143.103897094727 24153.390625
145.100005003000 305300.50375
```



# Let's play a bit with iTRAQ data

```
COM=Spectrumexported by MSnbase on Wed Mar 18
14:21:59 2026
BEGIN IONS
SCANS=44
TITLE=msLevel 2; retentionTime 2018.28; scanNum 44;
precMz 803.9468; precCharge 2
RTINSECONDS=2018.28
PEPMASS=803.9468
CHARGE=2+
113.749305725098 10169.232421875
114.111358642578 860852.875
115.108963012695 808308.5
116.112144470215 830120.875
117.115089416504 844034.5
143.103897094727 24153.390625
145.109085083008 395208.59375
151.451583862305 11307.22265625
154.042007446289 0
181.034927368164 11364.626953125
185.324508666992 12249.630859375
185.70637512207 13055.107421875
211.811340332031 12967.291015625
228.184799194336 20480.859375
230.198394775391 143064.90625
257.997619628906 11657.595703125
258.192108154297 230192.78125
259.198211669922 18034.021484375
264.3505859375 12893.40234375
291.214538574219 283843.625
292.216003417969 24142.6484375
296.197174072266 16457.451171875
314.205017089844 15040.384765625
329.235076904297 28056.5078125
346.234893798828 50148.5703125
348.239532470703 787729.4375
349.243469238281 57541.203125
359.248016357422 46445.28515625
362.261413574219 21057.57421875
376.263885498047 23513.486328125
406.248199462891 33569.0078125
428.305297851562 31943.1640625
430.240875244141 21248.79296875
447.136352539062 29850.234375
450.730163574219 21448.01171875
456.261566162109 18982.96875
```

Usually we would not work it one by one, but for this practical, we will save it into an mgf file (as a text file) and we load it from Mascot



# Let's play a bit with iTRAQ data

COM=Spectrumexported by MSnbase on Wed Mar 18  
14:21:59 2026  
BEGIN IONS  
SCANS=44  
TITLE=msLevel 2; retentionTime 2018.28; scanNum 44;  
precMz 803.9468; precCharge 2  
RTINSECONDS=2018.28  
PEPMASS=803.9468  
CHARGE=2+  
113.749305725098 10169.232421875  
114.111358642578 860852.875  
115.108963012695 808308.5  
116.112144470215 830120.875  
117.115089416504 844034.5  
143.103897094727 24153.390625  
145.109085083008 395208.59375  
151.451583862305 11307.22265625  
154.042007446289 0  
181.034927368164 11364.626953125  
185.324508666992 12249.630859375  
185.70637512207 13055.107421875  
211.811340332031 12967.291015625  
228.184799194336 20480.859375  
230.198394775391 143064.90625  
257.997619628906 11657.595703125  
258.192108154297 230192.78125  
259.198211669922 18034.021484375  
264.3505859375 12893.40234375  
291.214538574219 283843.625  
292.216003417969 24142.6484375  
296.197174072266 16457.451171875  
314.205017089844 15040.384765625  
329.235076904297 28056.5078125  
346.234893798828 50148.5703125  
348.239532470703 787729.4375  
349.243469238281 57541.203125  
359.248016357422 46445.28515625  
362.261413574219 21057.57421875  
376.263885498047 23513.486328125  
406.248199462891 33569.0078125  
428.305297851562 31943.1640625  
430.240875244141 21248.79296875  
447.136352539062 29850.234375  
450.730163574219 21448.01171875  
456.261566162109 18982.96875

### MASCOT MS/MS Ions Search

Your name:  Email:

Search title:

Database(s):

Amino acid (AA):

Taxonomy:

Enzyme:  Allow up to:  missed cleavages

Quantitation:

Crosslinking:

Fixed modifications:

Variable modifications:

Display all modifications: ☐

Error tolerant: ☐ Automatic second pass search of selected modification classes

Peptide tol.  ppm  MS/MS tol.  Da

Data file:

Data format:

Instrument:

Target FDR:

Machine learning: ☐ Refine results with machine learning (Percolator)

We update all  
the settings

# Let's play a bit with iTRAQ data

 **MASCOT SEARCH**

**Search running...** 

|

**Time now** Wed Mar 18 14:37:32 2026

Your Mascot search is running. This page will refresh at the end of the search to show the results report. [Refresh](#)  
If you close your browser window before the search ends, the search will finish in the background.

# Let's play a bit with iTRAQ data

**User** : Marco  
**E-mail** : marco.trevisan@newcastle.ac.uk  
**Search title** : Spectrumexported by MSnbase on Wed Mar 18 14:21:59 2026  
**MS data file** : spectrum\_X42.mgf  
**Databases** : 1: contaminants 20160129 (247 sequences; 128,130 residues)  
2: SwissProt 2026\_01 (574,627 sequences; 208,482,574 residues)  
**Timestamp** : 18 Mar 2026 at 14:37:34 GMT

[Re-search](#) ☒ All ☐ Non-significant ☐ Unassigned [\[help\]](#)

[Export](#) As XML

## Search parameters

## Score distribution

## Modification statistics for all protein families

## Legend

## Protein Family Summary

|                                       |                                     |                                      |                                          |
|---------------------------------------|-------------------------------------|--------------------------------------|------------------------------------------|
| Significance threshold p<             | <input type="text" value="0.6106"/> | Max. number of families              | <span>AUTO</span> <a href="#">[help]</a> |
| Target FDR (overrides sig. threshold) | <span>1%</span>                     | FDR type                             | <span>Sequence</span>                    |
| Display non-sig. matches              | <input type="checkbox"/>            | Min. number of sig. unique sequences | <span>1</span>                           |
|                                       |                                     | Dendrograms cut at                   | <span>0</span>                           |
| Preferred taxonomy                    | <span>All entries</span>            |                                      |                                          |
| <a href="#">Apply</a>                 |                                     |                                      |                                          |

## ▼ Sensitivity and FDR (reversed protein sequences)

|                                                    | Target | Decoy | FDR   |                                                                                         |
|----------------------------------------------------|--------|-------|-------|-----------------------------------------------------------------------------------------|
| Protein family members                             | 1      | 0     | 0.00% | Not enough decoy protein hits for a meaningful FDR calculation.                         |
| <span>Sequences</span> above <span>homology</span> | 1      | 0     | 0.00% | Not enough decoy matches above significance threshold for a meaningful FDR calculation. |

Decoy results are available in [the decoy report](#).

Proteins (1) [Report Builder](#)

## Protein family 1 (out of 1)

10 per page 1 [Expand all](#) [Collapse all](#)

Accession contains  [Find](#) [Clear](#)

► 1 **2::Y631\_PECAS**

# Let's play a bit with iTRAQ data

Re-search ☒ All ☐ Non-significant ☐ Unassigned [\[help\]](#) Export As XML

- Search parameters
- Score distribution
- Modification statistics for all protein families
- Legend

## Protein Family Summary

Significance threshold p< 0.6106 Max. number of families AUTO [\[help\]](#)  
Target FDR (overrides sig. threshold) 1% FDR type Sequence  
Display non-sig. matches ☐ Min. number of sig. unique sequences 1  
Dendrograms cut at 0  
Preferred taxonomy All entries  
Apply

## Sensitivity and FDR (reversed protein sequences)

|                          | Target | Decoy | FDR   |                                                                                         |
|--------------------------|--------|-------|-------|-----------------------------------------------------------------------------------------|
| Protein family members   | 1      | 0     | 0.00% | Not enough decoy protein hits for a meaningful FDR calculation.                         |
| Sequences above homology | 1      | 0     | 0.00% | Not enough decoy matches above significance threshold for a meaningful FDR calculation. |

Decoy results are available in [the decoy report](#).

Proteins (1) [Report Builder](#)

## Protein family 1 (out of 1)

10 per page 1 Expand all Collapse all

Accession contains Find Clear

## 1 2::Y631\_PECAS

77 UPF0337 protein ECA0631 OS=Pectobacterium atrosepticum (strain SCRI 1043 / ATCC BAA-672) OX=218491 GN=ECA0631 PE=

|                   | Score | Mass  | Matches | Sequences                                                                                                                     |
|-------------------|-------|-------|---------|-------------------------------------------------------------------------------------------------------------------------------|
| 1.1 2::Y631_PECAS | 77    | 10053 | 1 (1)   | 1 (1) UPF0337 protein ECA0631 OS=Pectobacterium atrosepticum (strain SCRI 1043 / ATCC BAA-672) OX=218491 GN=ECA0631 PE=3 SV=2 |

## 1 peptide matches (1 non-duplicate, 0 duplicate)

☒ Auto-fit to window

| Query Dups | Observed | Mr (expt) | Mr (calc) | ppm  | M | Score | Expect  | Rank | U | Peptide          |
|------------|----------|-----------|-----------|------|---|-------|---------|------|---|------------------|
| 1          | 803.9468 | 1605.8790 | 1605.8705 | 5.31 | 0 | 77    | 3.7e-06 | 1    | U | K.LTDDDLTVIEGK.R |

match quality looks good:

Score = 77

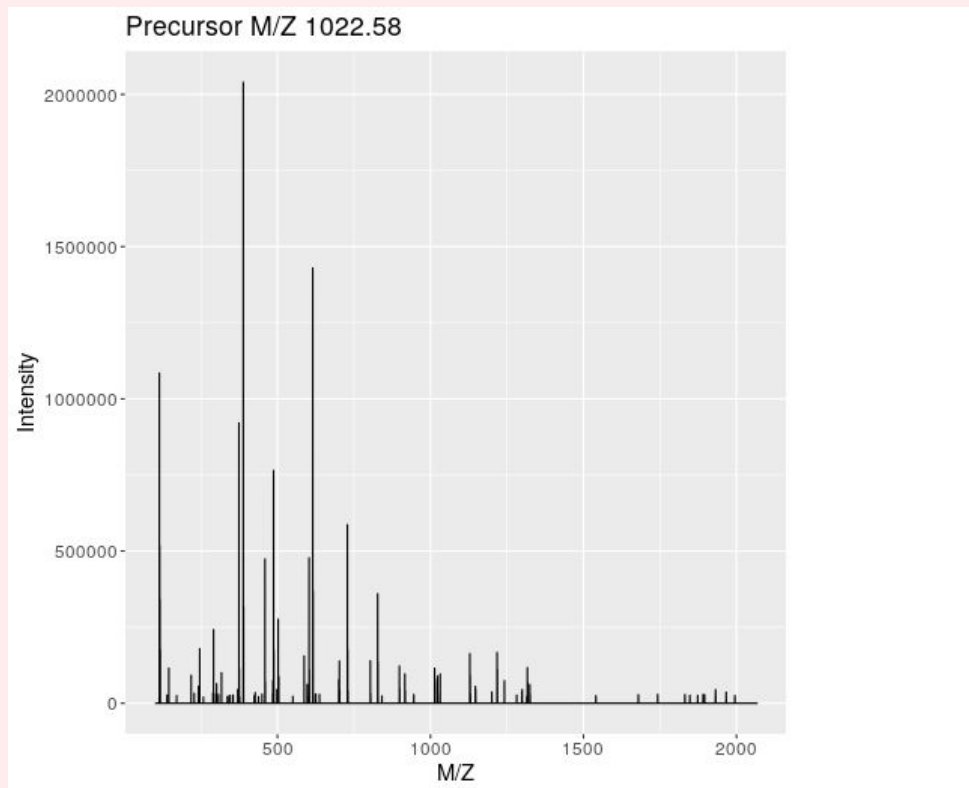
Expect = 3.7e-06

Mass error = 5.31 ppm

```
> fData(itraqdata)["X42", c("spectrum", "ProteinAccession", "ProteinDescription", "PeptideSequence")]
      spectrum ProteinAccession ProteinDescription PeptideSequence
X42         42          ECA0631 conserved hypothetical protein LTDDDLTVIEGK
```

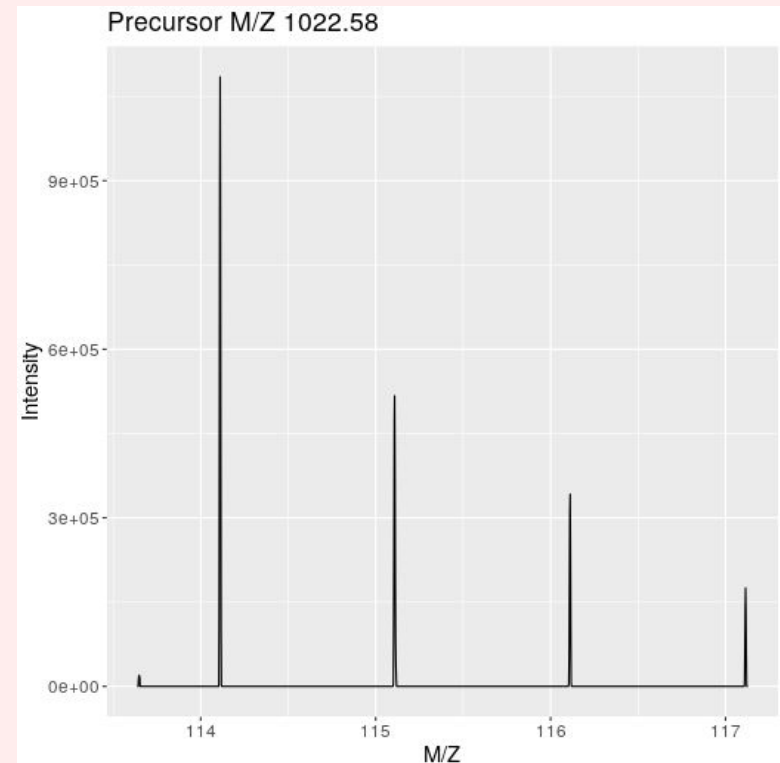
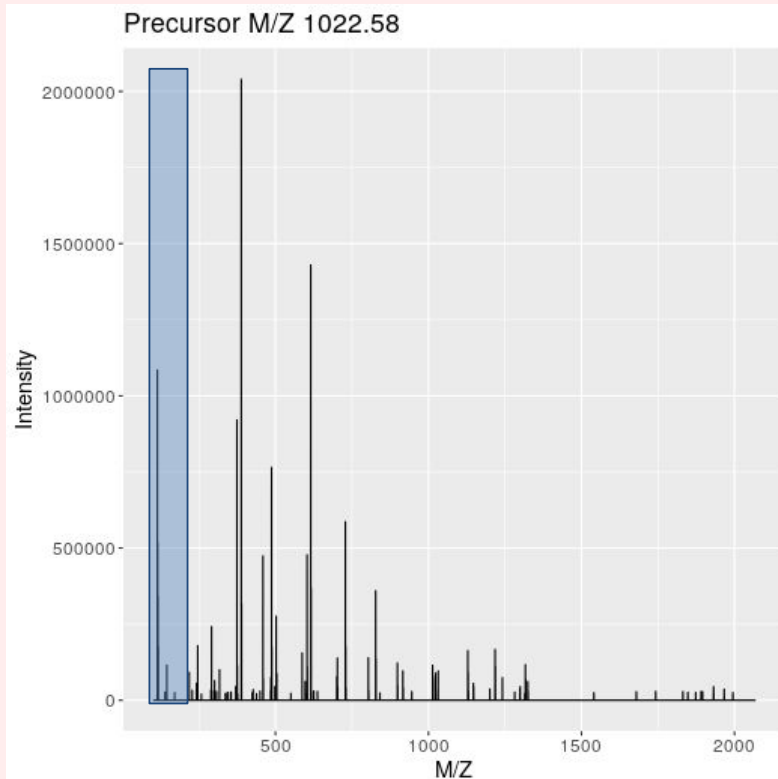
# Let's play a bit with iTRAQ data

```
> boxplot(  
+   log2(itraq + 1),  
+   main = "Reporter intensities by iTRAQ channel",  
+   ylab = "log2 intensity"  
+ )  
>  
>  
> # Let's have a look at X47, which is enolase, a spike in protein in this sample  
> sp = itraqdata[["X47"]]  
> plot(sp)
```



# Let's play a bit with iTRAQ data

```
> boxplot(  
+   log2(itraq + 1),  
+   main = "Reporter intensities by iTRAQ channel",  
+   ylab = "log2 intensity"  
+ )  
>  
> # Let's have a look at X47, which is enolase, a spike in protein in this sample  
> sp = itraqdata[["X47"]]  
> plot(sp)  
  
> plot(trimMz(sp, mzlim = c(110, 130)))
```





# Let's play a bit with iTRAQ data

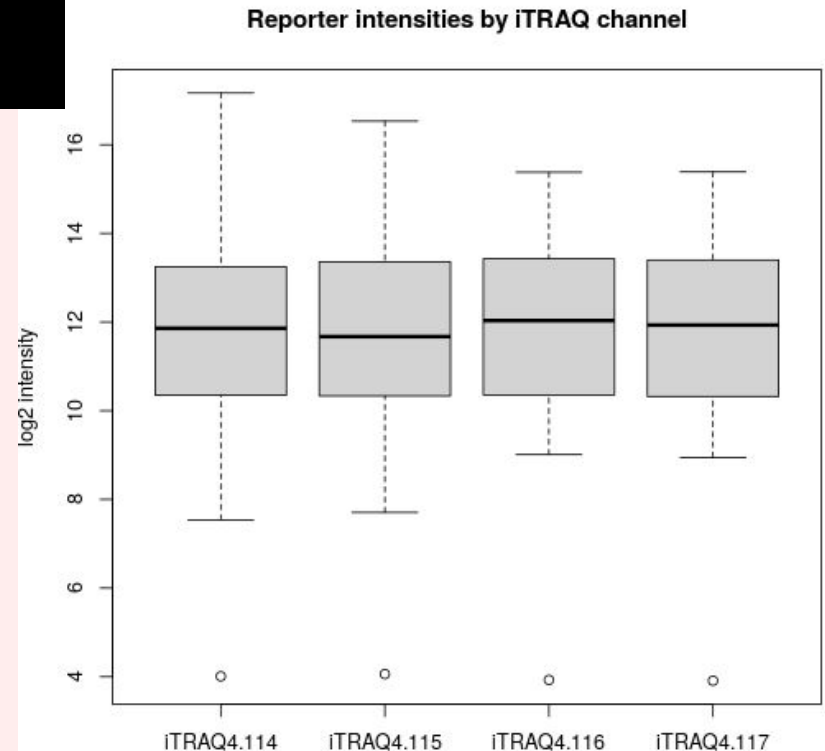
```
1259.65209960958 40929.578125
1260.66625976562 15503.82421875
1331.65002441406 20468.75390625
1349.67687988281 116093.8125
1350.7265625 81670.296875
1445.79638671875 19117.90234375
1462.81079101562 82077.0078125
1463.74133300781 47465.26953125
END IONS
> # But this practical is about quantification
> # So, let's quantify these spectra!
> msnset <- quantify(itraqdata, reporters = iTRAQ4, method = "trap")
> View(exprs(msnset))
```

|    | row.names | iTRAQ4.114   | iTRAQ4.115  | iTRAQ4.116  | iTRAQ4.117  |
|----|-----------|--------------|-------------|-------------|-------------|
| 1  | X1        | 1347.61584   | 2247.30975  | 3927.69307  | 7661.14628  |
| 2  | X10       | 739.98608    | 799.35012   | 712.59830   | 940.67930   |
| 3  | X11       | 27638.35819  | 33394.02524 | 32104.28786 | 26628.72776 |
| 4  | X12       | 31892.89279  | 33634.69801 | 37674.72721 | 37227.71193 |
| 5  | X13       | 26143.75416  | 29677.47813 | 29089.05931 | 27902.56076 |
| 6  | X14       | 6448.08287   | 6234.19568  | 6902.89029  | 6437.23029  |
| 7  | X15       | 722.29818    | 1058.87116  | 851.47016   | 944.62868   |
| 8  | X16       | 10154.95328  | 10486.94267 | 11018.19118 | 11289.55202 |
| 9  | X17       | 2800.39559   | 3244.97754  | 2965.31877  | 3184.19386  |
| 10 | X18       | 4458.77171   | 4786.51826  | 4350.78472  | 4303.02563  |
| 11 | X19       | 32838.04395  | 37066.05777 | 41429.62671 | 39700.47518 |
| 12 | X2        | 3715.08932   | 4254.32256  | 4748.46152  | 5249.90356  |
| 13 | X20       | 34509.68621  | 34928.74712 | 41911.03212 | 42843.83865 |
| 14 | X21       | 21262.14754  | 23168.72913 | 25407.06847 | 25949.95404 |
| 15 | X22       | 8635.31649   | 10036.52908 | 9254.43168  | 7769.74907  |
| 16 | X23       | 1566.74773   | 1444.27953  | 1455.89659  | 1557.19896  |
| 17 | X24       | 1101.06205   | 1124.16708  | 1140.09311  | 1191.80548  |
| 18 | X25       | 5135.56011   | 5519.05329  | 5828.37084  | 5583.72953  |
| 19 | X26       | 1524.14767   | 1399.89688  | 1547.21848  | 1563.22990  |
| 20 | X27       | 147093.25030 | 94770.28613 | 42616.07457 | 21259.42497 |
| 21 | X28       | 30047.24363  | 31498.21764 | 38489.80882 | 38275.26375 |
| 22 | X29       | 9339.07912   | 10509.31520 | 10824.73863 | 9504.38777  |
| 23 | X3        | 4483.31990   | 4873.99606  | 6743.44063  | 4601.37758  |
| 24 | X30       | 1147.03795   | 1281.25856  | 1278.70097  | 1175.08044  |
| 25 | X31       | 12682.55301  | 13586.68865 | 13579.17358 | 13067.36481 |

# Let's play a bit with iTRAQ data

```
> # But this practical is about quantification
> # So, let's quantify these spectra!
> msnsset <- quantify(itraqdata, reporters = iTRAQ4, method = "trap")

> View(exprs(msnsset))
> msnsset_log <- log(msnsset, base = 2)
>
> itraq <- as.data.frame(exprs(msnsset))
>
> # Let's check the channel distributions
> boxplot(
+   log2(itraq + 1),
+   main = "Reporter intensities by iTRAQ channel",
+   ylab = "log2 intensity"
+ )
```





# Let's play a bit with iTRAQ data

```
> # Let's check the channel distributions
> boxplot(
+   log2(itraq + 1),
+   main = "Reporter intensities by iTRAQ channel",
+   ylab = "log2 intensity"
+ )
> #####
> # Let's normalise the dataset
> msnset_n <- normalize(msnset_log, method = "center.median")
ProteinAccession = fData(msnset_n)$ProteinAccession,
PeptideSequence = fData(msnset_n)$PeptideSequence,
as.data.frame(exprs(msnset_n))
)

View(itraq_table)>
> # It would be nice to have the identification data on the same table
> itraq_table <- cbind(
+   ProteinAccession = fData(msnset_n)$ProteinAccession,
+   PeptideSequence = fData(msnset_n)$PeptideSequence,
+   as.data.frame(exprs(msnset_n))
+ )
>
> View(itraq_table)
```

ENO = enolase (spiked in at 4:3:2:1)

BSA = bovine serum albumin (spiked in at 1:2:3:4)

ECA\* = *Erwinia carotovora* background proteins

|    | row.names | ProteinAccession | PeptideSequence       | iTRAQ4.114  | iTRAQ4.115   | iTRAQ4.116  | iTRAQ4.117  |
|----|-----------|------------------|-----------------------|-------------|--------------|-------------|-------------|
| 1  | X1        | BSA              | NYQEAK                | -1.46298761 | -0.539511067 | -0.09268986 | 0.97200718  |
| 2  | X10       | ECA1422          | VTLVDTEGHSMTGGR       | -2.32782685 | -2.030810588 | -2.55521111 | -2.05377822 |
| 3  | X11       | ECA4030          | SPIWR                 | 2.89520312  | 3.353808047  | 2.93832204  | 2.76935851  |
| 4  | X12       | ECA3882          | TAIDALK               | 3.10176616  | 3.364168349  | 3.16915311  | 3.25275196  |
| 5  | X13       | ECA1364          | SILINK                | 2.81499752  | 3.183586547  | 2.79603269  | 2.83677254  |
| 6  | X14       | ECA0871          | DFEVVNNESDPR          | 0.79547339  | 0.932493388  | 0.72082850  | 0.72088700  |
| 7  | X15       | ECA4512          | EGNDFYHENTISNVIDK     | -2.36273046 | -1.625182996 | -2.29834418 | -2.04773384 |
| 8  | X16       | ECA4513          | VMLQAYDEGR            | 1.45071480  | 1.682812184  | 1.39544345  | 1.53136325  |
| 9  | X17       | ECA3969          | ANHWLVLATGAGR         | -0.40776626 | -0.009501559 | -0.49818485 | -0.29462491 |
| 10 | X18       | ECA3082          | VATPEMQSK             | 0.26324944  | 0.551266562  | 0.05490358  | 0.13979835  |
| 11 | X19       | ECA1032          | FEVGEIEK              | 3.14389939  | 3.504316727  | 3.30621887  | 3.34553128  |
| 12 | X2        | ECA1294          | TLAEGQR               | 0.00000000  | 0.381219371  | 0.18108811  | 0.42673783  |
| 13 | X20       | ECA4514          | VTLELK                | 3.21553255  | 3.418632939  | 3.32288609  | 3.45546275  |
| 14 | X21       | ECA1104          | SIGFEGDSIGR           | 2.51681852  | 2.685396954  | 2.60078997  | 2.73210699  |
| 15 | X22       | ECA3356          | VWVVEGSK              | 1.21685216  | 1.619478474  | 1.14377235  | 0.99231492  |
| 16 | X23       | ECA4037          | LANELSDRAENK          | -1.24562399 | -1.177360056 | -1.52446416 | -1.32659980 |
| 17 | X24       | ECA0621          | AVENFYISNNK           | -1.75450113 | -1.538853575 | -1.87722040 | -1.71240430 |
| 18 | X25       | ECA1093          | GVELALK               | 0.46712473  | 0.756710765  | 0.47672062  | 0.51567597  |
| 19 | X26       | ECA0452          | ADDFAEQLK             | -1.28539421 | -1.222389492 | -1.43669512 | -1.32102312 |
| 20 | X27       | ENO              | ARDALLLK              | 5.30719034  | 4.886852837  | 3.34695376  | 2.44447758  |
| 21 | X28       | ECA2391          | TLAEGQNVFEIQQGQK      | 3.01576385  | 3.269488238  | 3.20003255  | 3.29278733  |
| 22 | X29       | ECA0435          | ELSPVGSTHPLSAVDIER    | 1.32988340  | 1.369886708  | 1.36988824  | 1.28304061  |
| 23 | X3        | ECA1362          | AAKPEAPAPASAPALGAR    | 0.27117055  | 0.577395033  | 0.68711282  | 0.23651276  |
| 24 | X30       | ECA1363          | LHGGSANFLDVGGATK      | -1.69548377 | -1.350148403 | -1.71169313 | -1.73273956 |
| 25 | X31       | ECA3566          | IMIDLGTENK            | 1.77137639  | 2.056411929  | 1.69695173  | 1.74234324  |
| 26 | X32       | ECA0435          | ELSPVGSTHPLSAVDIER    | -2.86999354 | -2.429566717 | -2.52899372 | -2.57774372 |
| 27 | X33       | ECA0691          | DHTLFALTDGK           | -2.28926475 | -1.914537687 | -2.24440074 | -2.30320241 |
| 28 | X34       | ECA3566          | IMIDLGTENK            | -2.49540688 | -2.228971866 | -2.79583544 | -2.98696486 |
| 29 | X35       | ECA3377          | HIFGLK                | -0.26744857 | 0.000000000  | -0.03293028 | 0.00000000  |
| 30 | X36       | ECA0978          | FPEGTNVEDIDQLMDK      | -0.70539730 | -0.345896778 | -0.58840175 | -0.67465752 |
| 31 | X37       | ECA2831          | AAQLQGVAVVTSADALQK    | -1.56090207 | -1.173276456 | -1.34063180 | -1.63038487 |
| 32 | X38       | ECA4514          | GILDTFK               | 1.62624610  | 1.909816242  | 1.55874134  | 1.58120332  |
| 33 | X39       | ECA0469          | DNQSQSVGAYADDAVITSTIK | -1.91901187 | -1.545093209 | -2.24965086 | -1.82756158 |
| 34 | X4        | ECA4514          | QSVDEPVQTCYK          | -0.95373218 | -1.203812043 | -1.90596667 | -1.30384547 |
| 35 | X40       | ECA4514          | DSVGAVVMGPPYADLAEGHK  | -0.85636537 | -0.339102694 | -1.12633787 | -0.79027191 |
| 36 | X41       | ECA0172          | ECITSLVSR             | 2.24357762  | 2.505296521  | 2.20877186  | 2.23044299  |
| 37 | X42       | ECA0631          | LTDIDLTVIEGK          | 0.65566691  | 0.868665668  | 0.51144815  | 0.62105779  |
| 38 | X43       | ECA3349          | HDDMYAINELITK         | -2.09024318 | -1.772472472 | -1.67655034 | -1.58945826 |
| 39 | X44       | ECA0469          | DNQSQSVGAYADDAVITSTIK | -1.68240037 | -1.778582466 | -1.82986019 | -2.11912919 |
| 40 | X45       | ECA4514          | DSVGAVVMGPPYADLAEGHK  | -0.94321217 | -0.595850560 | -0.84773248 | -0.88472741 |
| 41 | X46       | ENO              | TAGIQIVADDLTVTNPK     | 0.53145331  | -1.508183172 | NA          | -1.57219621 |
| 42 | X47       | ENO              | TAGIQIVADDLTVTNPK     | 0.93116786  | 0.268763612  | -0.82146961 | -1.80753921 |
| 43 | X48       | ECA4026          | VSQLALEVLTYNK         | -2.22215784 | -2.287399656 | -2.09584220 | -2.19089463 |
| 44 | X49       | ECA3929          | LIDLGLK               | 1.19482641  | 1.603221935  | 1.26393765  | 1.29926322  |
| 45 | X5        | ECA4514          | QSVDEPVQTCYK          | 2.03364423  | 2.227396590  | 1.89643237  | 2.08324997  |
| 46 | X50       | ECA4026          | VSQLALEVLTYNK         | -1.40776500 | -1.328995905 | -1.48853901 | -1.38941612 |
| 47 | X51       | ECA4013          | ATLLGLGLR             | 1.02128955  | 1.233238002  | 0.85239262  | 1.03538013  |
| 48 | X52       | BSA              | QTALVELLK             | -1.22125608 | -0.158084605 | 0.26369635  | 1.40769710  |
| 49 | X53       | BSA              | QTALVELLK             | -4.33033041 | -3.974426157 | -3.02357145 | -2.17414471 |

# Let's play a bit with iTRAQ data

```
> #####
> # We filter out some bad data
> ok <- with(itraq_table,
+           !is.na(iTRAQ4.114) & !is.na(iTRAQ4.115))
>
> itraq_ok <- itraq_table[ok, ]
# And generate the MA plot of 115 vs 114
>
> # And generate the MA plot of 115 vs 114
> itraq_ok$logFC_115_114 <- itraq_ok$iTRAQ4.115 - itraq_ok$iTRAQ4.114
> itraq_ok$A_115_114 <- (itraq_ok$iTRAQ4.115 + itraq_ok$iTRAQ4.114) / 2
>
> plot(itraq_ok$A_115_114, itraq_ok$logFC_115_114,
+      pch = 16, cex = 0.8,
+      xlab = "A = average normalized log2-intensity (114,115)",
+      ylab = "M = normalized log2-difference (115 - 114)",
+      main = "iTRAQ MA plot: 115 vs 114")
> abline(h = 0, col = "red")
```

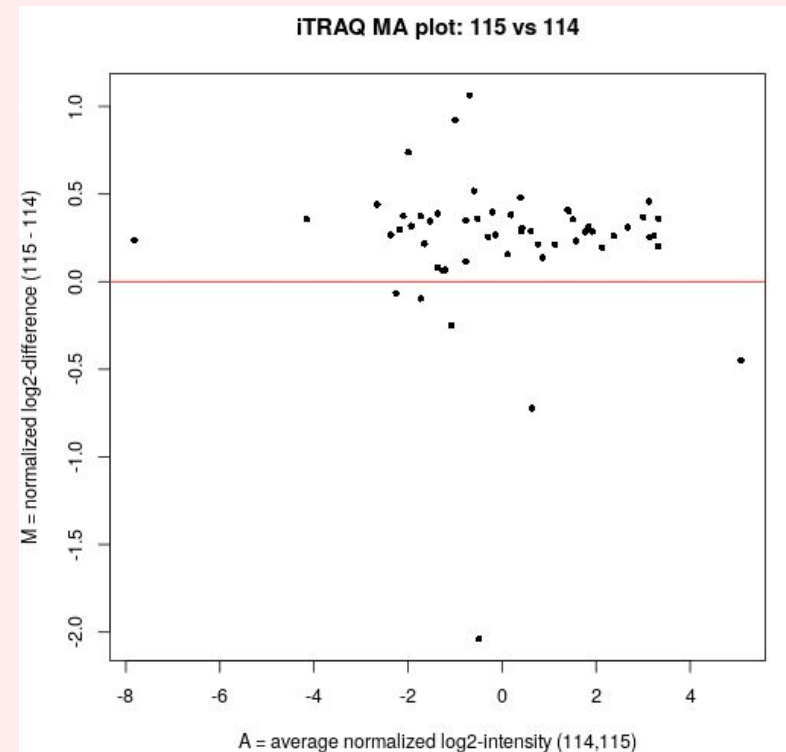
- Most proteins do not change
- Low-intensity measurements are noisier
- A few proteins are strongly shifted
  - are those the ones carrying the spike-in differences?

## MA plot

M (minus) =  $\log_2(115) - \log_2(114)$

A (average) =  $[\log_2(114) + \log_2(115)] / 2$

(imitating genomic analyses)



# Let's play a bit with iTRAQ data

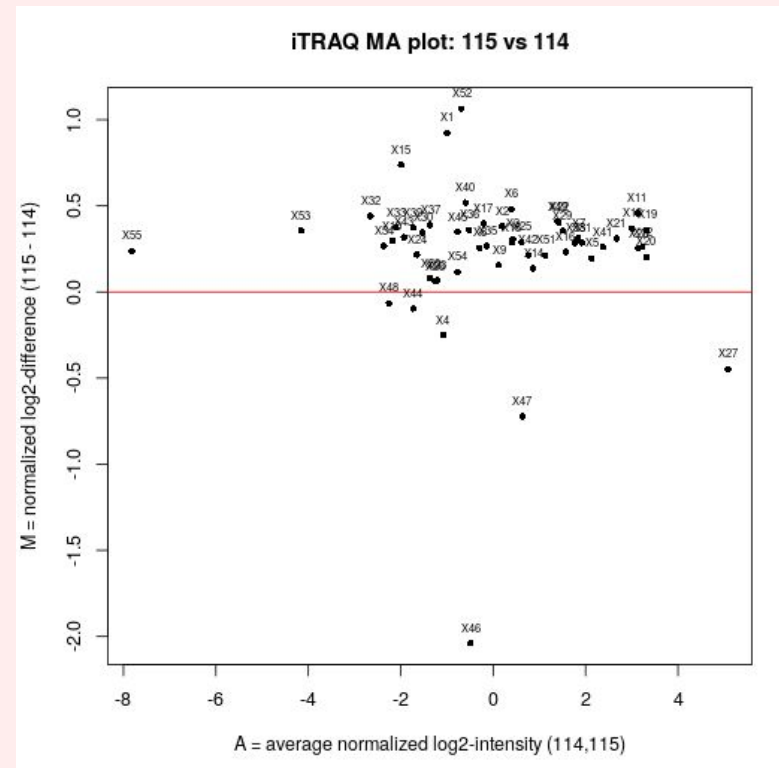
```
> #####
> # We filter out some bad data
> ok <- with(itraq_table,
+           !is.na(iTRAQ4.114) & !is.na(iTRAQ4.115))
>
> itraq_ok <- itraq_table[ok, ]
> # And generate the MA plot of 115 vs 114
>
> # And generate the MA plot of 115 vs 114
> itraq_ok$logFC_115_114 <- itraq_ok$iTRAQ4.115 - itraq_ok$iTRAQ4.114
> itraq_ok$A_115_114 <- (itraq_ok$iTRAQ4.115 + itraq_ok$iTRAQ4.114) / 2
>
> plot(itraq_ok$A_115_114, itraq_ok$logFC_115_114,
+      pch = 16, cex = 0.8,
+      xlab = "A = average normalized log2-intensity (114,115)",
+      ylab = "M = normalized log2-difference (115 - 114)",
+      main = "iTRAQ MA plot: 115 vs 114")
>
> abline(h = 0, col = "red")
>
> # Labelling the MA plot
> text(itraq_ok$A_115_114, itraq_ok$logFC_115_114,
+      labels = rownames(itraq_ok),
+      pos = 3, cex = 0.6)
```

## MA plot

M (minus) =  $\log_2(115) - \log_2(114)$

A (average) =  $[\log_2(114) + \log_2(115)] / 2$

(imitating genomic analyses)



- Most proteins do not change
- Low-intensity measurements are noisier
- A few proteins are strongly shifted
  - are those the ones carrying the spike-in differences?

# Let's play a bit with iTRAQ data

```
> plot(itraq_ok$A_115_114, itraq_ok$logFC_115_114,
+      pch = 16, cex = 0.8,
+      xlab = "A = average normalized log2-intensity (114,115)",
+      ylab = "M = normalized log2-difference (115 - 114)",
+      main = "iTRAQ MA plot: 115 vs 114")
> abline(h = 0, col = "red")

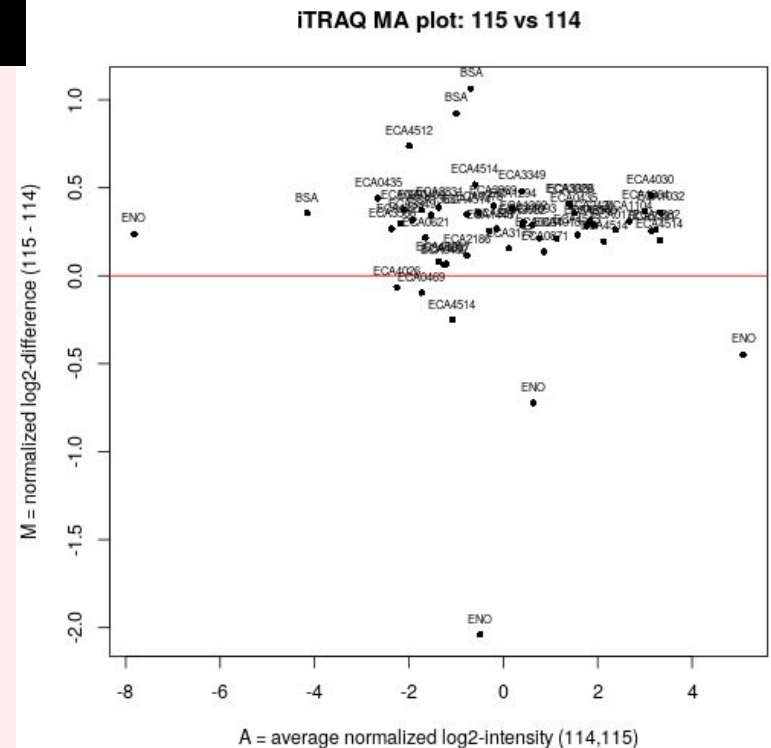
> text(itraq_ok$A_115_114, itraq_ok$logFC_115_114,
+      labels = itraq_ok$ProteinAccession,
+      pos = 3, cex = 0.6)
```

- Most proteins do not change
- Low-intensity measurements are noisier
- A few proteins are strongly shifted
  - are those the ones carrying the spike-in differences?
  - → yes, they are!

# MA plot

$$M(\text{minus}) = \log_2(115) - \log_2(114)$$
$$A(\text{average}) = [\log_2(114) + \log_2(115)] / 2$$

(imitating genomic analyses)





# Let's play a bit with iTRAQ data

```
> #####  
> # We will try to make a volcano plot at protein level  
> # In this case we will accept only one spectrum  
> # because this is a test dataset and we have almost nothing  
> prot_tab <- itraq_ok |>  
+   mutate(logratio = iTRAQ4.115 - iTRAQ4.114) |>  
+   group_by(ProteinAccession) |>  
+   summarize(  
+     n_spectra = n(),  
+     logFC = mean(logratio, na.rm = TRUE),  
+     pval = if (n() >= 2) t.test(logratio, mu = 0)$p.value else 1,  
+     .groups = "drop"  
+   ) |>  
+   mutate(neglog10p = -log10(pval))  
sel <- prot_tab$n_spectra >= 2  
  
plot(prot_tab$logFC, prot_tab$neglog10p,  
     pch = 16, cex = 0.8,  
     xlab = "Protein mean normalized log2-difference (115 - 114)",  
     ylab = "-log10 p-value",  
     main = "Protein-level volcano: 115 vs 114")  
abline(v = 0, col = "red")  
  
text(prot_tab$logFC[sel], prot_tab$neglog10p[sel],  
     labels = prot_tab$ProteinAccession[sel],  
     pos = 3, cex = 0.8)>  
> sel <- prot_tab$n_spectra >= 2  
>  
> plot(prot_tab$logFC, prot_tab$neglog10p,  
+     pch = 16, cex = 0.8,  
+     xlab = "Protein mean normalized log2-difference (115 - 114)",  
+     ylab = "-log10 p-value",  
+     main = "Protein-level volcano: 115 vs 114")  
+ abline(v = 0, col = "red")  
+  
+ text(prot_tab$logFC[sel], prot_tab$neglog10p[sel],  
+     labels = prot_tab$ProteinAccession[sel],  
+     pos = 3, cex = 0.8)
```

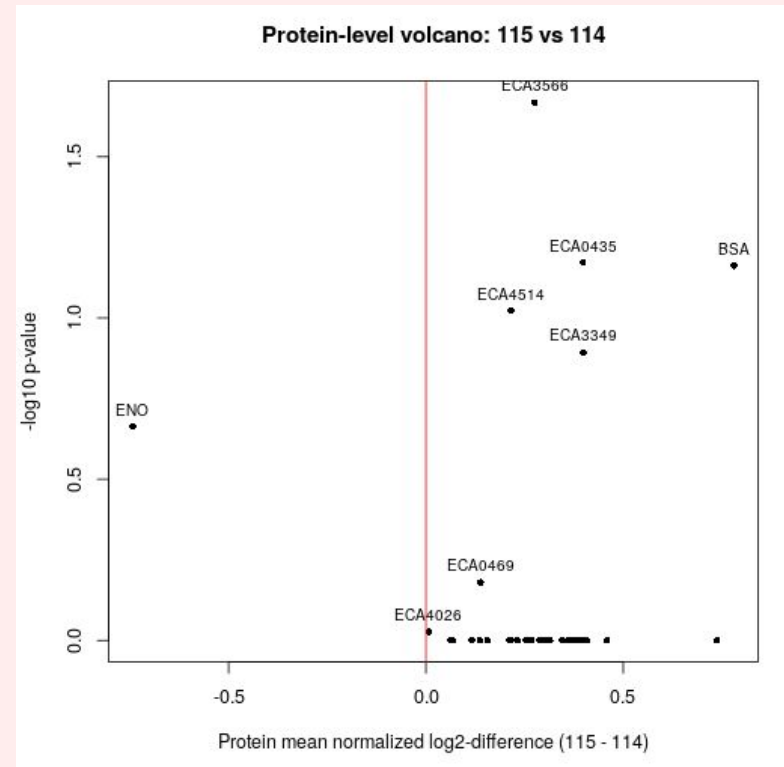
- proteins far to the left or right change more;
- proteins higher up are more statistically significant
- Note the spike in ENO and BSA

## Volcano plot

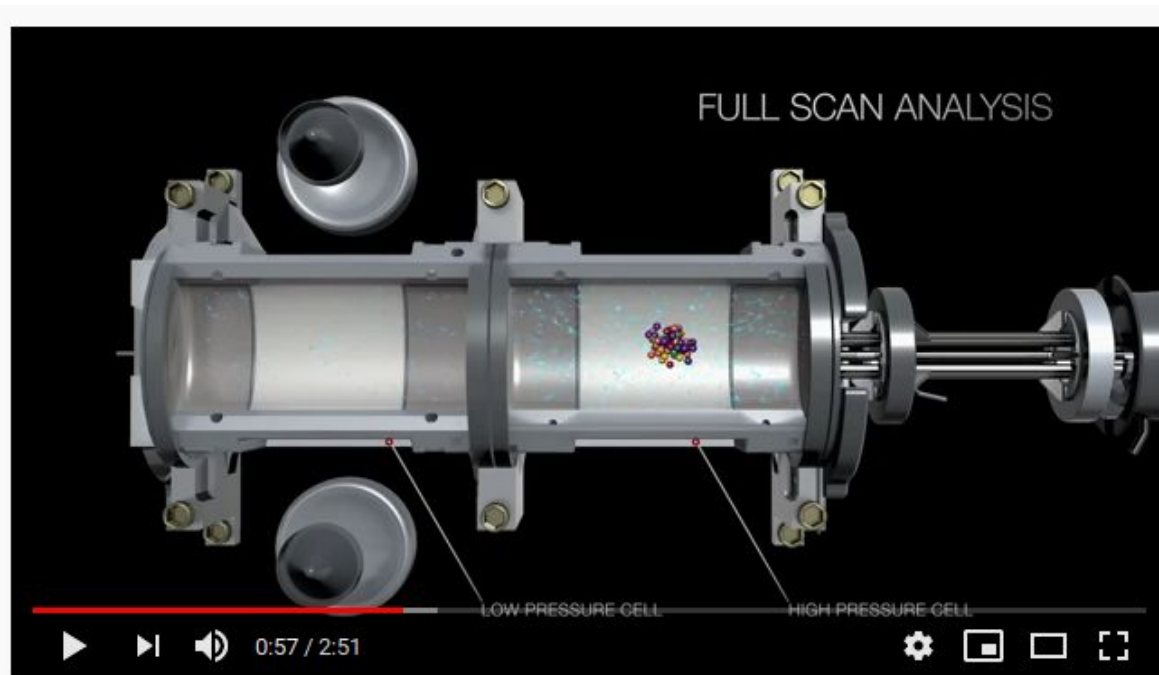
x = protein-level log2 fold-change  
=  $\text{mean}[\log_2(115) - \log_2(114)]$

y = statistical significance =  
 $-\log_{10}(\text{p-value})$

(imitating genomic analyses)



# Bonus video!



Search on YouTube *Itq velos*, or just use the link  
[https://youtu.be/EF\\_gW7jHEbY](https://youtu.be/EF_gW7jHEbY)